

MEDBRIDGE+ — DOCTOR DASHBOARD UI/UX BLUEPRINT

Clinical Consultation Workspace Design

1. DOCTOR MAIN DASHBOARD LAYOUT

|                                     |  |  |                          |  |  |
|-------------------------------------|--|--|--------------------------|--|--|
| MEDBRIDGE+   DOCTOR CLINICAL PORTAL |  |  | Dr. Jean Claude Habimana |  |  |
| District Hospital Kigali            |  |  | Cardiologist             |  |  |
| LEFT SIDEBAR                        |  |  | MAIN PATIENT WORKSPACE   |  |  |
| RIGHT SIDEBAR                       |  |  |                          |  |  |
| New Patients                        |  |  | Daily Reports            |  |  |
| Lab Exams Pending                   |  |  | • Total Seen: 24         |  |  |
| Lab Results Ready                   |  |  | • Admissions: 7          |  |  |
| Old Patients                        |  |  | • Transfers: 2           |  |  |
| Emergency Cases                     |  |  | • Deaths: 0              |  |  |
| Telehealth Consultations            |  |  | • Avg Wait: 13 mins      |  |  |
| Referrals                           |  |  |                          |  |  |

2. LEFT PANEL — NEW PATIENTS LIST

|              |  |
|--------------|--|
| NEW PATIENTS |  |
|--------------|--|

1. Uwimana Alice  
Patient ID: MBP-2026-000124  
TRIAGE: RED (Critical)  
Arrival: 08:12 AM

2. Niyonsenga Eric  
Patient ID: MBP-2026-000125  
TRIAGE: ● ORANGE (Urgent)  
Arrival: 08:16 AM

3. Mukamana Chantal  
Patient ID: MBP-2026-000126  
TRIAGE: ● YELLOW (Moderate)  
Arrival: 08:20 AM

4. Habte Jean Paul  
Patient ID: MBP-2026-000127  
TRIAGE: ● GREEN (Stable)  
Arrival: 08:26 AM

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3. TRIAGE COLOR STANDARDIZATION

| Color                                        | Priority | Meaning                |
|----------------------------------------------|----------|------------------------|
| <span style="color: red;">●</span> Red       | Critical | Immediate intervention |
| <span style="color: orange;">●</span> Orange | Urgent   | High-risk patient      |
| <span style="color: yellow;">●</span> Yellow | Moderate | Needs observation      |
| <span style="color: green;">●</span> Green   | Stable   | Non-urgent             |
| <span style="color: black;">●</span> Black   | Deceased | Mortality status       |

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4. WHEN PATIENT IS CLICKED — FULL CONSULTATION VIEW

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=====+

| PATIENT CLINICAL PROFILE |

=====+

=====+

FULL NAMES:       Uwimana Alice  
GENDER:            Female

DATE OF BIRTH: 12 March 1994  
AGE: 32 Years  
NATIONAL ID: 1199880076543210  
PATIENT ID: MBP-2026-000124  
PHONE NUMBER: +250 788 456 123  
ADDRESS: Gasabo District, Remera Sector  
HEALTH INSURANCE: RSSB / Mutuelle de Santé  
NEXT OF KIN: Uwase Marie  
EMERGENCY CONTACT: +250 788 112 334

TRIAGE STATUS:  RED — CRITICAL  
ARRIVAL MODE: Ambulance  
REFERRED FROM: Kibagabaga Health Center

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## 5. VITAL SIGNS PANEL

|             |  |
|-------------|--|
| +=====      |  |
| =====+      |  |
| VITAL SIGNS |  |
| +=====      |  |
| =====+      |  |

Blood Pressure: 85 / 50 mmHg  
Heart Rate: 128 bpm  
Respiratory Rate: 31 breaths/min  
Temperature: 39.2 °C  
Oxygen Saturation: 86%  
Blood Sugar: 142 mg/dL  
Weight: 64 Kg  
Height: 168 cm  
BMI: 22.7

### ALLERGIES:

- Penicillin
- Dust

### CHRONIC CONDITIONS:

- Hypertension
  - Asthma
-

6. CONSULTATION SUMMARY SECTION

|                      |  |  |
|----------------------|--|--|
|                      |  |  |
|                      |  |  |
| CONSULTATION SUMMARY |  |  |
|                      |  |  |
|                      |  |  |

Chief Complaint:

- Severe chest pain
- Shortness of breath
- Dizziness

History of Present Illness:

Patient reports acute chest discomfort beginning approximately 2 hours prior to arrival. Associated symptoms include diaphoresis and palpitations.

Physical Examination:

- Patient appears distressed
- Reduced oxygen saturation
- Tachycardia observed

Clinical Impression:

Possible Acute Coronary Syndrome (ACS)

7. LAB EXAMS & RESULTS TABLE

|                         |  |  |
|-------------------------|--|--|
|                         |  |  |
|                         |  |  |
| LABORATORY EXAMINATIONS |  |  |
|                         |  |  |
|                         |  |  |

|               |                   |  |
|---------------|-------------------|--|
|               |                   |  |
| EXAM          | RESULT            |  |
|               |                   |  |
| Troponin I    | Elevated          |  |
| CBC           | Mild Leukocytosis |  |
| Blood Glucose | 142 mg/dL         |  |
| ECG           | ST Elevation      |  |

|                  |          |  |
|------------------|----------|--|
| Serum Creatinine | Normal   |  |
| D-Dimer          | Negative |  |
| +-----+-----+    |          |  |

8. RADIOLOGY & IMAGING PANEL

|                     |  |
|---------------------|--|
| +=====+             |  |
| =====+              |  |
| RADIOLOGY & IMAGING |  |
| +=====+             |  |
| =====+              |  |

Chest X-Ray:

- Mild pulmonary congestion observed

Echocardiogram:

- Reduced left ventricular function

DICOM FILES:

[ View Images ] [ Download ]

9. PRESCRIBED MEDICATIONS TABLE

|                        |  |
|------------------------|--|
| +=====+                |  |
| =====+                 |  |
| PRESCRIBED MEDICATIONS |  |
| +=====+                |  |
| =====+                 |  |

|                           |         |       |               |            |
|---------------------------|---------|-------|---------------|------------|
| +-----+-----+-----+-----+ |         |       |               |            |
| MEDICATION                | DOSE    | ROUTE | FREQUENCY/DAY | TOTAL DAYS |
| +-----+-----+-----+-----+ |         |       |               |            |
| Aspirin                   | 75 mg   | Oral  | 1             | 30         |
| Clopidogrel               | 75 mg   | Oral  | 1             | 30         |
| Heparin                   | 5000 IU | IV    | 2             | 5          |
| Atorvastatin              | 20 mg   | Oral  | 1             | 30         |
| +-----+-----+-----+-----+ |         |       |               |            |

=====+  
=====+  
| ACTS PERFORMED |  
=====+  
=====+

=====+  
=====+  
| CONSUMED MATERIALS |  
=====+  
=====+

| FINAL PATIENT STATUS |  |
|----------------------|--|
|                      |  |

+=====+  
=====+

- ☐ Internal Transfer
- ☒ External Transfer
- ☒ Transfer with Ambulance
- ☐ Discharged
- ☐ Escaped
- ☐ Died

TRANSFER FACILITY:

King Faisal Hospital

TRANSFER REASON:

Advanced Cardiology Intervention Required

TRANSFER TIME:

11:48 AM

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### 13. RIGHT SIDEBAR — REAL-TIME REPORTS

+=====+  
|  REAL-TIME DOCTOR REPORTS |  
+=====+

#### TODAY SUMMARY

-----  
Patients Seen:           24  
Emergency Cases:       6  
Admissions:            7  
Transfers:              2  
Deaths:                 0

#### CONSULTATION ANALYTICS

-----  
Average Consultation Time:   14 mins  
Average Waiting Time:       11 mins  
Pending Results:            5  
Pending Referrals:           3

AI ALERTS

- ⚠ Rising respiratory cases detected
- ⚠ ICU occupancy above 82%
- ⚠ Delayed lab turnaround in Hematology

14. SMART AI CLINICAL ASSISTANT PANEL

|                       |  |
|-----------------------|--|
| +=====                |  |
| =====+                |  |
| AI CLINICAL ASSISTANT |  |
| +=====                |  |
| =====+                |  |

PREDICTED RISK SCORE:  
HIGH RISK — 87%

- AI SUGGESTIONS:
- Recommend immediate cardiology referral
  - High probability Acute Coronary Syndrome
  - Suggest Troponin repeat after 6 hours
  - Monitor oxygen saturation continuously

DRUG INTERACTION CHECK:  
 No dangerous interactions detected

15. TELEHEALTH CONSULTATION PANEL

|                    |  |
|--------------------|--|
| +=====             |  |
| =====+             |  |
| TELEHEALTH SESSION |  |
| +=====             |  |
| =====+             |  |

Patient Type: Remote Consultation  
Connection Quality: Excellent  
Language: Kinyarwanda  
AI Translator: ACTIVE



- [ Start Video ]
  - [ Voice Call ]
  - [ Share Screen ]
  - [ Upload Prescription ]
- 

## 16. NATIONAL INTEROPERABILITY VIEW

|                         |  |
|-------------------------|--|
| +=====                  |  |
| =====+                  |  |
| NATIONAL HEALTH NETWORK |  |
| +=====                  |  |
| =====+                  |  |

Previous Facilities Visited:

- Kibagabaga Hospital
- CHUK
- Rwanda Military Hospital

Previous Diagnoses:

- Hypertension
- Asthma

Cross-Hospital Records:

SYNCED SUCCESSFULLY 

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## 17. DOCTOR MOBILE TABLET VIEW

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## 18. FINAL MEDBRIDGE+ CLINICAL DASHBOARD CONCEPT

LEFT PANEL

-----

Patient Queues  
Lab Results  
Emergency Alerts  
Referrals  
Telehealth

CENTER PANEL

-----  
Patient Clinical Workspace  
Vitals  
Lab Results  
Prescriptions  
Clinical Notes

RIGHT PANEL

-----  
Reports  
Analytics  
AI Alerts  
Hospital Capacity  
Notifications

MEDBRIDGE+ — NURSE DASHBOARD UI/UX BLUEPRINT

Smart Nursing Clinical Workspace Design

1. NURSE MAIN DASHBOARD

|                                     |                                                                                                     |               |
|-------------------------------------|-----------------------------------------------------------------------------------------------------|---------------|
| +=====+<br>=====+                   |                                                                                                     |               |
| MEDBRIDGE+   NURSING PORTAL         | Nurse Diane Uwase                                                                                   |               |
| Kigali University Teaching Hospital | Emergency Unit                                                                                      |               |
| +=====+<br>=====+                   |                                                                                                     |               |
| +-----+-----+-----+                 |                                                                                                     |               |
| LEFT SIDEBAR                        | MAIN NURSING WORKSPACE                                                                              | RIGHT SIDEBAR |
| +-----+-----+-----+                 |                                                                                                     |               |
| ● New Patients                      |  Nursing Reports |               |
| ● Vitals Pending                    | • Patients Assigned: 18                                                                             |               |
| ● Medication Schedule               | • Critical Cases: 3                                                                                 |               |
| ● Emergency Patients                | • Pending Vitals: 5                                                                                 |               |
| ● Specimen Collection               | • Medications Due: 7                                                                                |               |
| ● Inpatients Monitoring             | • Transfers Today: 2                                                                                |               |

2. LEFT PANEL — NEW PATIENTS

+-----+

● NEW PATIENTS

+-----+

1. Uwimana Alice

Patient ID: MBP-2026-000124

TRIAGE: ● RED (Critical)

Bed: Emergency Room 2
2. Niyonsenga Eric

Patient ID: MBP-2026-000125

TRIAGE: ● ORANGE (Urgent)

Bed: Ward A-14
3. Mukamana Chantal

Patient ID: MBP-2026-000126

TRIAGE: ● YELLOW (Moderate)

Bed: Ward B-08

3. TRIAGE COLOR STANDARDIZATION

| Color    | Priority | Meaning                |
|----------|----------|------------------------|
| ● Red    | Critical | Immediate intervention |
| ● Orange | Urgent   | Requires rapid care    |
| ● Yellow | Moderate | Observation required   |
| ● Green  | Stable   | Routine care           |
| ● Black  | Deceased | Mortality status       |

#### 4. WHEN PATIENT IS CLICKED — NURSING PROFILE

|                         |  |
|-------------------------|--|
| =====                   |  |
| =====+                  |  |
| PATIENT NURSING PROFILE |  |
| =====                   |  |
| =====+                  |  |

FULL NAMES:       Uwimana Alice  
GENDER:            Female  
AGE:                32 Years  
PATIENT ID:        MBP-2026-000124  
NATIONAL ID:       1199880076543210  
ADDRESS:           Gasabo District, Remera  
PHONE NUMBER:     +250 788 456 123

HEALTH INSURANCE:   RSSB / Mutuelle de Santé  
WARD/BED:           Emergency Room 2  
ATTENDING DOCTOR:   Dr. Eric Murenzi

TRIAGE STATUS:       ● RED — CRITICAL  
ADMISSION STATUS:   Admitted

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#### 5. VITAL SIGNS PANEL

|             |  |
|-------------|--|
| =====       |  |
| =====+      |  |
| VITAL SIGNS |  |
| =====       |  |
| =====+      |  |

Blood Pressure:     85 / 50 mmHg  
Heart Rate:         128 bpm  
Respiratory Rate:   31 breaths/min  
Temperature:        39.2 °C  
Oxygen Saturation:   86 %  
Blood Sugar:         142 mg/dL  
Weight:             64 Kg  
Height:              168 cm

LAST UPDATED:  
08:42 AM by Nurse Diane Uwase

6. AI EARLY WARNING SYSTEM

|                         |  |
|-------------------------|--|
| +=====+                 |  |
| =====+                  |  |
| AI NURSING ALERT SYSTEM |  |
| +=====+                 |  |
| =====+                  |  |

 HIGH RISK ALERT  
Possible Sepsis Detected

- AI OBSERVATIONS:
- Elevated heart rate
  - Low blood pressure
  - Fever above 39°C
  - Reduced oxygen saturation

- AI RECOMMENDATIONS:
- ☒ Notify doctor immediately
  - ☒ Repeat vitals every 15 minutes
  - ☒ Prepare IV fluids
  - ☒ Monitor oxygen closely

7. MEDICATION ADMINISTRATION RECORD (MAR)

|                                  |  |
|----------------------------------|--|
| +=====+                          |  |
| =====+                           |  |
| MEDICATION ADMINISTRATION RECORD |  |
| +=====+                          |  |
| =====+                           |  |

|                                 |      |       |      |        |
|---------------------------------|------|-------|------|--------|
| +-----+-----+-----+-----+-----+ |      |       |      |        |
| MEDICATION                      | DOSE | ROUTE | TIME | STATUS |

|                |         |      |            |                                                                                   |              |  |
|----------------|---------|------|------------|-----------------------------------------------------------------------------------|--------------|--|
| Aspirin        | 75 mg   | Oral | 09:00 AM   | <input checked="" type="checkbox"/>                                               | Administered |  |
| Heparin        | 5000 IU | IV   | 09:15 AM   | <input checked="" type="checkbox"/>                                               | Administered |  |
| Oxygen Therapy | 5 L/min | Mask | Continuous | <input checked="" type="checkbox"/>                                               | Active       |  |
| Paracetamol    | 1 g     | IV   | 10:00 AM   |  | Pending      |  |

8. NURSING NOTES SECTION

|               |  |
|---------------|--|
| NURSING NOTES |  |
|---------------|--|

08:40 AM  
Patient received in critical condition with severe chest pain.

08:55 AM  
Oxygen therapy initiated.

09:10 AM  
IV access secured successfully.

09:18 AM  
Patient responding positively to stabilization measures.

[ ADD NEW NOTE ]

9. SPECIMEN COLLECTION PANEL

|                     |  |
|---------------------|--|
| SPECIMEN COLLECTION |  |
|---------------------|--|

| REQUESTED TEST | SAMPLE TYPE | STATUS      |
|----------------|-------------|-------------|
| CBC            | Blood       | ✓ Collected |
| Troponin I     | Blood       | ✓ Collected |
| Blood Glucose  | Blood       | ⌚ Pending   |

BARCODE GENERATED:  
LAB-2026-883712

10. PATIENT MONITORING PANEL

|                               |
|-------------------------------|
| CONTINUOUS PATIENT MONITORING |
|-------------------------------|

CARDIAC MONITOR:  
⚠ Tachycardia Detected

OXYGEN SATURATION TREND:  
86% → 90% → 93%

PAIN SCORE:  
8 / 10

FALL RISK:  
HIGH RISK ⚠

PRESSURE ULCER RISK:  
MODERATE

11. FLUIDS & INPUT/OUTPUT CHART

|                      |  |
|----------------------|--|
| INPUT / OUTPUT CHART |  |
|----------------------|--|

| TYPE         | AMOUNT  | TIME     |
|--------------|---------|----------|
| IV Fluids    | 1000 mL | 09:05 AM |
| Oral Intake  | 250 mL  | 10:00 AM |
| Urine Output | 400 mL  | 10:30 AM |

## 12. CONSUMABLE MATERIALS USED

|                    |  |
|--------------------|--|
| CONSUMED MATERIALS |  |
|--------------------|--|

| MATERIAL           | QUANTITY |
|--------------------|----------|
| IV Cannulas        | 2        |
| Syringes 10ml      | 5        |
| Examination Gloves | 10 pairs |
| Oxygen Mask        | 1        |
| ECG Electrodes     | 6        |

## 13. PATIENT STATUS & OUTCOME PANEL

|                |  |
|----------------|--|
| PATIENT STATUS |  |
|----------------|--|



+=====+  
=====+

- ☒ Admitted
- ☐ Internal Transfer
- ☐ External Transfer
- ☐ Transfer with Ambulance
- ☐ Discharged
- ☐ Escaped
- ☐ Died

CURRENT LOCATION:  
Emergency Ward — Bed 2

NEXT REVIEW:  
10:30 AM

\_\_\_\_\_

#### 14. SHIFT HANDOVER PANEL

+=====+  
=====+

| SHIFT HANDOVER SUMMARY |

+=====+  
=====+

Outgoing Nurse:  
Nurse Diane Uwase

Incoming Nurse:  
Nurse Alice Mukamana

PATIENT CONDITION:  
Improving but remains high-risk.

- IMPORTANT NOTES:
- Continue oxygen monitoring
  - Repeat vitals every 15 mins
  - Await repeat Troponin result

---

## 15. RIGHT SIDEBAR — NURSING ANALYTICS

|                                                                                                   |  |
|---------------------------------------------------------------------------------------------------|--|
| +=====+                                                                                           |  |
|  NURSING REPORTS |  |
| +=====+                                                                                           |  |

### SHIFT SUMMARY

|                           |    |
|---------------------------|----|
| -----                     |    |
| Patients Assigned:        | 18 |
| Critical Patients:        | 3  |
| Medications Administered: | 42 |
| Pending Medications:      | 7  |

### AI ALERTS

- 
- ⚠ Increased respiratory distress cases
  - ⚠ Ward A nearing full capacity
  - ⚠ Delayed specimen collection detected

### WORKLOAD STATUS

-----

Current Workload: HIGH  
Fatigue Risk Score: Moderate

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## 16. SMART NURSING ASSISTANT

|                            |  |
|----------------------------|--|
| +=====+                    |  |
| =====+                     |  |
| AI SMART NURSING ASSISTANT |  |
| +=====+                    |  |
| =====+                     |  |

PATIENT RISK SCORE:  
HIGH RISK — 84%

- AI SUGGESTIONS:
- Monitor BP every 10 mins

- Prepare emergency crash cart
- Consider ICU escalation
- Maintain continuous oxygen therapy

MEDICATION SAFETY:

 No adverse interaction detected

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## 17. MOBILE NURSING TABLET VIEW

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### 18. FINAL NURSE DASHBOARD STRUCTURE

#### LEFT PANEL

-----

New Patients  
Vitals Pending  
Medication Schedule  
Specimen Collection  
Emergency Cases  
Shift Handover

#### CENTER PANEL

-----

Patient Profile  
Vitals  
Medication Administration  
Nursing Notes  
Monitoring Charts  
Input/Output

#### RIGHT PANEL

-----

Nursing Reports  
AI Alerts  
Ward Capacity  
Workload Analytics  
Notifications

MEDBRIDGE+ — LAB TECHNICIAN DASHBOARD UI/UX BLUEPRINT

Laboratory Information System (LIS) Workspace Design

1. LAB TECHNICIAN MAIN DASHBOARD

|                                     |  |  |                           |  |  |
|-------------------------------------|--|--|---------------------------|--|--|
| MEDBRIDGE+   LABORATORY PORTAL      |  |  | Mr. Patrick Nshimiyimana  |  |  |
| Kigali University Teaching Hospital |  |  | Senior Lab Technician     |  |  |
| LEFT SIDEBAR                        |  |  | MAIN LAB WORKSPACE        |  |  |
| ● New Lab Orders                    |  |  | 📊 Lab Reports             |  |  |
| ● Samples Awaiting Results          |  |  | • Pending Tests: 38       |  |  |
| ● Completed Results                 |  |  | • Completed Today: 112    |  |  |
| ● Critical Samples                  |  |  | • Critical Alerts: 4      |  |  |
| ● Emergency STAT Orders             |  |  | • Avg Turnaround: 19 mins |  |  |
| ● Archived Exams                    |  |  | • Delayed Samples: 3      |  |  |

2. LEFT PANEL — NEW LAB ORDERS

|                  |  |
|------------------|--|
| ● NEW LAB ORDERS |  |
|------------------|--|

1. Uwimana Alice
- Patient ID: MBP-2026-000124

Ordered By: Dr. Eric Murenzi

Ordered By: Dr. Diane Uwase

Ordered By: Dr. Jean Bosco

| Color  | Priority         | Meaning              |
|--------|------------------|----------------------|
| Red    | STAT             | Immediate processing |
| Orange | Urgent           | Fast-track           |
| Yellow | Routine          | Standard queue       |
| Green  | Low Priority     | Non-urgent           |
| Black  | Rejected/Expired | Invalid sample       |

+=====+  
==+=+  
| LAB SAMPLE PROFILE |  
+=====+  
==+=+

FULL NAMES: Uwimana Alice

GENDER: Female  
AGE: 32 Years  
PATIENT ID: MBP-2026-000124  
NATIONAL ID: 1199880076543210  
HEALTH INSURANCE: RSSB / Mutuelle de Santé

REQUESTING DOCTOR: Dr. Eric Murenzi  
REQUESTING DEPARTMENT: Emergency Department  
REQUEST DATE: 22 May 2026  
REQUEST TIME: 08:22 AM

TRIAGE STATUS:  RED — STAT PRIORITY  
SAMPLE TYPE: Blood  
BARCODE ID: LAB-2026-883712

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## 5. PATIENT LIMITED CLINICAL SNAPSHOT

**(ONLY NECESSARY INFORMATION — PRIVACY PROTECTED)**

+=====+  
=====+  
| LIMITED CLINICAL INFORMATION |  
+=====+  
=====+

Chief Complaint:

- Severe chest pain
- Shortness of breath

Known Allergies:

- Penicillin

Infection Risk Alerts:

- ☒ Hepatitis B Precaution

Clinical Notes:

- △ Minimal summary only visible to lab personnel
-

6. LAB EXAMINATION TABLE

|                          |  |
|--------------------------|--|
| ORDERED LAB EXAMINATIONS |  |
|--------------------------|--|

|                            |           |          |
|----------------------------|-----------|----------|
| EXAM                       | STATUS    | PRIORITY |
| Troponin I                 | Pending   | STAT     |
| Complete Blood Count (CBC) | Running   | STAT     |
| Blood Glucose              | Completed | Routine  |
| Serum Creatinine           | Pending   | Urgent   |

7. LAB RESULT ENTRY PANEL

|              |
|--------------|
| RESULT ENTRY |
|--------------|

SELECT TEST:  
[ Troponin I ▼ ]

RESULT VALUE:  
[ 2.87 ng/mL ]

REFERENCE RANGE:  
0.00 — 0.04 ng/mL

RESULT STATUS:  
☒ Critical Value  
☐ Normal  
☐ Borderline

TECHNICIAN COMMENTS:  
Elevated cardiac biomarker detected.

[ SAVE RESULT ]  
[ VALIDATE ]  
[ SEND TO DOCTOR ]

8. CRITICAL VALUE ALERT SYSTEM

|                                                                                                         |  |
|---------------------------------------------------------------------------------------------------------|--|
|                                                                                                         |  |
|                                                                                                         |  |
|  CRITICAL RESULT ALERT |  |
|                                                                                                         |  |

PATIENT:  
Uwimana Alice

CRITICAL FINDING:  
Troponin I Extremely Elevated

- AUTO ACTIONS:
- ☒ Doctor notified immediately
  - ☒ Emergency dashboard updated
  - ☒ Case escalated to cardiology
  - ☒ RBC surveillance AI notified

9. SAMPLE TRACKING PANEL

|                 |  |
|-----------------|--|
|                 |  |
|                 |  |
| SAMPLE TRACKING |  |
|                 |  |

|       |        |           |  |
|-------|--------|-----------|--|
|       |        |           |  |
| STAGE | STATUS | TIMESTAMP |  |



|                     |           |          |
|---------------------|-----------|----------|
| Sample Collected    | Completed | 08:30 AM |
| Sample Received     | Completed | 08:36 AM |
| Sample Processing   | Running   | 08:40 AM |
| Validation          | Pending   | --       |
| Doctor Notification | Pending   | --       |

10. LAB INVENTORY & REAGENT MONITORING

|                    |
|--------------------|
| LAB REAGENT STATUS |
|--------------------|

| REAGENT            | STOCK LEVEL  | ALERT       |
|--------------------|--------------|-------------|
| Troponin Test Kits | 12 Remaining | ⚠ Low Stock |
| CBC Reagent        | 45 Remaining | ✅ Normal    |
| Glucose Strips     | 5 Remaining  | 🔴 Critical  |
| Malaria Test Kits  | 28 Remaining | ✅ Normal    |

11. AI LAB ASSISTANT PANEL

|                         |
|-------------------------|
| AI LABORATORY ASSISTANT |
|-------------------------|

AI ANALYSIS:  
 ⚠ Elevated Troponin Pattern Suggestive of Acute Coronary Syndrome

QUALITY CONTROL:

- ✓ Sample integrity acceptable
- ✓ No hemolysis detected

#### AI RECOMMENDATIONS:

- Repeat Troponin after 6 hours
- Suggest urgent cardiology review

#### OUTBREAK ALERTS:

⚠ Increased malaria positives detected in Gasabo District

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## 12. LAB PERFORMANCE REPORTS

|                                                                                                 |  |
|-------------------------------------------------------------------------------------------------|--|
| =====+                                                                                          |  |
|  LAB ANALYTICS |  |
| =====+                                                                                          |  |

#### TODAY SUMMARY

-----

|                     |         |
|---------------------|---------|
| Samples Processed:  | 112     |
| Critical Results:   | 8       |
| Rejected Samples:   | 3       |
| Average Turnaround: | 19 mins |

#### MOST REQUESTED TESTS

-----

|               |    |
|---------------|----|
| CBC           | 38 |
| Malaria       | 26 |
| Blood Glucose | 22 |
| Troponin I    | 14 |

#### AI SYSTEM ALERTS

- 
- ⚠ Rising respiratory infections
  - ⚠ Low stock on glucose strips
  - ⚠ Increased emergency STAT tests
-

### 13. SAMPLE REJECTION PANEL

|                  |  |
|------------------|--|
| +=====           |  |
| =====+           |  |
| SAMPLE REJECTION |  |
| +=====           |  |
| =====+           |  |

REJECTION REASON:

- ☒ Hemolyzed Sample
- ☐ Insufficient Volume
- ☐ Wrong Patient Label
- ☐ Expired Collection Time

TECHNICIAN COMMENT:

Sample integrity compromised.

AUTO ACTION:

- ☒ Doctor notified
- ☒ Repeat collection requested

---

### 14. LAB SECURITY & PRIVACY MODEL

|                    |  |
|--------------------|--|
| +=====             |  |
| =====+             |  |
| ACCESS LIMITATIONS |  |
| +=====             |  |
| =====+             |  |

ALLOWED:

- ☒ View ordered laboratory requests
- ☒ Enter and validate results
- ☒ View minimal patient clinical context
- ☒ Track sample workflow

PROHIBITED:

- ☒ Access psychiatric records
- ☒ Modify prescriptions
- ☒ Access billing data

- ☒ View unrelated patient history
- ☒ Delete finalized results

---

## 15. NATIONAL LAB INTEROPERABILITY

|                      |  |
|----------------------|--|
| +=====               |  |
| =====+               |  |
| NATIONAL LAB NETWORK |  |
| +=====               |  |
| =====+               |  |

### CONNECTED FACILITIES:

- CHUK Central Laboratory
- Rwanda Biomedical Center
- King Faisal Hospital Lab
- National Reference Laboratory

### SYNC STATUS:

- ☒ National synchronization successful

---

## 16. MOBILE LAB TABLET INTERFACE

## 17. FINAL LAB TECHNICIAN DASHBOARD STRUCTURE

### LEFT PANEL

-----

New Lab Orders  
STAT Cases  
Pending Samples  
Completed Results  
Archived Exams

### CENTER PANEL

-----

Patient Sample Profile

Lab Orders  
Result Entry  
Sample Tracking  
Validation Workspace

RIGHT PANEL

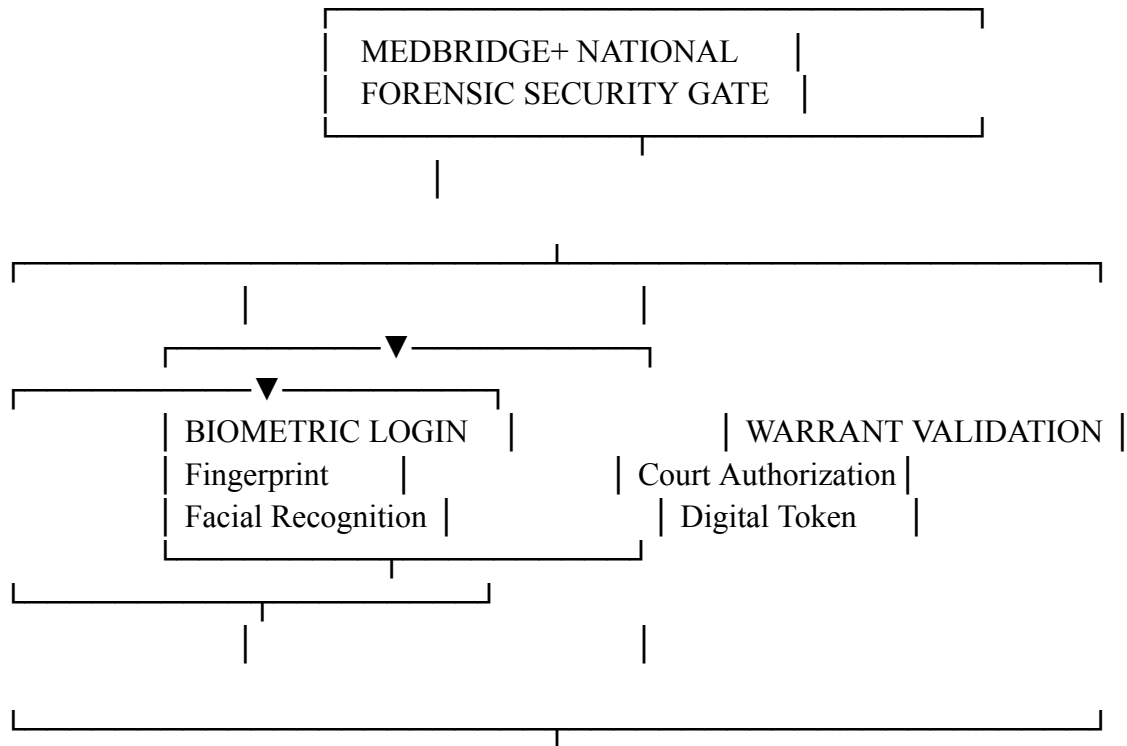
-----  
Lab Analytics  
AI Alerts  
Reagent Stock  
Turnaround Reports  
Outbreak Notifications

 **MEDBRIDGE+ — RIB FORENSIC INVESTIGATIVE SYSTEM**

**FULL GRAPHICAL WORKFLOW & DASHBOARD STRUCTURE**

---

**1. NATIONAL FORENSIC ENTRY FLOW**



## 2. RIB FORENSIC HOME PAGE (FULL GRAPHIC)

MEDBRIDGE+ — RIB NATIONAL FORENSIC INVESTIGATIVE TERMINAL

Investigator: Emmanuel Habiyambere

WARRANT STATUS: ACTIVE

Badge ID: RIB-44291

Session Encryption: ENABLED

Death Case Audits | Violence & Assault | Privacy Breach Tracking |  
Suspicious deaths | GBV / Assault / Trauma | Insider leaks & invasions |  
Mortality investigations | Court evidence | Unauthorized access tracing |

Malpractice Reports | Isange Child Protection | Evidence Vault |  
Negligence investigations | Under-18 abuse protection | Blockchain legal storage |

### RIGHT PANEL

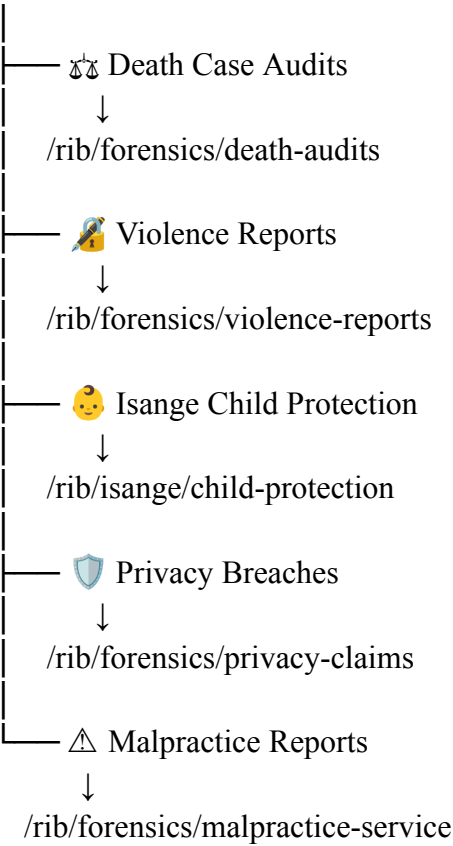
|                         |    |
|-------------------------|----|
| Active Cases:           | 14 |
| Death Audits:           | 3  |
| GBV Cases:              | 4  |
| Child Protection Cases: | 5  |
| Privacy Breaches:       | 2  |

### AI ALERTS

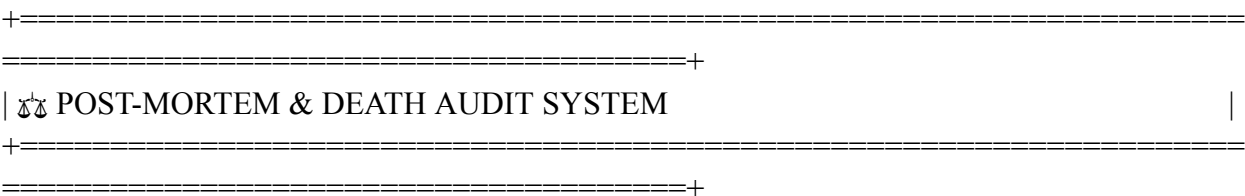
- △ ICU mortality spike detected
- △ Unauthorized midnight access
- △ High-risk GBV recurrence

3. CLICK FLOW NAVIGATION

HOME PAGE



4. DEATH CASE AUDIT DASHBOARD



LEFT PANEL

CENTER PANEL

RIGHT PANEL

 Critical Deaths  
ALERTS

 Toxicology Cases

 VITAL COLLAPSE TIMELINE

BP Drop vs Medication Delay

AI FORENSIC

 Delayed response

Pending Autopsies

Closed Cases

Oxygen Saturation Collapse

Cardiac Arrest Progression

⚠

Missing escalation

⚠

Medication omission

CASE: RIB-2026-00321

PATIENT:  
Jean Claude Ndayisaba  
Male | 47 Years  
Patient ID: MBP-7789912

| TIME  | VITAL STATUS   | MEDICATION EVENT     |
|-------|----------------|----------------------|
| 03:00 | Stable         | Aspirin Ordered      |
| 04:00 | Oxygen Falling | ❌ Medication Delayed |
| 05:00 | Critical       | Heparin Administered |
| 06:00 | Cardiac Arrest | ⚠ Emergency Gap      |

TOXICOLOGY:

- Troponin: HIGH
- Acidosis detected
- Potassium critically low

FINAL STATUS:  
🚒 UNDER FORENSIC REVIEW

5. VIOLENCE & GBV INVESTIGATION PANEL

|                                             |
|---------------------------------------------|
| 🔪 FORENSIC VIOLENCE & ASSAULT INVESTIGATION |
|---------------------------------------------|

CASE ID: GBV-2026-1177



---

---

---

COURT STATUS  
⚖️ Active Trial  
✓ Evidence Signed  
✓ Witness Secured

Uwimana Alice  
Female | 26 Years

- Arm bruising
- Rib tenderness
- Defensive injuries

- Severe distress
- Fear response
- PTSD indicators

## ACTIVE GBV INVESTIGATION

=====+

| 🧠 ISANGE ONE STOP CENTER — CHILD FORENSIC PROTECTION

|

=====+

=====+

RIGHT PANEL

---

---

 Emergency Cases  
STATUS

-  Sexual Abuse
-  Physical Violence
-  Neglect Cases
-  Safe Cases

 CHILD PROTECTION PROFILE

Identity Hidden  
Trauma Documentation  
Psychological Assessment  
Guardian Validation

PROTECTION

- ✓ Shelter Activated
- ✓ Psychologist Assigned
- ✓ Court Protection
- ✓ Medical Care Active

CASE ID: ISG-2026-00411

CHILD:

[ IDENTITY PROTECTED ]

Female | 13 Years

FORENSIC FINDINGS:

- Soft tissue trauma
- Emotional distress
- Emergency prophylaxis provided

BODY MAP:

---

Head ✓  
Arms ✓  
Torso ✓  
Lower Body ✓

---

LEGAL STATUS:

 HIGH PRIORITY CHILD PROTECTION

---

## 7. PRIVACY BREACH INVESTIGATION TERMINAL

+=====

=====+

|  DATA PRIVACY BREACH & CYBER FORENSICS

|

+=====

=====+

LEFT PANEL

- Insider Threats
- Unauthorized Access
- Export Attempts
- Resolved Cases

CENTER PANEL

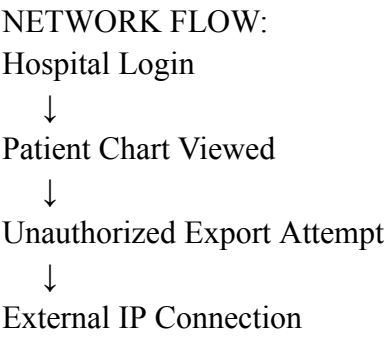
- ACCESS NODE GRAPH
- User Tracking
- IP Correlation
- Device Trace

RIGHT PANEL

- AI CYBER ENGINE
- Suspicious behavior
- Midnight intrusion
- Data exfiltration

TARGET RECORD:  
Patient ID: MBP-998211

| USER ACCOUNT | LOCATION | STATUS       |
|--------------|----------|--------------|
| USER-7721    | Kigali   | Unauthorized |
| USER-1183    | Musanze  | Suspicious   |
| USER-0091    | CHUK     | Violation    |



FINAL STATUS:  
ACTIVE CYBER FORENSIC INVESTIGATION

8. MALPRACTICE INVESTIGATION DASHBOARD

|                                          |
|------------------------------------------|
| CLINICAL NEGLIGENCE & MALPRACTICE SYSTEM |
|------------------------------------------|

FACILITY:  
District Hospital

LEFT PANEL

CENTER PANEL

RIGHT PANEL

|                                             |                                             |                                            |
|---------------------------------------------|---------------------------------------------|--------------------------------------------|
| <div><div></div>Critical Delays</div>       | <div><div>🕒</div>SERVICE TIMELINE</div>     | <div><div></div>NEGLIGENCE SCORE</div>     |
| <div><div></div>Medication Omissions</div>  | <div><div></div>Triage Delay Analysis</div> | <div><div></div>HIGH RISK</div>            |
| <div><div></div>Attendance Gaps</div>       | <div><div></div>Provider Attendance</div>   | <div><div>⚠️</div>Staffing shortage</div>  |
| <div><div></div>Closed Investigations</div> | <div><div></div>Emergency Response</div>    | <div><div>⚠️</div>Protocol deviation</div> |

|                     |           |           |  |
|---------------------|-----------|-----------|--|
| +-----+-----+-----+ |           |           |  |
| EVENT               | EXPECTED  | ACTUAL    |  |
| +-----+-----+-----+ |           |           |  |
| Triage              | 5 min     | 42 min ❌  |  |
| Doctor Review       | 15 min    | 1h 15m ❌  |  |
| Antibiotics         | Immediate | Delayed ❌ |  |
| +-----+-----+-----+ |           |           |  |

FINAL STATUS:  
🚨 NATIONAL QUALITY ESCALATION

9. FORENSIC EVIDENCE VAULT

|   |                                   |  |
|---|-----------------------------------|--|
| + | =====+                            |  |
| + | =====+                            |  |
|   | 🔒 NATIONAL DIGITAL EVIDENCE VAULT |  |
| + | =====+                            |  |
| + | =====+                            |  |

EVIDENCE STORAGE STRUCTURE

- 📁 Medical Records
- 📁 Toxicology Reports
- 📁 Radiology Images
- 📁 Psychosocial Statements
- 📁 Audio Interviews
- 📁 CCTV Evidence
- 📁 Provider Audit Logs

---

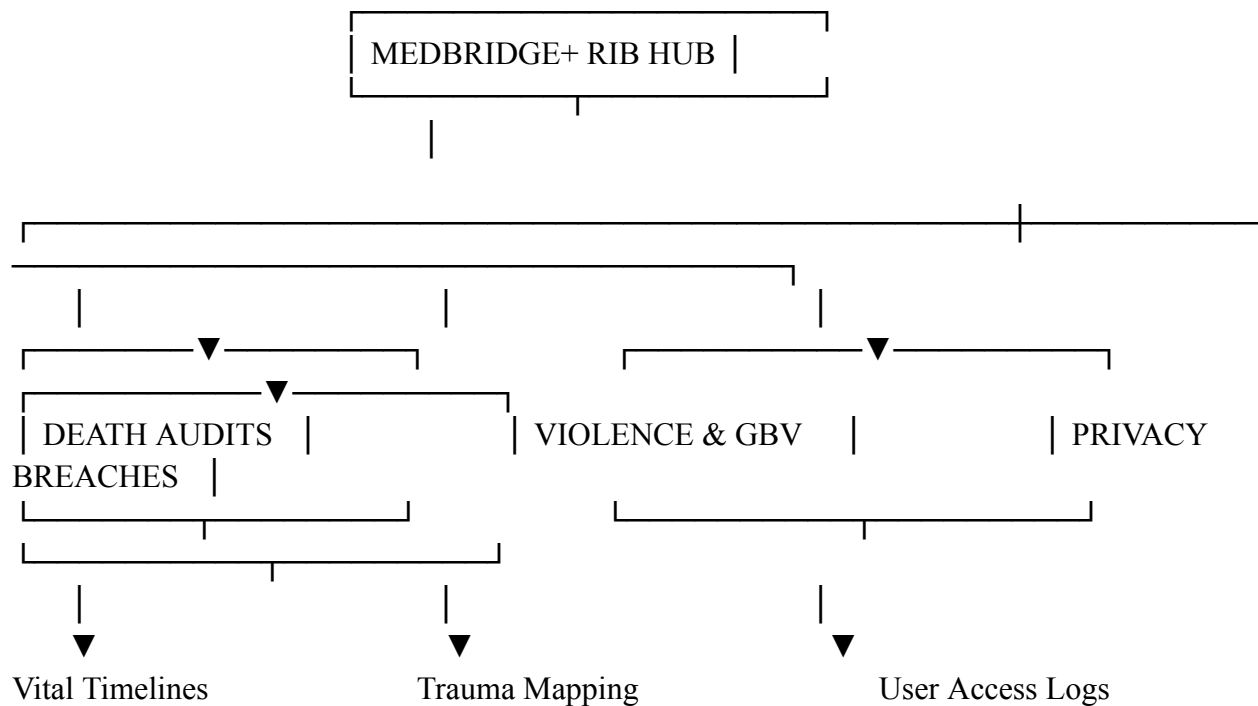
## INTEGRITY VERIFICATION

- ✓ Blockchain Signed
  - ✓ Hash Protected
  - ✓ Tamper Proof
  - ✓ Court Admissible
  - ✓ Time Stamped
- 

## EXPORT POLICY

- ✗ No public downloads
  - ✗ No unrestricted browsing
  - ✓ Warrant-based viewing only
  - ✓ Every click permanently logged
- 

## 10. FULL NATIONAL FORENSIC ECOSYSTEM



IP Tracking  
AI Threat Detection



**National Nursing & Midwifery Regulation, Licensing & Workforce Governance Command Center**

MEDBRIDGE+ — RWANDA NATIONAL COUNCIL OF NURSES & MIDWIVES (NCNM)  
National Licensing, Professional Regulation & Workforce Governance Platform

Registrar General: Prof. Chantal Mukarugwiza  
Council ID: NCNM-RW-0001  
National Workforce Hub:  ACTIVE

AI Compliance Engine:  ENABLED

|                                                                                                        |                                                                                                           |                                                                                                            |  |
|--------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|--|
| +-----+-----+-----+                                                                                    |                                                                                                           |                                                                                                            |  |
|  National Licensing   |  Professional Discipline |  Training & Accreditation |  |
| Nurse registration                                                                                     | Malpractice investigations                                                                                | CPD compliance tracking                                                                                    |  |
| Midwife licensing                                                                                      | Scope-of-practice violations                                                                              | School accreditation                                                                                       |  |
| License renewals                                                                                       | Ethical misconduct audits                                                                                 | Internship supervision                                                                                     |  |
| +-----+-----+-----+                                                                                    |                                                                                                           |                                                                                                            |  |
| +-----+-----+-----+                                                                                    |                                                                                                           |                                                                                                            |  |
|  Workforce Governance |  National Analytics      |  Ministerial Reporting  |  |
| Staffing compliance                                                                                    | National workforce heatmaps                                                                               | Direct MoH escalation                                                                                      |  |
| Rural deployment tracking                                                                              | Nurse-midwife ratios                                                                                      | RBC synchronization                                                                                        |  |
| Workforce shortages                                                                                    | AI workforce forecasting                                                                                  | National emergency alerts                                                                                  |  |
| +-----+-----+-----+                                                                                    |                                                                                                           |                                                                                                            |  |

2. FULL NCNM DASHBOARD LAYOUT

|                                                                                                         |                                                                                                                     |                                                                                                              |  |
|---------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|--|
| +=====+                                                                                                 |                                                                                                                     |                                                                                                              |  |
| =====+                                                                                                  |                                                                                                                     |                                                                                                              |  |
| LEFT SIDEBAR                                                                                            | CENTRAL GOVERNANCE WORKSPACE                                                                                        | RIGHT                                                                                                        |  |
| ANALYTICS PANEL                                                                                         |                                                                                                                     |                                                                                                              |  |
| +=====+                                                                                                 |                                                                                                                     |                                                                                                              |  |
| =====+                                                                                                  |                                                                                                                     |                                                                                                              |  |
|  Dashboard Home      |  National Workforce Registry     |  National Reports       |  |
|  Licensing & Renewal |  Midwifery Practice Monitoring   |  AI Compliance Alerts   |  |
|  Discipline & Ethics |  Facility Staffing Compliance    |  Workforce Trends       |  |
|  CPD & Accreditation |  Internship & Training Oversight |  Minister Notifications |  |
|  National Analytics  |  Predictive Workforce Models     |  Regional Heatmaps      |  |
|  Minister Reporting  |                                                                                                                     |                                                                                                              |  |

3. NATIONAL LICENSE REGISTRY

|         |  |  |  |
|---------|--|--|--|
| +=====+ |  |  |  |
| =====+  |  |  |  |

| 📖 NATIONAL NURSE & MIDWIFE LICENSE REGISTRY

|

+=====+

=====+

ACTIVE LICENSES

| PROFESSIONAL         | LICENSE TYPE       | STATUS          | EXPIRY DATE |  |
|----------------------|--------------------|-----------------|-------------|--|
| Chantal Uwimana      | Registered Nurse   | ✓ ACTIVE        | 14 Jan 2027 |  |
| Alice Mukandayisenga | Registered Midwife | ✓ ACTIVE        | 03 Mar 2026 |  |
| Eric Habimana        | ICU Nurse          | ✓ ACTIVE        | 11 Jul 2027 |  |
| Olive Uwase          | Pediatric Midwife  | ⚠ EXPIRING SOON | 22 Jun 2025 |  |

- AI LICENSING ALERTS:
- ⚠ 221 licenses expiring within 60 days
  - ⚠ Rural facilities using temporary staff licenses
  - ⚠ Internship supervision gaps identified

4. MIDWIFERY PRACTICE MONITORING

+=====+

=====+

| 📊 NATIONAL MIDWIFERY PRACTICE MONITOR

|

+=====+

=====+

| FACILITY            | MONTHLY DELIVERIES | MIDWIVES AVAILABLE | STATUS     |
|---------------------|--------------------|--------------------|------------|
| CHUK                | 1,201              | 42                 | 🟢 STABLE   |
| Kibagabaga Hospital | 821                | 24                 | 🟢 STABLE   |
| Nyagatare District  | 611                | 11                 | 🟡 SHORTAGE |
| Ruhango Hospital    | 411                | 7                  | 🔴 CRITICAL |



MATERNAL CARE ALERTS:

- ⚠ Midwife-to-delivery ratio unsafe in Ruhango
- ⚠ Rural emergency obstetric coverage declining
- ⚠ Night shift shortages increasing

5. PROFESSIONAL DISCIPLINE & ETHICS CENTER

🚒 PROFESSIONAL DISCIPLINE & ETHICS INVESTIGATION CENTER

ACTIVE INVESTIGATIONS

| CASE ID   | FACILITY          | INCIDENT              | STATUS          |
|-----------|-------------------|-----------------------|-----------------|
| NCNM-8821 | District Hospital | Medication Negligence | 🔴 ACTIVE        |
| NCNM-7721 | Maternity Ward    | Scope Violation       | 🟡 REVIEW        |
| NCNM-9912 | Health Center     | Ethical Misconduct    | 🟡 INVESTIGATION |

FORENSIC FINDINGS:

- ⚠ Delayed emergency maternal response detected
- ⚠ Unauthorized clinical procedure identified
- ⚠ Documentation inconsistencies flagged

6. CPD & TRAINING ACCREDITATION CENTER

📖 CONTINUOUS PROFESSIONAL DEVELOPMENT (CPD) & TRAINING ACCREDITATION

=====

=====+

| TRAINING PROGRAM       | PARTICIPANTS | COMPLIANCE RATE | STATUS            |
|------------------------|--------------|-----------------|-------------------|
| Emergency Obstetrics   | 2,114        | 91%             | GOOD              |
| Neonatal Resuscitation | 1,821        | 84%             | MODERATE          |
| Infection Prevention   | 4,422        | 96%             | EXCELLENT         |
| Mental Health Care     | 811          | 62%             | LOW PARTICIPATION |

#### AI TRAINING ALERTS:

- △ Rural CPD participation declining
- △ Maternal emergency training gaps detected
- △ High neonatal risk facilities need refresher courses

## 7. NATIONAL WORKFORCE HEATMAP

=====

=====+

|  NATIONAL NURSING & MIDWIFERY WORKFORCE HEATMAP

|

=====

=====+

KIGALI  ADEQUATE STAFFING

NORTHERN PROVINCE  MODERATE SHORTAGE

SOUTHERN PROVINCE  MIDWIFE DEFICIT

EASTERN PROVINCE  CRITICAL SHORTAGE

WESTERN PROVINCE  HIGH WORKLOAD

#### LEGEND:

-  Adequate Coverage
-  Moderate Pressure
-  Workforce Deficit
-  Critical Staffing Crisis

8. INTERNSHIP & STUDENT OVERSIGHT CENTER

🎓 NURSING & MIDWIFERY INTERNSHIP OVERSIGHT

ACTIVE INTERNS

| STUDENT        | TRAINING FACILITY   | SUPERVISOR           | STATUS            |
|----------------|---------------------|----------------------|-------------------|
| Aline Mukamana | CHUK                | Chantal Uwimana      | ✓ ACTIVE          |
| Jean Bosco     | Kibagabaga Hospital | Alice Mukandayisenga | ✓ ACTIVE          |
| Sandrine Uwera | Ruhango Hospital    | Eric Habimana        | ⚠ SUPERVISION GAP |

OVERSIGHT ALERTS:

- ⚠ Internship supervision ratio exceeded
- ⚠ Rural student mentorship shortages
- ⚠ Clinical competency delays identified

9. RIGHT PANEL — NCNM EXECUTIVE ANALYTICS

📊 NCNM EXECUTIVE INTELLIGENCE

TOTAL LICENSED NURSES: 18,422  
TOTAL LICENSED MIDWIVES: 4,211  
ACTIVE INVESTIGATIONS: 28  
CPD COMPLIANCE RATE: 89%

AI DETECTIONS:

- ⚠ Rural staffing crisis rising

- ⚠ Midwife burnout increasing
- ⚠ Maternal emergency skill gaps identified

## MINISTER REPORTS:

- ✓ Workforce shortage escalation sent
- ✓ RBC synchronized
- ✓ Emergency staffing alert active

## 10. DIRECT REPORTING TO MINISTER

# NATIONAL PROFESSIONAL GOVERNANCE ESCALATION CHANNEL

### DIRECT CONNECTIONS:

- ✓ Ministry of Health
- ✓ Rwanda Biomedical Center
- ✓ National Licensing Authorities
- ✓ Hospital Governance Units

REPORT TYPES:

- ✓ National Workforce Shortages
- ✓ Maternal Care Emergencies
- ✓ Ethical Misconduct Investigations
- ✓ Licensing Compliance Gaps
- ✓ Rural Midwife Deficits
- ✓ Emergency Workforce Deployment

## LATEST REPORT:

-----

Subject:

# Eastern Province Midwife Workforce Crisis

Affected Districts:

- Nyagatare
- Kirehe

- Gatsibo

Priority:

 NATIONAL MATERNAL HEALTH ALERT

Timestamp:

12:11 PM

## 11. AI WORKFORCE GOVERNANCE ENGINE

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# | 🧠 AI NURSING & MIDWIFERY GOVERNANCE ENGINE

---

---

## ACTIVE AI DETECTIONS

- ⚠ Midwife burnout risk identified
- ⚠ Unsafe staffing ratios forecasted
- ⚠ Rural deployment imbalance detected
- ⚠ Maternal emergency skill gaps predicted
- ⚠ Internship supervision deficits identified
- ⚠ CPD non-compliance clusters detected

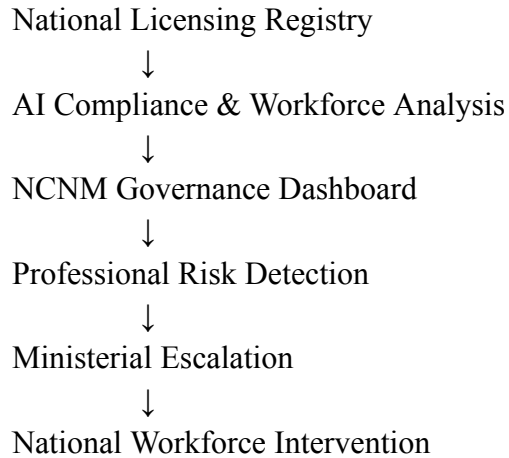
## AI SYSTEMS:

- ✓ Workforce Forecasting ACTIVE
- ✓ Ethical Compliance ACTIVE
- ✓ Licensing Intelligence ACTIVE
- ✓ Maternal Safety Analytics ACTIVE
- ✓ Rural Deployment Modeling ACTIVE

## 12. NATIONAL WORKFORCE GOVERNANCE FLOW

Health Facility Workforce Data





### 13. SMART PROFESSIONAL PROFILE

|                                                                                                                |  |
|----------------------------------------------------------------------------------------------------------------|--|
| =====                                                                                                          |  |
| =====+                                                                                                         |  |
|  PROFESSIONAL LICENSE PROFILE |  |
| =====                                                                                                          |  |
| =====+                                                                                                         |  |

FULL NAME: Alice Mukandayisenga  
PROFESSION: Registered Midwife  
LICENSE NUMBER: MW-RW-88211  
FACILITY: Kibagabaga Hospital

#### QUALIFICATIONS:

- ✓ Bachelor of Midwifery
- ✓ Neonatal Resuscitation Certified
- ✓ Emergency Obstetrics Certified

#### CPD STATUS:

- ✓ Completed 2025 Requirements

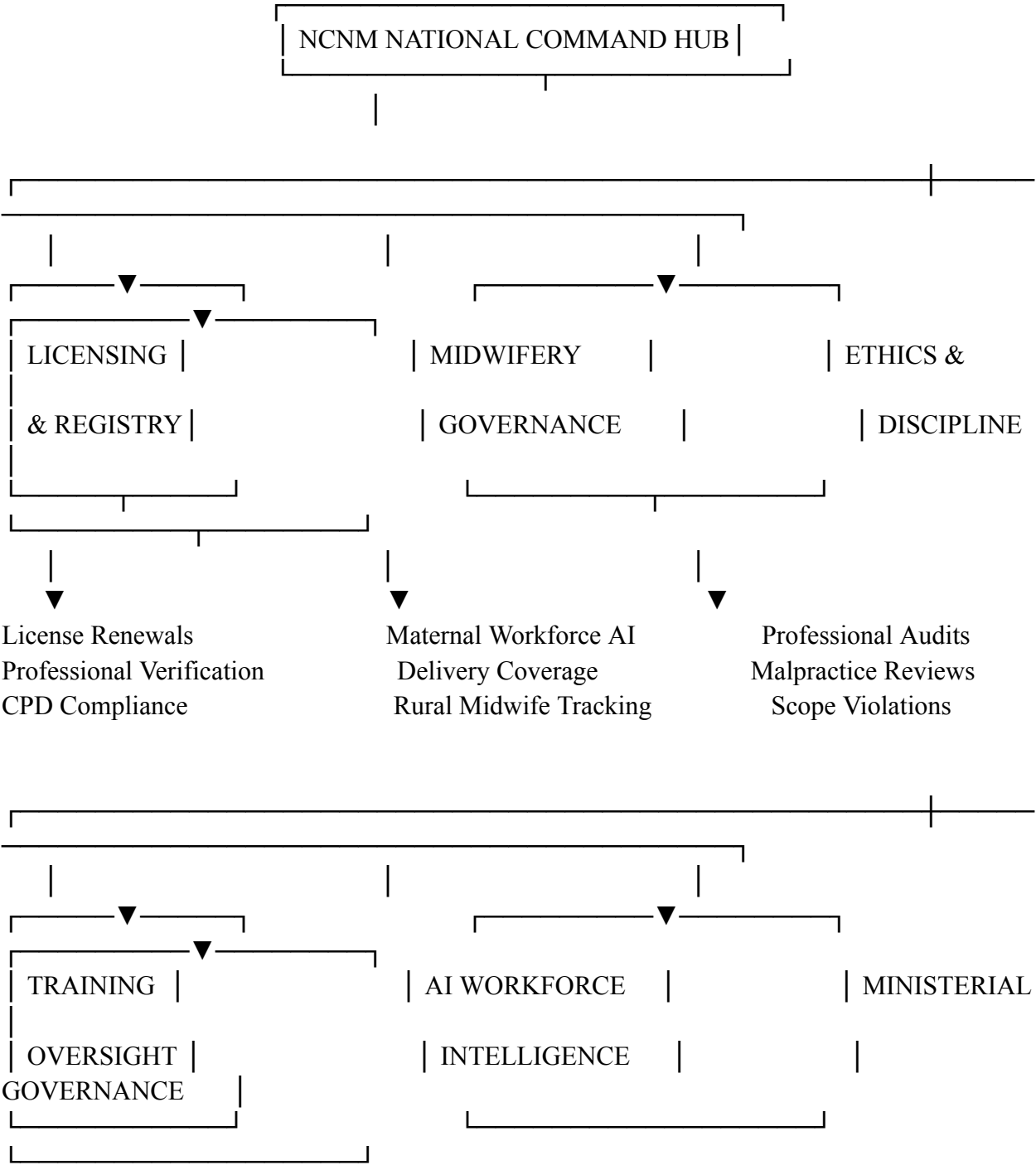
#### DISCIPLINE STATUS:

- ✓ No Active Investigations

#### AI RISK PROFILE:

 SAFE PRACTICE STATUS

14. FULL NCNM NATIONAL ARCHITECTURE



MEDBRIDGE+ — LAB TECHNICIAN DASHBOARD UI/UX BLUEPRINT

Laboratory Information System (LIS) Workspace Design

1. LAB TECHNICIAN MAIN DASHBOARD

|                                     |  |  |                           |  |  |
|-------------------------------------|--|--|---------------------------|--|--|
| MEDBRIDGE+   LABORATORY PORTAL      |  |  | Mr. Patrick Nshimiyimana  |  |  |
| Kigali University Teaching Hospital |  |  | Senior Lab Technician     |  |  |
| LEFT SIDEBAR                        |  |  | MAIN LAB WORKSPACE        |  |  |
| ● New Lab Orders                    |  |  | 📊 Lab Reports             |  |  |
| ● Samples Awaiting Results          |  |  | • Pending Tests: 38       |  |  |
| ● Completed Results                 |  |  | • Completed Today: 112    |  |  |
| ● Critical Samples                  |  |  | • Critical Alerts: 4      |  |  |
| ● Emergency STAT Orders             |  |  | • Avg Turnaround: 19 mins |  |  |
| ● Archived Exams                    |  |  | • Delayed Samples: 3      |  |  |

2. LEFT PANEL — NEW LAB ORDERS

|                  |  |
|------------------|--|
| ● NEW LAB ORDERS |  |
|------------------|--|

1. Uwimana Alice
- Patient ID: MBP-2026-000124
- Requested Exam: Troponin I + CBC
- TRIAGE: ● RED (Critical)
- Ordered By: Dr. Eric Murenzi



## 2. Niyonsenga Eric

Patient ID: MBP-2026-000125

Requested Exam: Malaria Test

TRIAGE:  ORANGE (Urgent)

Ordered By: Dr. Diane Uwase

### 3. Mukamana Chantal

Patient ID: MBP-2026-000126

Requested Exam: Blood Glucose

TRIAGE:  YELLOW (Moderate)

Ordered By: Dr. Jean Bosco

### 3. LAB TRIAGE PRIORITY COLORS

| Color  | Priority         | Meaning              |
|--------|------------------|----------------------|
| Red    | STAT             | Immediate processing |
| Orange | Urgent           | Fast-track           |
| Yellow | Routine          | Standard queue       |
| Green  | Low Priority     | Non-urgent           |
| Black  | Rejected/Expired | Invalid sample       |

#### 4. WHEN PATIENT SAMPLE IS CLICKED

=====+  
=====+  
| LAB SAMPLE PROFILE |  
=====+  
=====+

FULL NAMES: Uwimana Alice

GENDER: Female

AGE: 32 Years

PATIENT ID: MBP-2026-000124

NATIONAL ID: 1199880076543210  
HEALTH INSURANCE: RSSB / Mutuelle de Santé

REQUESTING DOCTOR: Dr. Eric Murenzi  
REQUESTING DEPARTMENT: Emergency Department  
REQUEST DATE: 22 May 2026  
REQUEST TIME: 08:22 AM

TRIAGE STATUS: ● RED — STAT PRIORITY  
SAMPLE TYPE: Blood  
BARCODE ID: LAB-2026-883712

5. PATIENT LIMITED CLINICAL SNAPSHOT

(ONLY NECESSARY INFORMATION — PRIVACY PROTECTED)

|                              |  |
|------------------------------|--|
| +=====                       |  |
| =====+                       |  |
| LIMITED CLINICAL INFORMATION |  |
| +=====                       |  |
| =====+                       |  |

Chief Complaint:  
• Severe chest pain  
• Shortness of breath

Known Allergies:  
• Penicillin

Infection Risk Alerts:  
☒ Hepatitis B Precaution

Clinical Notes:  
△ Minimal summary only visible to lab personnel

6. LAB EXAMINATION TABLE

|                          |  |
|--------------------------|--|
| ORDERED LAB EXAMINATIONS |  |
|--------------------------|--|

| EXAM                       | STATUS    | PRIORITY               |
|----------------------------|-----------|------------------------|
| Troponin I                 | Pending   | <span>●</span> STAT    |
| Complete Blood Count (CBC) | Running   | <span>●</span> STAT    |
| Blood Glucose              | Completed | <span>●</span> Routine |
| Serum Creatinine           | Pending   | <span>●</span> Urgent  |

7. LAB RESULT ENTRY PANEL

|              |  |
|--------------|--|
| RESULT ENTRY |  |
|--------------|--|

SELECT TEST:  
[ Troponin I ▼ ]

RESULT VALUE:  
[ 2.87 ng/mL ]

REFERENCE RANGE:  
0.00 — 0.04 ng/mL

RESULT STATUS:  
☒ Critical Value  
☐ Normal  
☐ Borderline

TECHNICIAN COMMENTS:  
Elevated cardiac biomarker detected.

[ SEND TO DOCTOR ]

=====+  
=====+  
| 🚨 CRITICAL RESULT ALERT |  
=====+  
=====+

- ☑ Doctor notified immediately
- ☑ Emergency dashboard updated
- ☑ Case escalated to cardiology
- ☑ RBC surveillance AI notified

=====+  
=====+  
| SAMPLE TRACKING |  
=====+  
=====+

|                  |           |           |
|------------------|-----------|-----------|
| STAGE            | STATUS    | TIMESTAMP |
| Sample Collected | Completed | 08:30 AM  |

|                     |           |          |  |
|---------------------|-----------|----------|--|
| Sample Received     | Completed | 08:36 AM |  |
| Sample Processing   | Running   | 08:40 AM |  |
| Validation          | Pending   | --       |  |
| Doctor Notification | Pending   | --       |  |
| +-----+-----+-----+ |           |          |  |

10. LAB INVENTORY & REAGENT MONITORING

|                    |  |
|--------------------|--|
| =====              |  |
| =====+             |  |
| LAB REAGENT STATUS |  |
| =====              |  |
| =====+             |  |

|                    |              |             |
|--------------------|--------------|-------------|
| +-----+            | +-----+      | +-----+     |
| REAGENT            | STOCK LEVEL  | ALERT       |
| +-----+            |              |             |
| Troponin Test Kits | 12 Remaining | ⚠ Low Stock |
| CBC Reagent        | 45 Remaining | ✅ Normal    |
| Glucose Strips     | 5 Remaining  | ❌ Critical  |
| Malaria Test Kits  | 28 Remaining | ✅ Normal    |
| +-----+            |              |             |

11. AI LAB ASSISTANT PANEL

|                         |  |
|-------------------------|--|
| =====                   |  |
| =====+                  |  |
| AI LABORATORY ASSISTANT |  |
| =====                   |  |
| =====+                  |  |

AI ANALYSIS:  
⚠ Elevated Troponin Pattern Suggestive of Acute Coronary Syndrome

QUALITY CONTROL:  
✅ Sample integrity acceptable  
✅ No hemolysis detected

AI RECOMMENDATIONS:

- Repeat Troponin after 6 hours
- Suggest urgent cardiology review

OUTBREAK ALERTS:

⚠ Increased malaria positives detected in Gasabo District

---

12. LAB PERFORMANCE REPORTS

|                                                                                                 |  |
|-------------------------------------------------------------------------------------------------|--|
| +=====+                                                                                         |  |
|  LAB ANALYTICS |  |
| +=====+                                                                                         |  |

TODAY SUMMARY

|                     |         |
|---------------------|---------|
| -----               |         |
| Samples Processed:  | 112     |
| Critical Results:   | 8       |
| Rejected Samples:   | 3       |
| Average Turnaround: | 19 mins |

MOST REQUESTED TESTS

|               |    |
|---------------|----|
| -----         |    |
| CBC           | 38 |
| Malaria       | 26 |
| Blood Glucose | 22 |
| Troponin I    | 14 |

AI SYSTEM ALERTS

- |                                  |  |
|----------------------------------|--|
| -----                            |  |
| ⚠ Rising respiratory infections  |  |
| ⚠ Low stock on glucose strips    |  |
| ⚠ Increased emergency STAT tests |  |

---

13. SAMPLE REJECTION PANEL

|                  |  |
|------------------|--|
| +=====           |  |
| =====+           |  |
| SAMPLE REJECTION |  |
| +=====           |  |
| =====+           |  |

REJECTION REASON:

- ☒ Hemolyzed Sample
- ☐ Insufficient Volume
- ☐ Wrong Patient Label
- ☐ Expired Collection Time

TECHNICIAN COMMENT:

Sample integrity compromised.

AUTO ACTION:

- ☒ Doctor notified
- ☒ Repeat collection requested

---

**14. LAB SECURITY & PRIVACY MODEL**

|                    |  |
|--------------------|--|
| +=====             |  |
| =====+             |  |
| ACCESS LIMITATIONS |  |
| +=====             |  |
| =====+             |  |

ALLOWED:

- ☒ View ordered laboratory requests
- ☒ Enter and validate results
- ☒ View minimal patient clinical context
- ☒ Track sample workflow

PROHIBITED:

- ☒ Access psychiatric records
- ☒ Modify prescriptions
- ☒ Access billing data
- ☒ View unrelated patient history
- ☒ Delete finalized results

---

## 15. NATIONAL LAB INTEROPERABILITY

|                      |  |
|----------------------|--|
| +=====               |  |
| =====+               |  |
| NATIONAL LAB NETWORK |  |
| +=====               |  |
| =====+               |  |

### CONNECTED FACILITIES:

- CHUK Central Laboratory
- Rwanda Biomedical Center
- King Faisal Hospital Lab
- National Reference Laboratory

### SYNC STATUS:

☒ National synchronization successful

---

## 16. MOBILE LAB TABLET INTERFACE

## 17. FINAL LAB TECHNICIAN DASHBOARD STRUCTURE

### LEFT PANEL

-----

New Lab Orders  
STAT Cases  
Pending Samples  
Completed Results  
Archived Exams

### CENTER PANEL

-----

Patient Sample Profile  
Lab Orders  
Result Entry



Sample Tracking  
Validation Workspace

RIGHT PANEL

-----

Lab Analytics  
AI Alerts  
Reagent Stock  
Turnaround Reports  
Outbreak Notifications

MEDBRIDGE+ — MINISTERIAL EXECUTIVE PORTAL

National Healthcare Governance & Intelligence Dashboard

-----

1. MINISTERIAL SYSTEM ORIENTATION

+=====+  
=====+

| MEDBRIDGE+ — NATIONAL HEALTH INTELLIGENCE GRID  
|  
| MINISTERIAL EXECUTIVE CONTROL PORTAL |  
| Republic of Rwanda | Ministry of Health | National Command Infrastructure |  
+=====+  
=====+

CONNECTED SYSTEMS:

-----

- ✓ National EMR/EHR Exchange
- ✓ Rwanda Pandemic Surveillance Grid
- ✓ National Ambulance Coordination Engine
- ✓ RMS Supply Chain Intelligence
- ✓ RSSB Insurance Verification Layer
- ✓ National Workforce Governance
- ✓ AI Predictive Epidemiology Engine
- ✓ Emergency Communication Network
- ✓ National Healthcare Analytics Warehouse

SECURITY STATUS:

-  Zero-Trust Security Active
-  RBAC Governance Tokens Verified
-  End-to-End Encryption Enabled
-  National Audit Integrity Confirmed
-  Biometric Executive Authentication Active

2. MINISTERIAL LOGIN & NATIONAL IDENTITY GATE

|                                     |  |
|-------------------------------------|--|
| MEDBRIDGE+                          |  |
| MINISTERIAL EXECUTIVE ACCESS PORTAL |  |

 NATIONAL HEALTH GRID

|                                                                                                                   |   |
|-------------------------------------------------------------------------------------------------------------------|---|
| NATIONAL ID / GOVERNMENT SSO                                                                                      |   |
| [ MINISANTE-EXEC-00014                                                                                            | ] |
| SECURE PIN                                                                                                        |   |
| [ *****                                                                                                           | ] |
| BIOMETRIC AUTHENTICATION                                                                                          |   |
| ◉ FINGERPRINT VERIFIED                                                                                            |   |
| SYSTEM AUDIT STATUS: VERIFIED  |   |

| NATIONAL CYBER DEFENSE LINK: ACTIVE|

| ZERO-PHANTOM SESSION CHECK: PASSED|

+-----+

[ AUTHORIZED EXECUTIVE SIGN-IN ]

3. NATIONAL MINISTERIAL DASHBOARD — MAIN LAYOUT

+=====

+=====

| MEDBRIDGE+ — MINISTER OF HEALTH EXECUTIVE COMMAND CENTER

|

| Minister: Dr. Emmanuel NkurunzizaNATIONAL STATUS: OPERATIONAL 

|

+=====

+=====

+-----+-----+-----+

| LEFT NAVIGATION| MAIN NATIONAL WORKSPACE| RIGHT

| LIVE MONITOR|

+-----+-----+-----+

|  Epidemiological Grid| |  Critical Alerts|

|  Emergency Coordination| |  Pending Incidents|

|  National Hospitals| |  System Integrity|

|  RMS Supply Chain| |  API Exchange Status|

|  Workforce Governance| |  Surveillance Network|

|  National Broadcasts| |

|  Audit Gateway| |

|  Security & Compliance| |

+-----+-----+-----+

4. NATIONAL HEALTH OVERVIEW CARDS

+=====

+=====

| NATIONAL HEALTH OVERVIEW|

=====+  
=====+

+-----+-----+-----+-----+  
| TOTAL ACTIVE | PANDEMIC | ICU NATIONAL | AMBULANCES |  
| FACILITIES | HOTSPOTS | AVAILABILITY | ACTIVE |  
| | | | |  
| 612 | 18 | 71% | 482 |  
| ● +12 Operational | ● Moderate Alert | ● ICU Pressure | ● 97% Online |  
+-----+-----+-----+-----+

+-----+-----+-----+-----+  
| ACTIVE CLINICIANS | DAILY CONSULTATIONS | NATIONAL STOCK | STAFF  
ATTENDANCE |  
| ONLINE | | STATUS | RATE |  
| | | | |  
| 14,280 | 48,912 | 88% Stable | 94.2% |  
| ● Stable Workforce | ● High Throughput | ● Replenishment | ● Excellent |  
+-----+-----+-----+-----+

---

## 5. NATIONAL GIS PANDEMIC SURVEILLANCE MAP

=====+  
=====+  
| NATIONAL PANDEMIC & EPIDEMIOLOGICAL SURVEILLANCE GRID  
|  
+-----+-----+  
=====+

[ INTERACTIVE GIS HEATMAP OF RWANDA ]

WESTERN      NORTHERN      KIGALI CITY      EASTERN  
● Moderate Risk    ● Stable      ● High Cluster    ● Watch Zone  
  
SOUTHERN  
● Stable

OUTBREAK CLUSTERS:

-----

- Respiratory Syndromic Spike — Kigali City
- Malaria Case Increase — Eastern Province
- Gastrointestinal Cluster — Western Province

LIVE SURVEILLANCE ENGINE:

- ✓ Telehealth Symptom Streams
- ✓ Hospital Intake Analytics
- ✓ Lab Confirmation Correlation
- ✓ AI Outbreak Prediction Active
- ✓ RBC Surveillance Synchronization

6. AI PREDICTIVE OUTBREAK ENGINE

| AI PREDICTIVE EPIDEMIOLOGY ENGINE |

PREDICTIVE OUTBREAK TRENDS — NEXT 14 DAYS

|                   |             |                 |
|-------------------|-------------|-----------------|
| Kigali City       | <div></div> | 82% Probability |
| Eastern Province  | <div></div> | 58% Probability |
| Northern Province | <div></div> | 24% Probability |
| Western Province  | <div></div> | 37% Probability |
| Southern Province | <div></div> | 18% Probability |

AI RECOMMENDATIONS:

- ⚠ Increase respiratory specimen collection in Kigali
- ⚠ Prepare ICU surge capacity at CHUK and CHUB
- ⚠ Accelerate malaria medicine redistribution eastward
- ⚠ Activate targeted public awareness campaign

=====+  
=====+  
| NATIONAL EMERGENCY RESPONSE & BROADCAST CENTER  
|  
=====+  
=====+

-----+  
| EXECUTE MULTI-CHANNEL NATIONAL BROADCAST  
+-----+

+

+

| NATIONAL STAFF GOVERNANCE & OPERATIONAL AUDIT GRID

|

+=====+  
=====+

+-----+-----+-----+-----+

| STAFF ATTENDANCE | ACTIVE CLINICIANS | CLIENTS TREATED | TRAINING COMPLIANCE |

| RATE | ONLINE | TODAY |

| | | |

| 94.2% | 14,280 | 48,912 | 88.7% |

| ● +2.4% Increase | ● Stable | ● Efficient | ● In Progress |

+-----+-----+-----+-----+

=====

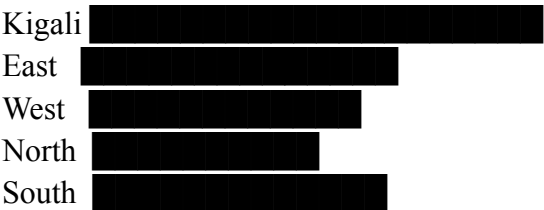
9. NATIONAL WORKFORCE ANALYTICS

+=====+  
=====+

| DAILY SERVICE ATTENDANCE BY PROFESSION |

+=====+  
=====+

DOCTORS



TRIAGE NURSES



LAB TECHNICIANS



|       |  |
|-------|--|
| West  |  |
| North |  |
| South |  |

10. CLINICAL THROUGHPUT vs WORKFORCE STRAIN

| CLINICAL THROUGHPUT & WORKFORCE STRAIN ANALYSIS

| REGION            | REPORTED CASES | COMPLETED CONSULTATIONS |
|-------------------|----------------|-------------------------|
| Kigali City       | 12,842         | 11,901                  |
| Eastern Province  | 9,112          | 8,520                   |
| Western Province  | 7,430          | 7,114                   |
| Northern Province | 5,921          | 5,702                   |
| Southern Province | 6,310          | 6,122                   |

AI DETECTED ALERTS:

- ⚠ Kigali workload pressure increasing
- ⚠ Eastern malaria screening demand rising
- ⚠ Northern Province maintaining stable workforce equilibrium

11. TRAINING & PANDEMIC COMPLIANCE TRACKER

| PANDEMIC SAFETY & SURVEILLANCE TRAINING TRACKER

|                    |  |     |
|--------------------|--|-----|
| REFERRAL HOSPITALS |  | 96% |
|--------------------|--|-----|



|                    |             |     |
|--------------------|-------------|-----|
| DISTRICT HOSPITALS | <div></div> | 82% |
| HEALTH CENTERS     | <div></div> | 64% |
| PRIVATE FACILITIES | <div></div> | 78% |

ACTIVE NATIONAL TRAINING:

- ✓ Emergency Pandemic Response
- ✓ AI Surveillance Interpretation
- ✓ Digital Incident Reporting
- ✓ Emergency Telehealth Coordination
- ✓ National Triage Standardization

12. NATIONAL INCIDENT & AUDIT LOGS

|                                       |                    |                    |               |
|---------------------------------------|--------------------|--------------------|---------------|
| NATIONAL INCIDENT & SYSTEM AUDIT LOGS |                    |                    |               |
| TIME                                  | FACILITY           | INCIDENT TYPE      | STATUS        |
| 08:14 AM                              | CHUK               | ICU Capacity Alert | Investigating |
| 08:27 AM                              | Ruhengeri Hospital | Lab Delay          | Resolved      |
| 08:41 AM                              | Kibungo Hospital   | Medicine Shortage  | Critical      |
| 09:02 AM                              | CHUB               | Oxygen Pressure    | Monitoring    |

13. NATIONAL SUPPLY CHAIN INTELLIGENCE

|                                                |  |
|------------------------------------------------|--|
| NATIONAL MEDICINE & RESOURCE DISTRIBUTION GRID |  |
|                                                |  |

|                      |     |
|----------------------|-----|
| Malaria Medication   | 78% |
| Antibiotics          | 91% |
| ICU Oxygen           | 69% |
| Vaccination Supplies | 96% |

- ⚠ Eastern Province malaria stock may reduce within 5 days
- ⚠ Kigali oxygen consumption increasing by 14%
- ⚠ Southern Province supply chain stable

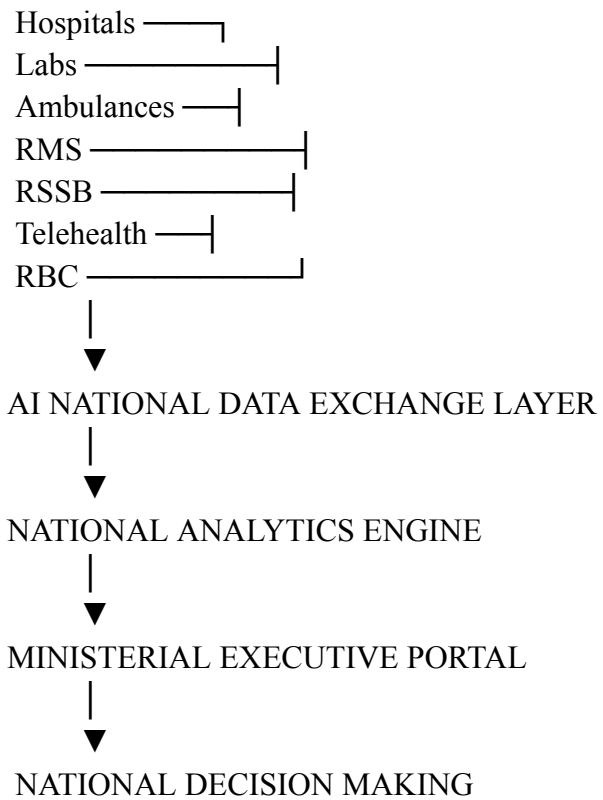
=====+  
=====+  
| NATIONAL CYBER DEFENSE & SYSTEM INTEGRITY CENTER  
|  
+=====+  
=====+

- 🔒 Zero-Pii Anonymization Engine: ACTIVE
- 🔒 National RBAC Governance Tokens: VERIFIED
- 🔒 Cross-Facility API Exchange: OPERATIONAL
- 🔒 Threat Detection AI: ACTIVE
- 🔒 National Audit Trails: IMMUTABLE

- ⚠ Suspicious login attempts blocked: 14
- ⚠ Unauthorized export requests denied: 3
- ⚠ Infrastructure integrity score: 99.4%

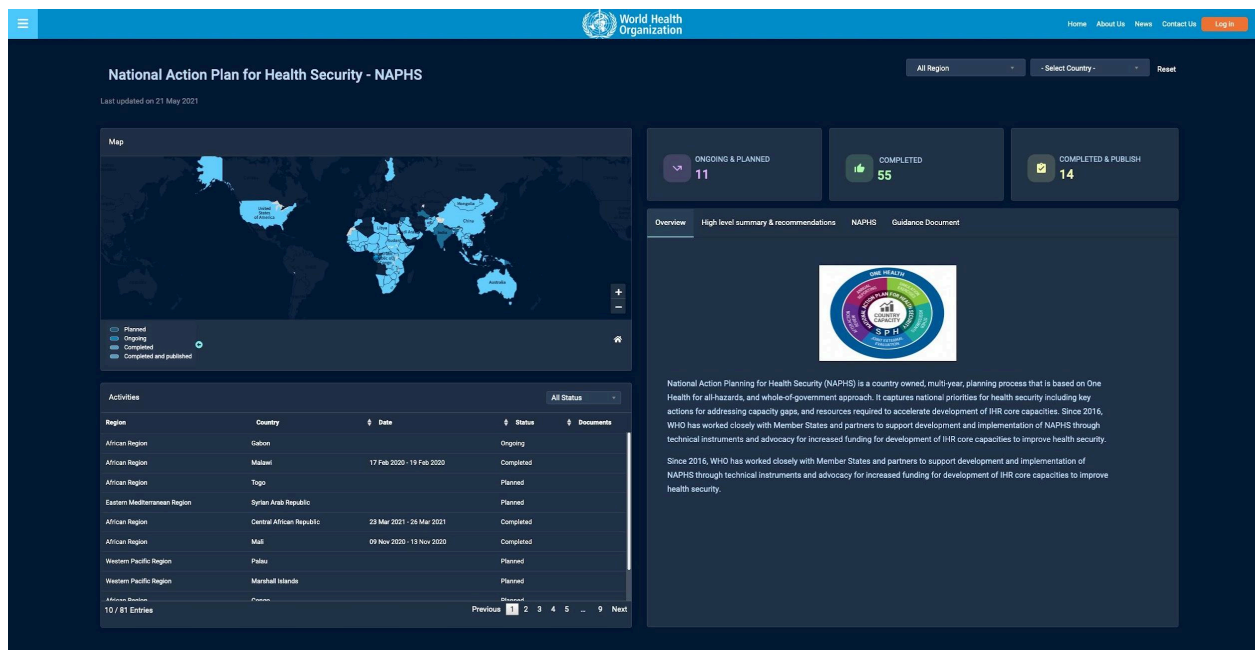
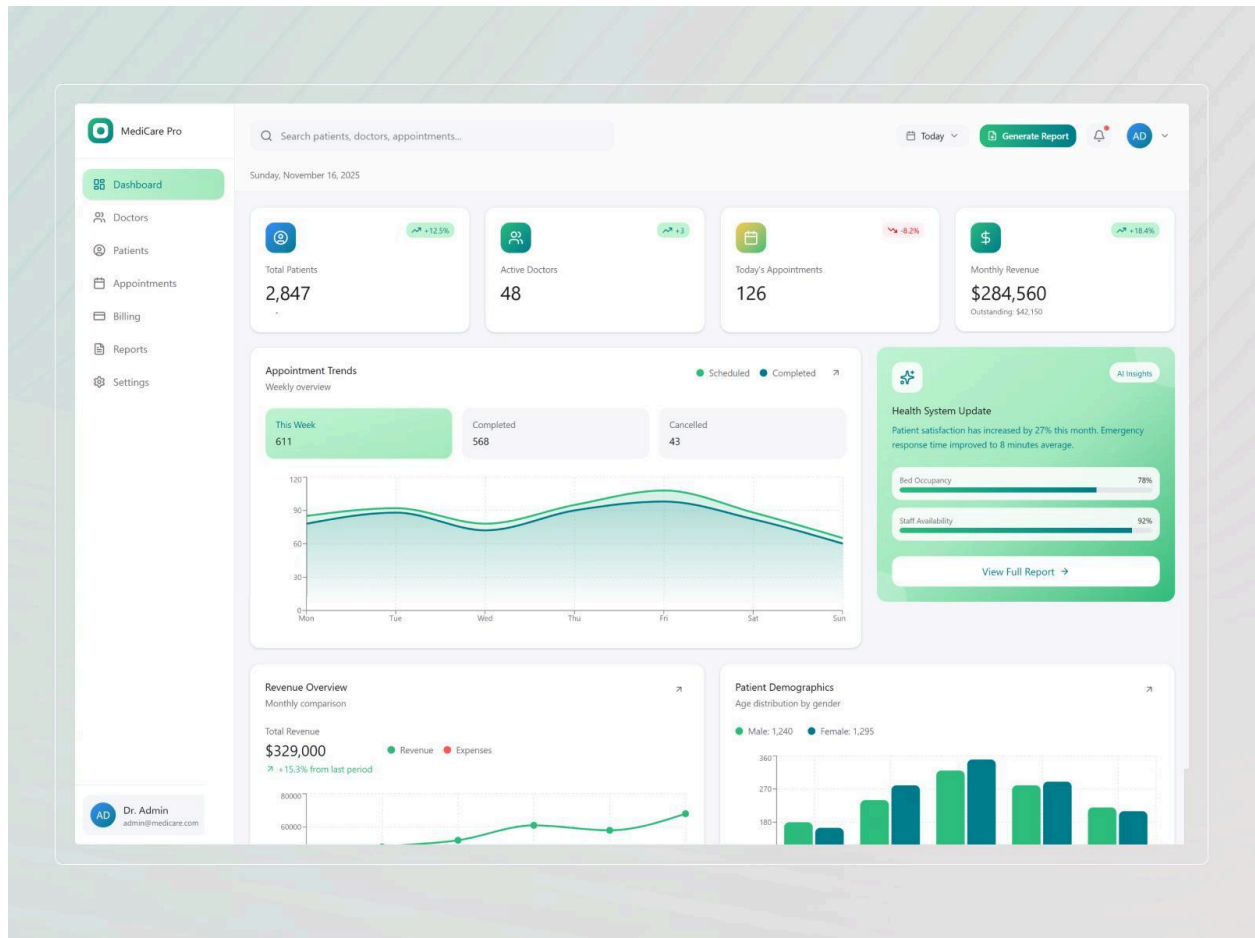
## 15. NATIONAL HEALTH GOVERNANCE FLOW

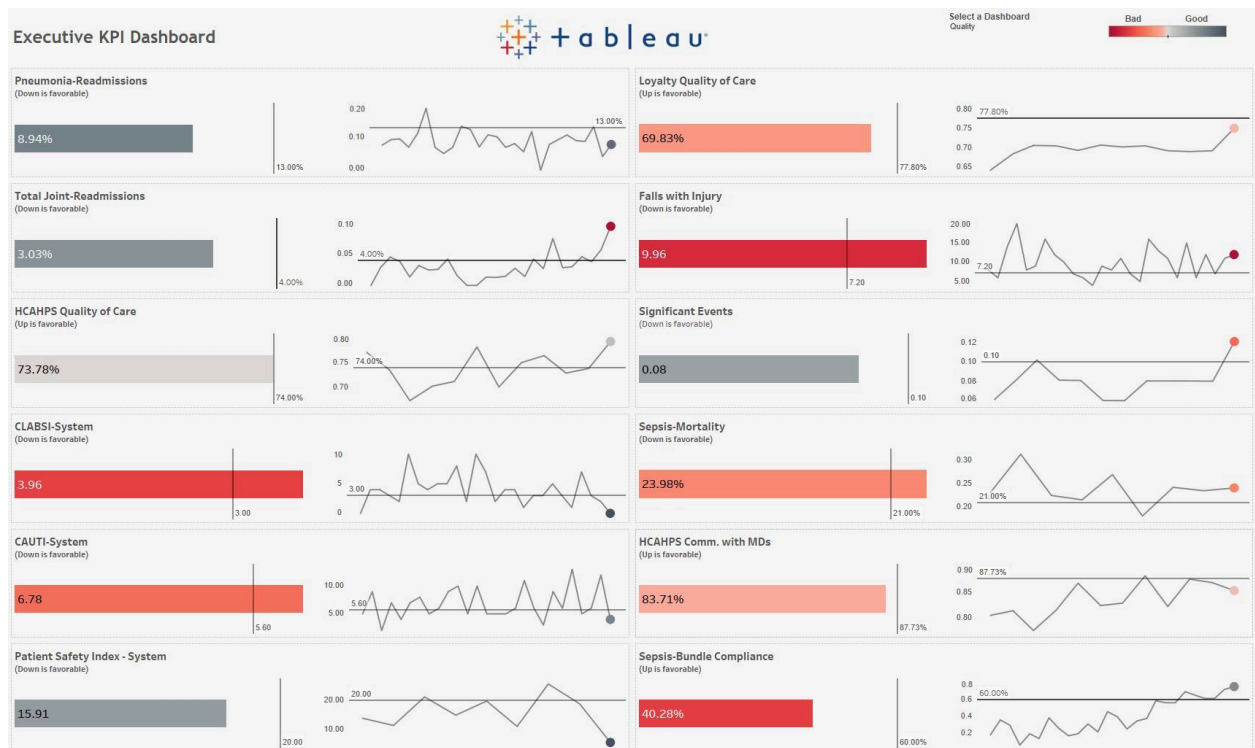
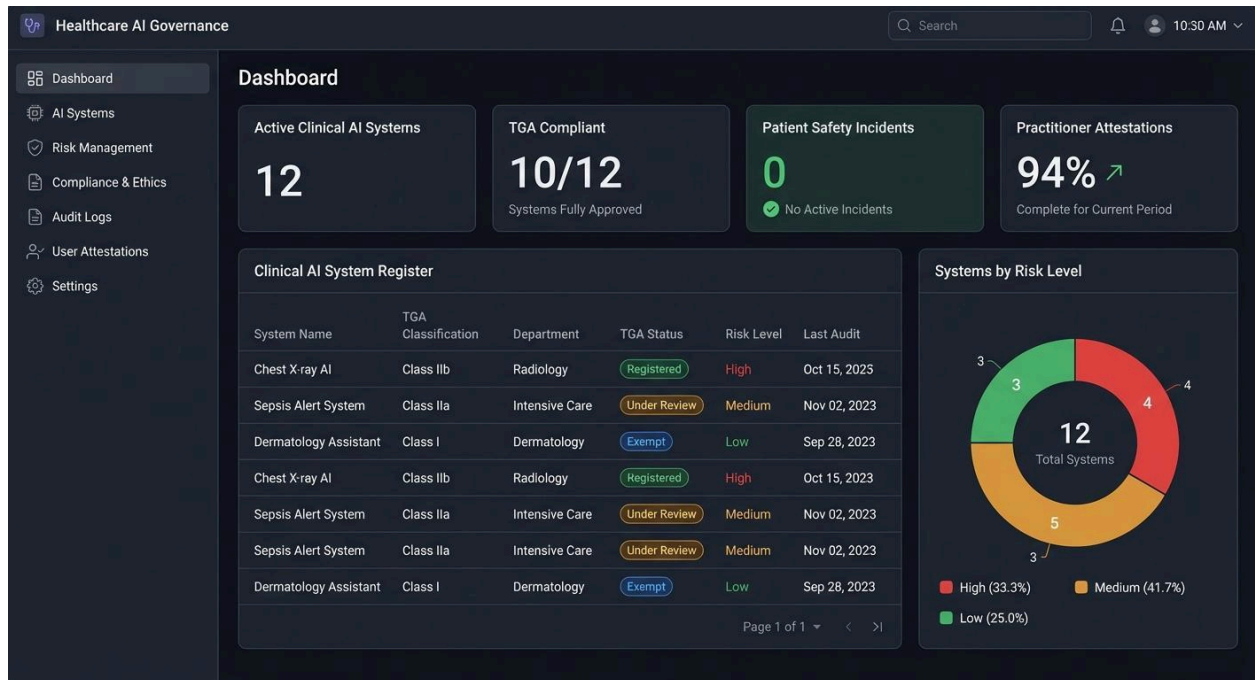
### NATIONAL HEALTH GRID



---

## 16. MINISTERIAL MOBILE/TABLET VIEW





## 17. FINAL MINISTERIAL EXECUTIVE ORIENTATION

### MEDBRIDGE+ MINISTERIAL PORTAL

## LEFT PANEL

-----

National Modules  
Emergency Navigation  
Audit Gateway  
Broadcast Center  
Security Systems

## CENTER PANEL

-----

National GIS Heatmaps  
Predictive Analytics  
Operational Intelligence  
Workforce Monitoring  
Emergency Coordination

## RIGHT PANEL

-----

Critical Alerts  
AI Recommendations  
Infrastructure Status  
National Security Logs  
Real-Time Notifications

## MEDBRIDGE — DIRECTOR OF NURSING SERVICES DASHBOARD

### National Smart Nursing Operations & Workforce Governance Center

#### 1. DIRECTOR OF NURSING — MAIN COMMAND DASHBOARD

+=====+  
=====+  
| MEDBRIDGE+ — DIRECTOR OF NURSING SERVICES PORTAL  
|  
| University Teaching Hospital Kigali |  
+=====+  
=====+

Director of Nursing: Beatha Mukamazimpaka  
License ID: RN-DNS-00211  
Hospital Status: ● Operational  
Direct Minister Link: 🛡️ ENABLED

|                         |                            |                              |
|-------------------------|----------------------------|------------------------------|
| 👩🏾‍⚕️ Nursing Workforce | 🚨 Critical Alerts          | 📊 National Reporting         |
| Staff attendance        | ICU staffing shortage      | Direct reporting to Minister |
| Shift coverage          | Medication omission alerts | RBC synchronization          |
| Nursing performance     | Delayed emergency triage   | MoH quality metrics          |

|                   |                            |                          |
|-------------------|----------------------------|--------------------------|
| 🏥 Ward Operations | 💊 Medication Safety        | 🚑 Emergency Coordination |
| Bed occupancy     | Missed administration logs | Ambulance escalations    |
| Ward nurse ratios | High-risk medications      | Emergency staffing       |
| Shift turnover    | MAR compliance tracking    | Critical response teams  |

2. FULL DASHBOARD LAYOUT STRUCTURE

|                 |                              |       |
|-----------------|------------------------------|-------|
| LEFT SIDEBAR    | CENTRAL OPERATIONS WORKSPACE | RIGHT |
| ANALYTICS PANEL |                              |       |

- Dashboard Home

Staff Attendance

Shift Scheduling

Notifications

Ward Management

Benchmarks

Medication Safety

Warnings

Critical Incident Reports

Quality Assurance
- Nursing Workforce Monitor

Ward Coverage Heatmap

Medication Administration Logs

Critical Escalation Queue

Nursing Performance Metrics

Emergency Coordination Center
- Real-Time Reports

AI Alerts

Minister

National

Emergency

### 3. LEFT SIDEBAR — STAFF ATTENDANCE

|                                                                                                                |  |
|----------------------------------------------------------------------------------------------------------------|--|
| +=====                                                                                                         |  |
| =====+                                                                                                         |  |
|  ACTIVE PROFESSIONALS — TODAY |  |
| +=====                                                                                                         |  |
| =====+                                                                                                         |  |

#### PRESENT STAFF

#### 1. Uwimana Chantal

Role: Senior ICU Nurse

Ward: ICU

Shift: Morning

Status: ✓ ACTIVE

#### 2. Habimana Eric

Role: Emergency Nurse

Ward: Emergency

Shift: Night

Status: ✓ ACTIVE

#### 3. Mukandayisenga Olive

Role: Pediatric Nurse

Ward: Pediatrics

Shift: Morning

Status: ✓ ACTIVE

#### 4. Nsengiyumva Alice

Role: Theater Nurse

Ward: Surgical Unit

Shift: Afternoon

Status: ⚠ LATE ARRIVAL

---

### 4. CENTRAL PANEL — NURSING OPERATIONS MONITOR



NATIONAL NURSING OPERATIONS CENTER

| WARD       | REQUIRED NURSES |    | PRESENT    | STATUS |
|------------|-----------------|----|------------|--------|
| ICU        | 8               | 5  | ● SHORTAGE |        |
| Emergency  | 10              | 9  | ● MODERATE |        |
| Pediatrics | 6               | 6  | ● STABLE   |        |
| Maternity  | 12              | 11 | ● MODERATE |        |
| Surgical   | 7               | 7  | ● STABLE   |        |

- AI STAFFING ANALYSIS:
- ⚠ ICU overload risk detected
  - ⚠ Emergency unit burnout probability high
  - ⚠ Maternity night shift fatigue rising

5. SHIFT MANAGEMENT SYSTEM

SHIFT SCHEDULING & ROTATION

TODAY SHIFT STATUS

| STAFF                | UNIT       | SHIFT     | STATUS    |
|----------------------|------------|-----------|-----------|
| Chantal Uwimana      | ICU        | Morning   | ✓ PRESENT |
| Eric Habimana        | Emergency  | Night     | ✓ PRESENT |
| Olive Mukandayisenga | Pediatrics | Morning   | ✓ PRESENT |
| Alice Nsengiyumva    | Surgical   | Afternoon | ⚠ DELAYED |

SHIFT ALERTS:

- 2 nurses absent in ICU
- 1 emergency nurse exceeded workload threshold
- 3 staff approaching overtime risk

6. MEDICATION ADMINISTRATION MONITORING

|                                                                                                                                |  |
|--------------------------------------------------------------------------------------------------------------------------------|--|
| +=====+                                                                                                                        |  |
| =====+                                                                                                                         |  |
|  MEDICATION ADMINISTRATION RECORD (MAR) AUDIT |  |
|                                                                                                                                |  |
| +=====+                                                                                                                        |  |
| =====+                                                                                                                         |  |

|                |             |                      |           |  |
|----------------|-------------|----------------------|-----------|--|
| +-----+        | +-----+     | +-----+              | +-----+   |  |
| PATIENT        | MEDICATION  | NURSE                | STATUS    |  |
| +-----+        |             |                      |           |  |
| Uwase Marie    | Ceftriaxone | Chantal Uwimana      | ✓ GIVEN   |  |
| Jean Bosco     | Morphine    | Eric Habimana        | ⚠ DELAYED |  |
| Mukamana Alice | Insulin     | Olive Mukandayisenga | ✓ GIVEN   |  |
| Habineza Paul  | Heparin     | Alice Nsengiyumva    | ✗ MISSED  |  |
| +-----+        |             |                      |           |  |

AI SAFETY ENGINE:

- ⚠ Missed anticoagulant administration detected
- ⚠ Delayed analgesia identified
- ⚠ Medication omission escalation triggered

7. CRITICAL INCIDENT REPORTING SYSTEM

|                                                                                                                 |  |
|-----------------------------------------------------------------------------------------------------------------|--|
| +=====+                                                                                                         |  |
| =====+                                                                                                          |  |
|  CRITICAL INCIDENT REPORTING |  |
| +=====+                                                                                                         |  |
| =====+                                                                                                          |  |

ACTIVE INCIDENTS

**1. ICU Staffing Collapse**

Severity:  **CRITICAL**

Unit: ICU

Escalated To: Hospital Director + Minister

**2. Missed Heparin Administration**

Severity:  **HIGH RISK**

Patient Outcome Risk: Severe

Investigation: ACTIVE

**3. Emergency Triage Delay**

Severity:  **HIGH**

Delay Time: 47 minutes

Escalation: Quality Assurance

---

**8. DIRECT REPORTING TO MINISTER**

+=====+  
=====+  
|  **DIRECT NATIONAL REPORTING CHANNEL**  
|  
+=====+  
=====+

**MINISTERIAL REPORT STATUS:**

✓ **CONNECTED TO NATIONAL HEALTH COMMAND CENTER**

**REPORT TYPES:**

- ✓ **Staffing Emergencies**
- ✓ **Mass Casualty Events**
- ✓ **ICU Collapse Risks**
- ✓ **National Nursing Shortages**
- ✓ **High Mortality Escalations**
- ✓ **Medication Safety Breaches**

**LATEST REPORT SENT:**

-----

**Subject:**

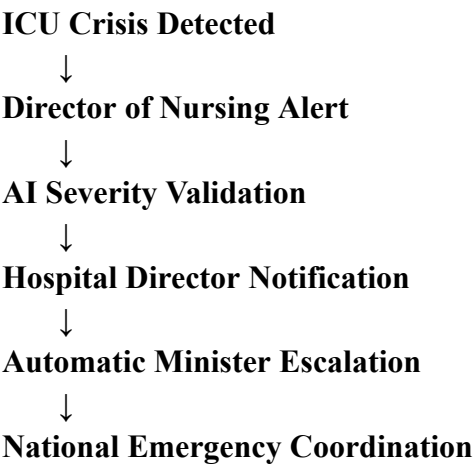
**Critical ICU Nurse Shortage — Kigali Referral Hospital**

Sent To:  
Minister of Health  
RBC Emergency Coordination  
National Nursing Council

Priority:  
 NATIONAL CRITICAL

Timestamp:  
08:41 AM

9. MINISTER ESCALATION FLOW



10. NURSING QUALITY ASSURANCE PANEL

|                                                                                     |                                         |  |  |  |
|-------------------------------------------------------------------------------------|-----------------------------------------|--|--|--|
| +=====+                                                                             |                                         |  |  |  |
| =====+                                                                              |                                         |  |  |  |
|  | NURSING QUALITY & PERFORMANCE ANALYTICS |  |  |  |
|                                                                                     |                                         |  |  |  |
| +=====+                                                                             |                                         |  |  |  |
| =====+                                                                              |                                         |  |  |  |

QUALITY INDICATORS

|               |        |  |
|---------------|--------|--|
| +-----+-----+ |        |  |
| INDICATOR     | STATUS |  |
| +-----+-----+ |        |  |

|                              |                  |  |
|------------------------------|------------------|--|
| Medication Compliance        | 94%              |  |
| Shift Attendance             | 91%              |  |
| ICU Nurse-to-Patient Ratio   | BELOW STANDARD ⚠ |  |
| Emergency Response Time      | 13 mins          |  |
| Patient Satisfaction         | 88%              |  |
| Critical Incident Resolution | 79%              |  |
| +-----+-----+                |                  |  |

- AI QUALITY INSIGHTS:
  - ⚠ Rising ICU fatigue indicators
  - ⚠ Burnout risk in Emergency Department
  - ⚠ Increased omission trends on night shifts

11. WARD HEATMAP VISUALIZATION

🏥

LIVE HOSPITAL WARD STATUS MAP

- ICU

● HIGH PRESSURE

Emergency

● OVERLOADED

Pediatrics

● STABLE

Maternity

● MODERATE

Surgery

● STABLE

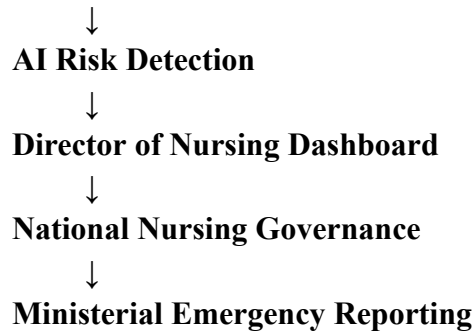
Isolation Unit

● HIGH ALERT

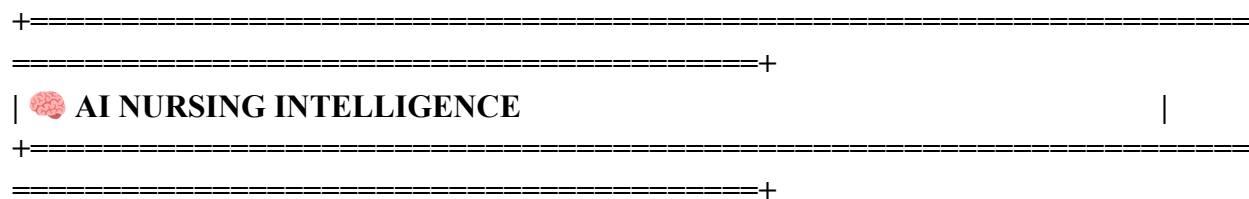
- LEGEND:
  - Stable
  - Moderate Pressure
  - High Pressure
  - Critical Pressure

12. NATIONAL NURSING GOVERNANCE FLOW





### 13. AI NURSING INTELLIGENCE ENGINE



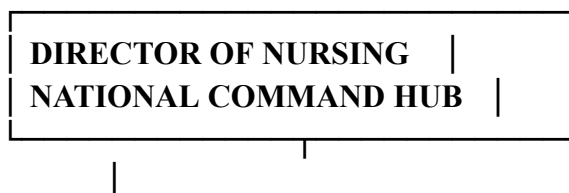
#### ACTIVE AI DETECTIONS

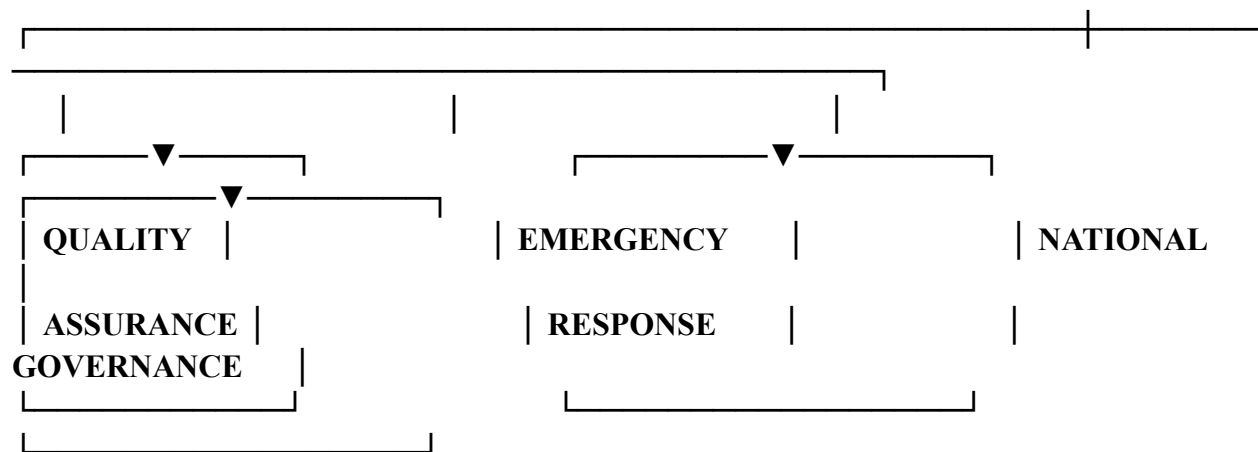
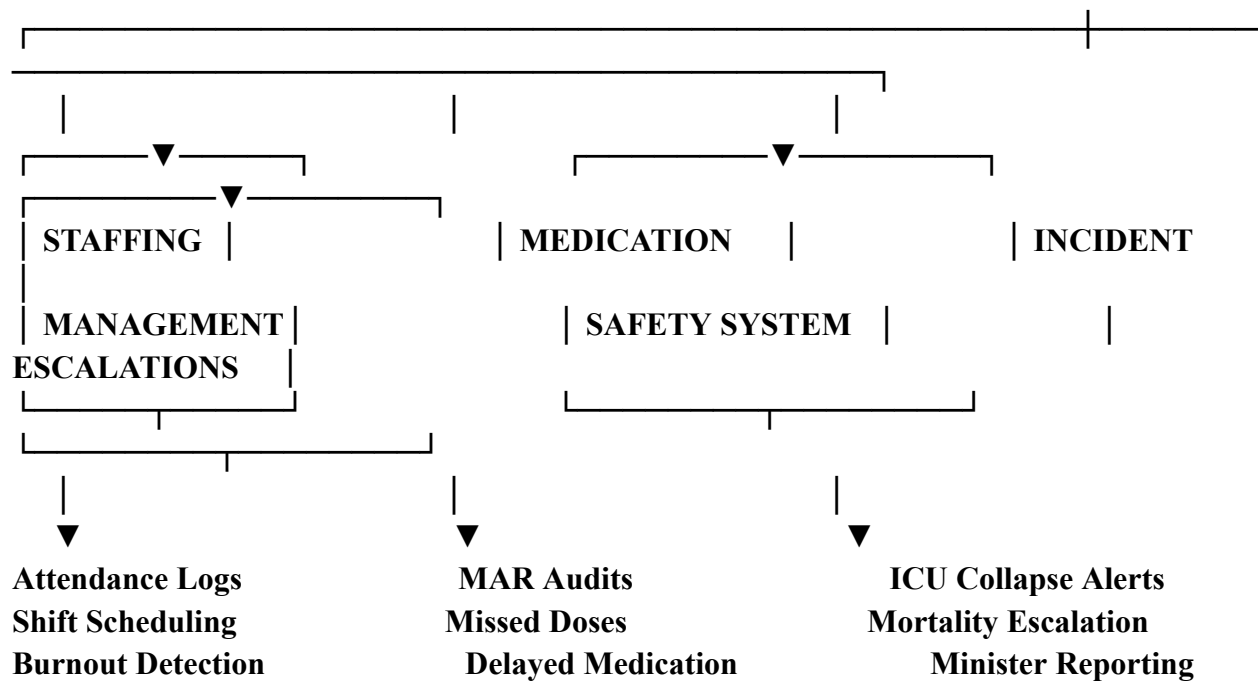
- ⚠ **Burnout prediction in ICU staff**
- ⚠ **Medication omission risk detected**
- ⚠ **Shift imbalance analysis active**
- ⚠ **Emergency overcrowding forecast**
- ⚠ **Staff absenteeism spike predicted**
- ⚠ **Delayed patient monitoring identified**

#### AI STATUS:

- ✓ **Workforce Analytics ACTIVE**
  - ✓ **Medication Safety AI ACTIVE**
  - ✓ **Fatigue Monitoring ACTIVE**
  - ✓ **Predictive Staffing ACTIVE**
- 

### 14. FULL MEDBRIDGE+ NURSING ARCHITECTURE





## MEDBRIDGE+ — RMS DIRECTOR NATIONAL SUPPLY CHAIN COMMAND PORTAL

### Rwanda Medical Supply (RMS) Executive Intelligence Dashboard

#### 1. RMS DIRECTOR — MAIN ENTRY PORTAL

## National Pharmaceutical & Medical Logistics Intelligence Platform

**AI Forecast Engine:**  **ENABLED**

**| Drug consumption analytics      | ICU supply alerts      | District supply coordination**

| Vaccine cold chain                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Counterfeit risk detection                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | RBC integration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>1. <b>Temperature Monitoring:</b> Real-time monitoring of temperature throughout the supply chain using IoT sensors.</p> <p>2. <b>Chain of Custody:</b> Tracking the movement of vaccines from manufacturer to end-user.</p> <p>3. <b>Authentication:</b> Verifying the authenticity of vaccine vials using digital signatures or QR codes.</p> <p>4. <b>Inventory Management:</b> Maintaining accurate records of vaccine stock levels and expiration dates.</p> <p>5. <b>Regulatory Compliance:</b> Ensuring adherence to international and national regulatory standards.</p> | <p>1. <b>Visual Inspection:</b> Identifying physical signs of tampering or damage on packaging.</p> <p>2. <b>Barcode Verification:</b> Using barcodes to cross-check product information against a secure database.</p> <p>3. <b>Chemical Analysis:</b> Employing laboratory tests to detect counterfeit ingredients.</p> <p>4. <b>Blockchain:</b> Utilizing distributed ledger technology to create an immutable record of product provenance.</p> <p>5. <b>AI-Powered Detection:</b> Using machine learning algorithms to analyze patterns and identify potential counterfeit risks.</p> | <p>1. <b>Integration with Existing Systems:</b> Seamlessly connecting the RBC platform with current supply chain management software.</p> <p>2. <b>Standardization:</b> Establishing common protocols and data formats for interoperability.</p> <p>3. <b>Security Measures:</b> Implementing robust encryption and access controls to protect sensitive data.</p> <p>4. <b>Scalability:</b> Designing the system to handle increasing volumes of transactions and data.</p> <p>5. <b>User Training:</b> Providing comprehensive training for all stakeholders involved in the supply chain.</p> |

| LEFT SIDEBAR | CENTRAL SUPPLY CHAIN WORKSPACE | RIGHT  
ANALYTICS PANEL |



-  Dashboard Home

 Medicine Stock

 Shortage Detection

 Delivery Operations

 Lab Supply Chain

 National Analytics

 Minister Reporting
-  National Inventory Monitor

 Fleet Distribution Tracking

 Lab & Vaccine Monitoring

 Warehouse Capacity Mapping

 Emergency Supply Coordination
-  Live National

 AI Supply Alerts

 Consumption

 Minister

 Regional

3. LEFT SIDEBAR — NATIONAL MEDICINE STOCK

 NATIONAL MEDICINE INVENTORY

 ADEQUATE STOCK

1. Paracetamol  
National Stock: 1,200,000  
Status:  STABLE

2. Ceftriaxone  
National Stock: 420,000  
Status:  STABLE

3. Artemether-Lumefantrine  
National Stock: 710,000  
Status:  STABLE

---

 CRITICAL SHORTAGES

1. Oxygen Cylinders

Available: 112  
Threshold: 400  
Status: ✖ CRITICAL SHORTAGE

2. ICU Ventilator Tubing  
Available: 82  
Threshold: 300  
Status: ✖ NATIONAL ALERT

3. Insulin  
Available: LOW  
Status: ⚠ DISTRIBUTION PRIORITY

4. CENTRAL PANEL — NATIONAL INVENTORY MONITOR

💊 NATIONAL PHARMACEUTICAL INTELLIGENCE CENTER

| REGION            | STOCK LEVEL | MONTHLY USAGE | STATUS                                            |
|-------------------|-------------|---------------|---------------------------------------------------|
| Kigali            | 89%         | HIGH          | <span style="color: orange;">●</span> HIGH DEMAND |
| Northern Province | 67%         | MODERATE      | <span style="color: green;">●</span> STABLE       |
| Southern Province | 51%         | HIGH          | <span style="color: orange;">●</span> LOW STOCK   |
| Eastern Province  | 42%         | CRITICAL      | <span style="color: red;">●</span> SHORTAGE       |
| Western Province  | 71%         | MODERATE      | <span style="color: green;">●</span> STABLE       |

- AI SUPPLY ANALYSIS:
- ⚠ Oxygen shortage predicted within 48 hours
  - ⚠ Malaria drug demand spike expected
  - ⚠ Southern Province vaccine depletion risk detected

5. NATIONAL WAREHOUSE OPERATIONS

|                             |  |  |  |
|-----------------------------|--|--|--|
| 📦 NATIONAL WAREHOUSE STATUS |  |  |  |
|-----------------------------|--|--|--|

WAREHOUSE NETWORK

| WAREHOUSE      | CAPACITY | OCCUPANCY | STATUS      |
|----------------|----------|-----------|-------------|
| Kigali Central | 100%     | 92%       | 🟡 HIGH LOAD |
| Huye Storage   | 100%     | 68%       | 🟢 STABLE    |
| Musanze Depot  | 100%     | 74%       | 🟢 STABLE    |
| Rubavu Depot   | 100%     | 89%       | 🟡 HIGH LOAD |

AI STORAGE ALERTS:

- ⚠️ Kigali central warehouse nearing overload
- ⚠️ Temperature fluctuations detected in vaccine room
- ⚠️ Expiry-risk medicines identified

6. DELIVERY & DISTRIBUTION OPERATIONS

|                                             |  |  |  |
|---------------------------------------------|--|--|--|
| 🚚 NATIONAL DISTRIBUTION & DELIVERY TRACKING |  |  |  |
|---------------------------------------------|--|--|--|

ACTIVE DELIVERY FLEET

| VEHICLE ID  | DESTINATION        | CARGO          | STATUS       |
|-------------|--------------------|----------------|--------------|
| RMS-TRK-001 | CHUK               | ICU Supplies   | 🚚 IN TRANSIT |
| RMS-TRK-014 | Nyagatare Hospital | Vaccines       | 🚚 DELIVERING |
| RMS-TRK-021 | Ruhengeri Hospital | Blood Products | ⚠️ DELAYED   |

DRONE DELIVERY STATUS

Drone ID: DRN-8821  
Destination: Remote Health Center  
Payload: Emergency Blood Units  
ETA: 14 minutes  
Status: 🚁 ACTIVE FLIGHT

7. LABORATORY & VACCINE SUPPLY CHAIN

🧪 LABORATORY & VACCINE LOGISTICS

| ITEM               | STOCK       | TEMPERATURE | STATUS      |
|--------------------|-------------|-------------|-------------|
| COVID PCR Reagents | 8,200 Kits  | 4°C         | ● STABLE    |
| Blood Bags         | 2,100 Units | Controlled  | ● STABLE    |
| Childhood Vaccines | LOW         | 2°C         | ● REPLENISH |
| Rabies Vaccines    | CRITICAL    | 2°C         | ● SHORTAGE  |

COLD CHAIN ALERTS:

- ⚠ Vaccine freezer instability detected
- ⚠ Rabies vaccine emergency procurement recommended

8. RIGHT PANEL — RMS EXECUTIVE ANALYTICS

NATIONAL RMS EXECUTIVE INTELLIGENCE

TOTAL ACTIVE FACILITIES: 612  
NATIONAL STOCK COVERAGE: 78%

**ACTIVE DELIVERIES:** 214  
**EMERGENCY ALERTS:** 7  
**DRONES ACTIVE:** 11

**AI PREDICTIONS:**

- ⚠ ICU oxygen depletion risk
- ⚠ Rising malaria drug demand
- ⚠ Counterfeit medicine risk detected

**MINISTER REPORTS:**

- ✓ National shortage escalation sent
  - ✓ RBC synchronized
  - ✓ Emergency logistics activated
- 

**9. DIRECT REPORTING TO MINISTER**

+=====+  
=====+  
|  **NATIONAL MINISTERIAL SUPPLY ESCALATION**  
|  
+=====+  
=====+

**DIRECT CHANNEL:**

- ✓ Ministry of Health Connected
- ✓ RBC Emergency Coordination Connected
- ✓ National Disaster Unit Connected

**REPORT TYPES:**

- ✓ National Medicine Shortages
- ✓ Oxygen Emergencies
- ✓ Vaccine Cold Chain Failures
- ✓ Counterfeit Medicine Detection
- ✓ Epidemic Supply Surges
- ✓ ICU Capacity Supply Collapse

**LATEST REPORT:**

-----

**Subject:**

**National Oxygen Crisis Escalation**

- **CHUK**
- **King Faisal Hospital**
- **Rwanda Military Hospital**

**CRITICAL NATIONAL ALERT**

**07:41 AM**

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# | 🧠 AI SUPPLY CHAIN INTELLIGENCE |

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- ⚠️ Counterfeit medicine pattern detected
- ⚠️ Unusual consumption spikes identified
- ⚠️ Expiry-risk medicines flagged
- ⚠️ Vaccine cold-chain instability predicted
- ⚠️ Regional malaria outbreak demand forecasted
- ⚠️ Emergency oxygen depletion risk detected

- ✓ Predictive Demand ACTIVE
- ✓ Route Optimization ACTIVE
- ✓ Counterfeit Detection ACTIVE
- ✓ Smart Procurement ACTIVE
- ✓ Epidemic Forecasting ACTIVE

## 11. NATIONAL MEDICINE HEATMAP

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|  **NATIONAL MEDICINE DEMAND HEATMAP**

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**KIGALI**  **HIGH CONSUMPTION**  
**NORTHERN PROVINCE**  **STABLE**  
**SOUTHERN PROVINCE**  **RISING DEMAND**  
**EASTERN PROVINCE**  **CRITICAL SHORTAGE**  
**WESTERN PROVINCE**  **STABLE**

**LEGEND:**

-  **Stable**
  -  **Moderate Demand**
  -  **High Demand**
  -  **Critical Shortage**
- 

## 12. RMS NATIONAL OPERATIONS FLOW

**Hospital Consumption Data**



**Regional Warehouse Monitoring**



**AI Supply Forecasting**



**RMS Director Dashboard**



**National Emergency Escalation**



**Ministerial Coordination**

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## 13. COUNTERFEIT MEDICINE DETECTION CENTER

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|  **COUNTERFEIT MEDICINE FORENSIC UNIT**

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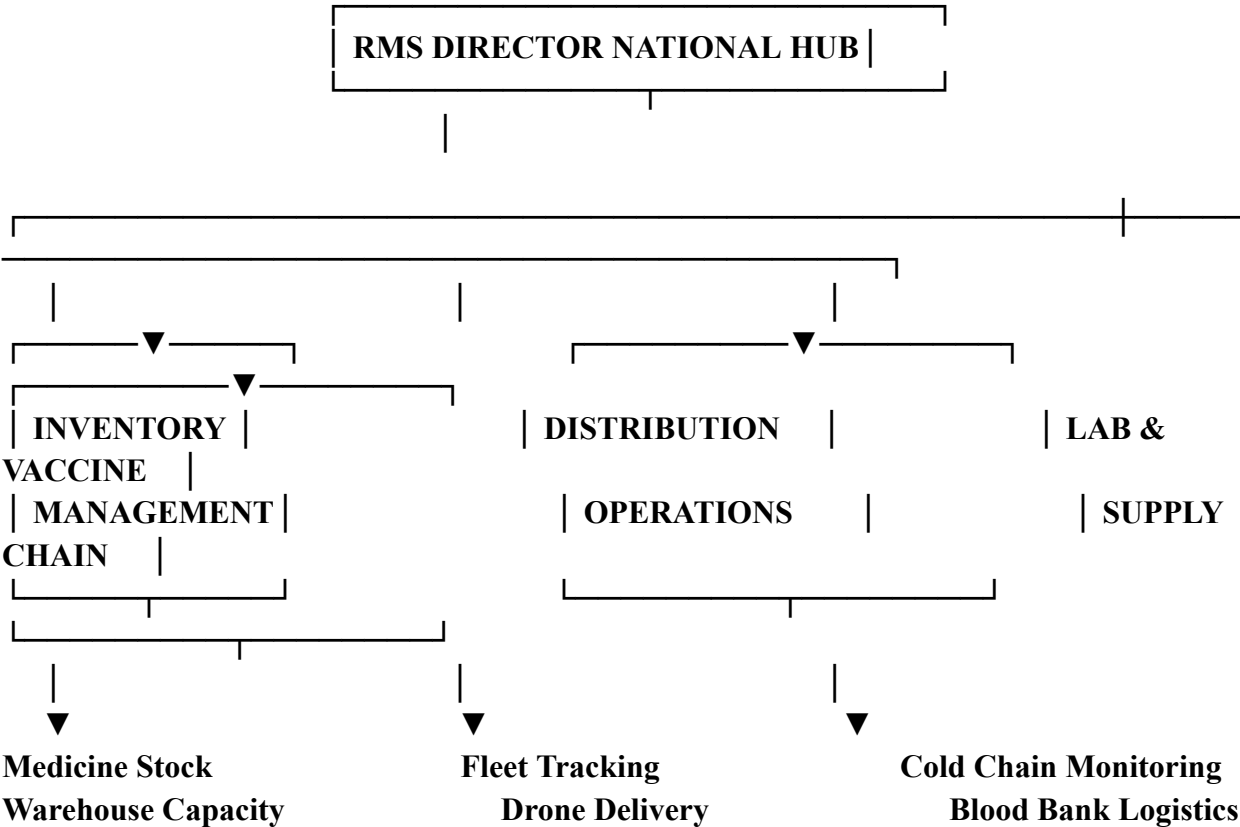
|  |
|--|
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|  |

SUSPICIOUS PRODUCTS

| MEDICATION           | REGION           | RISK LEVEL                                   | STATUS        |
|----------------------|------------------|----------------------------------------------|---------------|
| Amoxicillin Batch A2 | Eastern Province | <span style="color:red">●</span> HIGH        | INVESTIGATION |
| Insulin Batch X9     | Kigali           | <span style="color:orange">●</span> MODERATE | LAB TESTING   |

- AI FORENSIC FINDINGS:
- ⚠ Barcode mismatch detected
  - ⚠ Distribution anomaly identified
  - ⚠ Unauthorized supplier route flagged

14. FULL RMS NATIONAL ARCHITECTURE





Reagent



|                                                                                                          |                                                                                                            |                                                                                                            |
|----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
|  National Claims Center |  Fraud Detection          |  Insurance Verification |
| Real-time claim approvals                                                                                | Suspicious billing detection                                                                               | Mutuelle eligibility                                                                                       |
| Hospital reimbursement                                                                                   | Duplicate claims analysis                                                                                  | National coverage validation                                                                               |
| Payment reconciliation                                                                                   | Ghost patient identification                                                                               | Biometric verification                                                                                     |
| +-----+-----+-----+                                                                                      |                                                                                                            |                                                                                                            |
| +-----+-----+-----+                                                                                      |                                                                                                            |                                                                                                            |
|  National Analytics     |  Hospital Billing Monitor |  Ministerial Reporting  |
| Claim heatmaps                                                                                           | Facility financial tracking                                                                                | Direct MoH escalation                                                                                      |
| Disease expenditure trends                                                                               | Outstanding reimbursements                                                                                 | National financial alerts                                                                                  |
| AI cost forecasting                                                                                      | Billing code audits                                                                                        | RBC integration                                                                                            |
| +-----+-----+-----+                                                                                      |                                                                                                            |                                                                                                            |

2. FULL RSSB DASHBOARD LAYOUT

|                                                                                                            |                                                                                                                   |                                                                                                              |
|------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| +=====+<br>=====+                                                                                          |                                                                                                                   |                                                                                                              |
| LEFT SIDEBAR                                                                                               | CENTRAL FINANCIAL WORKSPACE                                                                                       | RIGHT                                                                                                        |
| ANALYTICS PANEL                                                                                            |                                                                                                                   |                                                                                                              |
| +=====+<br>=====+                                                                                          |                                                                                                                   |                                                                                                              |
|  Dashboard Home         |  National Claims Processing    |  Financial Reports      |
|  Insurance Verification |  Hospital Reimbursement Center |  AI Fraud Alerts        |
|  Billing Audits         |  Expenditure Analytics         |  Claim Trends            |
|  Fraud Detection        |  Predictive Cost Forecasting   |  Minister Notifications |
|  National Analytics     |  National Coverage Heatmap     |  Regional Spending      |
|  Minister Reporting     |                                                                                                                   |                                                                                                              |

3. LEFT SIDEBAR — INSURANCE VERIFICATION

|                                                                                                                     |  |  |
|---------------------------------------------------------------------------------------------------------------------|--|--|
| +=====+<br>=====+                                                                                                   |  |  |
|  NATIONAL INSURANCE VERIFICATION |  |  |
|                                                                                                                     |  |  |

+=====+  
=====+

## ACTIVE VERIFICATIONS

### 1. Patient: Uwimana Chantal

**RSSB Number: RSSB-009821**

**Insurance Type: Mutuelle**

**Status: ✓ ACTIVE**

### 2. Patient: Habimana Eric

**RSSB Number: RSSB-110922**

**Insurance Type: RAMA**

**Status: ✓ ACTIVE**

### 3. Patient: Mukamana Alice

**RSSB Number: RSSB-889211**

**Insurance Type: Military Insurance**

**Status: ✓ VERIFIED**

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## ⚠ INVALID / SUSPICIOUS

### 1. Jean Bosco

**RSSB Number: RSSB-771201**

**Status: ✗ EXPIRED COVERAGE**

### 2. Claudette Uwera

**RSSB Number: RSSB-112933**

**Status: ⚠ DUPLICATE REGISTRATION**

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## 4. CENTRAL PANEL — NATIONAL CLAIMS PROCESSING

+=====+  
=====+

|  NATIONAL HEALTHCARE CLAIMS PROCESSING CENTER  
|

| HOSPITAL             | CLAIMS TODAY | APPROVED | STATUS                     |
|----------------------|--------------|----------|----------------------------|
| CHUK                 | 2,182        | 2,001    | <span>●</span> STABLE      |
| King Faisal          | 1,144        | 1,003    | <span>●</span> REVIEW      |
| Rwanda Military Hosp | 912          | 870      | <span>●</span> STABLE      |
| Butare Hospital      | 780          | 642      | <span>●</span> AUDIT ALERT |

#### AI CLAIMS ANALYSIS:

- ⚠ Duplicate billing trend detected
- ⚠ ICU expenditure spike identified
- ⚠ Unusual oncology claims increase detected

## 5. HOSPITAL REIMBURSEMENT CENTER

### 💰 HOSPITAL REIMBURSEMENT & PAYMENT MONITOR

| FACILITY            | PENDING PAYMENT | APPROVED | STATUS                      |
|---------------------|-----------------|----------|-----------------------------|
| CHUK                | 482M RWF        | 400M RWF | <span>●</span> PROCESSING   |
| Kibagabaga Hospital | 119M RWF        | 117M RWF | <span>●</span> CLEARED      |
| Ruhengeri Hospital  | 88M RWF         | 61M RWF  | <span>●</span> UNDER REVIEW |
| Nyagatare Hospital  | 44M RWF         | 42M RWF  | <span>●</span> CLEARED      |

#### FINANCIAL ALERTS:

- ⚠ Delayed reimbursement risk
- ⚠ Billing inconsistencies identified
- ⚠ Emergency epidemic expenditure rising

6. FRAUD DETECTION & FORENSIC BILLING

AI FRAUD DETECTION & FORENSIC CLAIM ANALYSIS

SUSPICIOUS CLAIMS

| CLAIM ID   | FACILITY        | RISK LEVEL           | STATUS        |
|------------|-----------------|----------------------|---------------|
| CLM-009211 | Butare Hospital | <div></div> HIGH     | INVESTIGATION |
| CLM-110022 | Private Clinic  | <div></div> MODERATE | AUDIT REVIEW  |
| CLM-771003 | CHUK            | <div></div> LOW      | MONITORING    |

AI FORENSIC DETECTIONS:

- ⚠ Duplicate patient billing
- ⚠ Ghost patient suspicion
- ⚠ Inflated laboratory pricing
- ⚠ Unusual surgical coding patterns

7. BILLING AUDIT TERMINAL

HOSPITAL BILLING AUDIT SYSTEM

AUDITED CLAIM:

Claim ID: CLM-009211

PATIENT:

Name: Jean Claude Ndayisaba  
Insurance: Mutuelle

| SERVICE          | AMOUNT CLAIMED | APPROVED      | STATUS       |
|------------------|----------------|---------------|--------------|
| ICU Admission    | 1,200,000 RWF  | 1,200,000 RWF | ✓ APPROVED   |
| CT Scan          | 450,000 RWF    | 450,000 RWF   | ✓ APPROVED   |
| Laboratory Tests | 980,000 RWF    | ⚠ REVIEW      | 🟡 HIGH PRICE |
| Oxygen Therapy   | 1,400,000 RWF  | ✗ BLOCKED     | 🔴 DUPLICATE  |

AUDIT STATUS:  
🚨 FORENSIC BILLING REVIEW ACTIVE

### 8. NATIONAL COVERAGE HEATMAP

| NATIONAL HEALTH INSURANCE COVERAGE |
|------------------------------------|
|                                    |

- KIGALI 🟢 HIGH COVERAGE
- NORTHERN PROVINCE 🟢 STABLE
- SOUTHERN PROVINCE 🟡 MODERATE
- EASTERN PROVINCE 🟠 LOW ENROLLMENT
- WESTERN PROVINCE 🟢 STABLE

- LEGEND:
- 🟢 High Coverage
  - 🟡 Moderate Coverage
  - 🟠 Low Enrollment
  - 🔴 Critical Insurance Gap

### 9. RIGHT PANEL — RSSB EXECUTIVE ANALYTICS

| RSSB EXECUTIVE INTELLIGENCE |
|-----------------------------|
|                             |

**TOTAL ACTIVE BENEFICIARIES: 12.4M**

**CLAIMS TODAY: 18,422**

**APPROVED CLAIMS: 16,911**

**FRAUD ALERTS: 29**

**ACTIVE AUDITS: 14**

**AI FORECASTS:**

- ⚠ **Malaria treatment costs rising**
- ⚠ **Oncology expenditure increasing**
- ⚠ **Emergency epidemic funding risk**

**MINISTER REPORTS:**

- ✓ **National expenditure escalation sent**
  - ✓ **RBC synchronized**
  - ✓ **Fraud investigation active**
- 

**10. DIRECT REPORTING TO MINISTER**

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|  **NATIONAL FINANCIAL ESCALATION CHANNEL**

|

+=====

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**DIRECT CONNECTIONS:**

- ✓ **Ministry of Health**
- ✓ **Ministry of Finance**
- ✓ **RBC National Surveillance**
- ✓ **National Emergency Coordination**

**REPORT TYPES:**

- ✓ **National Healthcare Spending**
- ✓ **Epidemic Financial Burden**
- ✓ **Insurance Fraud Escalations**
- ✓ **Hospital Reimbursement Crises**
- ✓ **ICU Expenditure Emergencies**
- ✓ **National Insurance Coverage Gaps**

**LATEST REPORT:**

## Emergency Oncology Expenditure Escalation

- **CHUK**
- **King Faisal Hospital**
- **Cancer Referral Unit**

**NATIONAL FINANCIAL ALERT**

**09:12 AM**

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# AI HEALTH FINANCING INTELLIGENCE

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- ⚠ Duplicate claim submission detected
- ⚠ Ghost patient fraud suspected
- ⚠ Epidemic expenditure surge predicted
- ⚠ Oncology treatment costs rising
- ⚠ ICU reimbursement overload forecasted
- ⚠ Suspicious private clinic billing identified

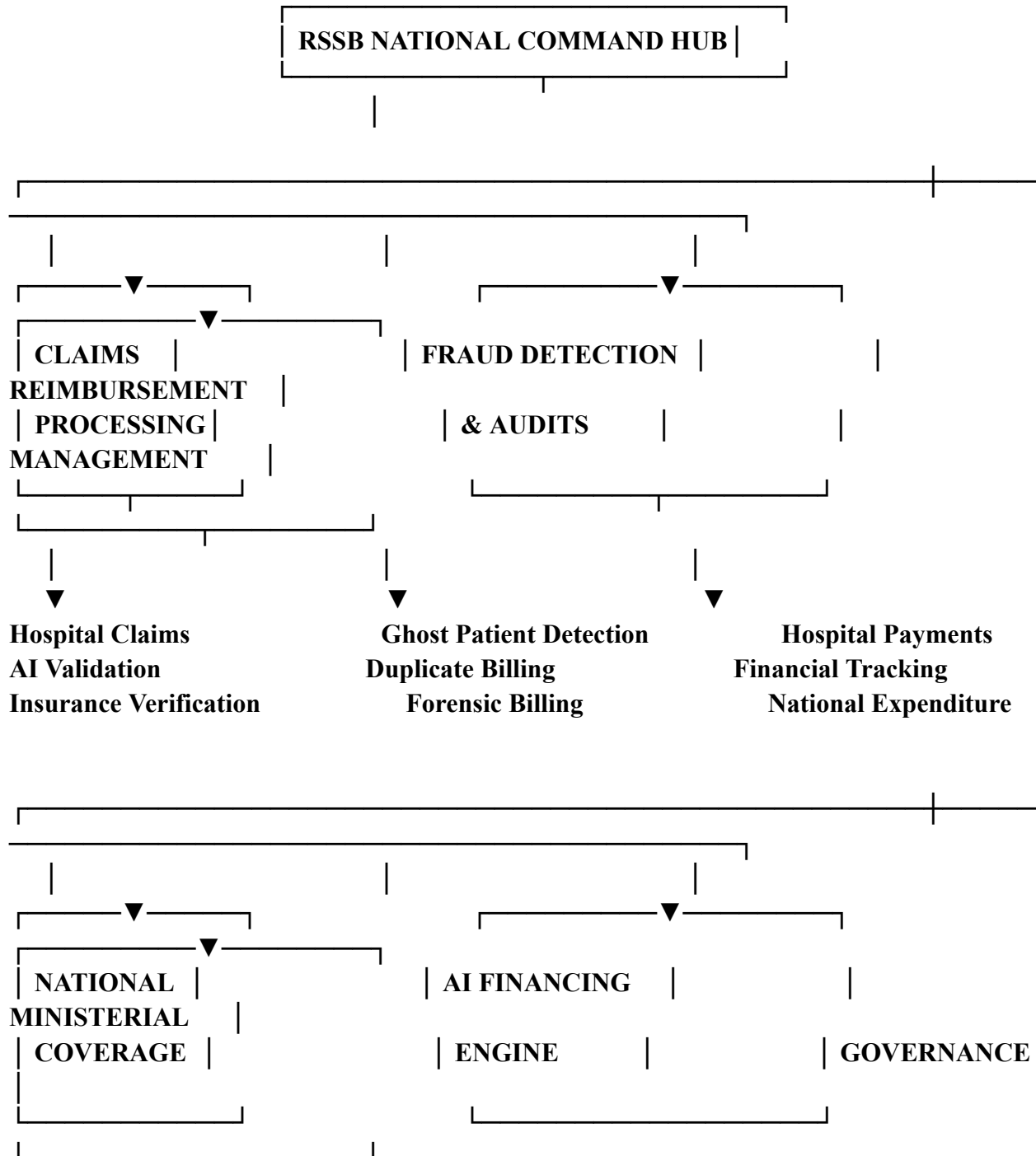
- ✓ **Fraud Detection ACTIVE**
- ✓ **Predictive Cost Forecasting ACTIVE**
- ✓ **National Spending Analytics ACTIVE**
- ✓ **Smart Claims Verification ACTIVE**
- ✓ **Epidemic Financial Modeling ACTIVE**





- ✓ Fingerprint Verified
- ✓ Facial Match Confirmed

## 14. FULL RSSB NATIONAL ARCHITECTURE



## Rwanda Social Security Board (RSSB) Smart Healthcare Financing & Claims Intelligence System

### 1. RSSB DIRECTOR — MAIN ENTRY PORTAL

| MEDBRIDGE+ — RSSB NATIONAL HEALTH INSURANCE COMMAND CENTER

| Smart Claims, Mutuelle Management & National Healthcare Financing Platform

Director General: Claudine Uwase

Director ID: RSSB-DIR-0007

Insurance Grid:  ACTIVE

AI Fraud Engine:  ENABLED

|  National Claims Center |  Fraud Detection |  Insurance Verification

| Real-time claim approvals | Suspicious billing detection | Mutuelle eligibility |  
| Hospital reimbursement | Duplicate claims analysis | National coverage validation

| Payment reconciliation | Ghost patient identification | Biometric verification |

|  National Analytics |  Hospital Billing Monitor |  Ministerial Reporting

| Claim heatmaps | Facility financial tracking | Direct MoH escalation |  
| Disease expenditure trends | Outstanding reimbursements | National financial alerts

| AI cost forecasting | Billing code audits | RBC integration |

The diagram illustrates a three-pane financial dashboard layout. It consists of a top header bar, a central workspace, and a bottom footer bar. The top header bar is divided into three sections: 'LEFT SIDEBAR' (containing 'ANALYTICS PANEL'), 'CENTRAL FINANCIAL WORKSPACE', and 'RIGHT' (containing 'MARKET DATA'). The central workspace is the largest area, currently displaying a 'FINANCIAL STATEMENT' table with columns for Date, Revenue, Expenses, Profit, Assets, Liabilities, and Equity. The bottom footer bar contains a 'TOTAL REVENUE' section with a value of '\$1,234,567' and a 'TOTAL EXPENSES' section with a value of '\$876,543'.

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## NATIONAL INSURANCE VERIFICATION

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- 1. Patient: Uwimana Chantal**  
**RSSB Number: RSSB-009821**  
**Insurance Type: Mutuelle**  
**Status: ✓ ACTIVE**
- 2. Patient: Habimana Eric**  
**RSSB Number: RSSB-110922**  
**Insurance Type: RAMA**  
**Status: ✓ ACTIVE**
- 3. Patient: Mukamana Alice**  
**RSSB Number: RSSB-889211**  
**Insurance Type: Military Insurance**

Status: ✓ VERIFIED

⚠ INVALID / SUSPICIOUS

1. Jean Bosco  
RSSB Number: RSSB-771201  
Status: ✗ EXPIRED COVERAGE

2. Claudette Uwera  
RSSB Number: RSSB-112933  
Status: ⚠ DUPLICATE REGISTRATION

4. CENTRAL PANEL — NATIONAL CLAIMS PROCESSING

=====+  
=====+  
| 🏥 NATIONAL HEALTHCARE CLAIMS PROCESSING CENTER  
|  
+=====+  
=====+

| HOSPITAL             | CLAIMS TODAY | APPROVED | STATUS        |
|----------------------|--------------|----------|---------------|
| CHUK                 | 2,182        | 2,001    | 🟢 STABLE      |
| King Faisal          | 1,144        | 1,003    | 🟡 REVIEW      |
| Rwanda Military Hosp | 912          | 870      | 🟢 STABLE      |
| Butare Hospital      | 780          | 642      | 🟠 AUDIT ALERT |

- AI CLAIMS ANALYSIS:
- ⚠ Duplicate billing trend detected
  - ⚠ ICU expenditure spike identified
  - ⚠ Unusual oncology claims increase detected

5. HOSPITAL REIMBURSEMENT CENTER

|                                          |  |  |  |
|------------------------------------------|--|--|--|
| HOSPITAL REIMBURSEMENT & PAYMENT MONITOR |  |  |  |
|------------------------------------------|--|--|--|

| FACILITY            | PENDING PAYMENT | APPROVED | STATUS       |
|---------------------|-----------------|----------|--------------|
| CHUK                | 482M RWF        | 400M RWF | PROCESSING   |
| Kibagabaga Hospital | 119M RWF        | 117M RWF | CLEARED      |
| Ruhengeri Hospital  | 88M RWF         | 61M RWF  | UNDER REVIEW |
| Nyagatare Hospital  | 44M RWF         | 42M RWF  | CLEARED      |

- FINANCIAL ALERTS:
- Delayed reimbursement risk
  - Billing inconsistencies identified
  - Emergency epidemic expenditure rising

6. FRAUD DETECTION & FORENSIC BILLING

|                                              |  |  |  |
|----------------------------------------------|--|--|--|
| AI FRAUD DETECTION & FORENSIC CLAIM ANALYSIS |  |  |  |
|----------------------------------------------|--|--|--|

SUSPICIOUS CLAIMS

| CLAIM ID   | FACILITY        | RISK LEVEL | STATUS        |
|------------|-----------------|------------|---------------|
| CLM-009211 | Butare Hospital | HIGH       | INVESTIGATION |
| CLM-110022 | Private Clinic  | MODERATE   | AUDIT REVIEW  |
| CLM-771003 | CHUK            | LOW        | MONITORING    |

+-----+-----+-----+-----+

**AI FORENSIC DETECTIONS:**

- ⚠ Duplicate patient billing
- ⚠ Ghost patient suspicion
- ⚠ Inflated laboratory pricing
- ⚠ Unusual surgical coding patterns

**7. BILLING AUDIT TERMINAL**

+=====+

=====+

 HOSPITAL BILLING AUDIT SYSTEM

+=====+

=====+

**AUDITED CLAIM:**  
Claim ID: CLM-009211

**PATIENT:**  
Name: Jean Claude Ndayisaba  
Insurance: Mutuelle

|                           |                |               |              |
|---------------------------|----------------|---------------|--------------|
| +-----+-----+-----+-----+ |                |               |              |
| SERVICE                   | AMOUNT CLAIMED | APPROVED      | STATUS       |
| +-----+-----+-----+-----+ |                |               |              |
| ICU Admission             | 1,200,000 RWF  | 1,200,000 RWF | ✓ APPROVED   |
| CT Scan                   | 450,000 RWF    | 450,000 RWF   | ✓ APPROVED   |
| Laboratory Tests          | 980,000 RWF    | ⚠ REVIEW      | ● HIGH PRICE |
| Oxygen Therapy            | 1,400,000 RWF  | ✗ BLOCKED     | ● DUPLICATE  |
| +-----+-----+-----+-----+ |                |               |              |

**AUDIT STATUS:**  
 FORENSIC BILLING REVIEW ACTIVE

**8. NATIONAL COVERAGE HEATMAP**

+=====+

=====+

## | NATIONAL HEALTH INSURANCE COVERAGE

|

+=====+

=====+

KIGALI  HIGH COVERAGE  
NORTHERN PROVINCE  STABLE  
SOUTHERN PROVINCE  MODERATE  
EASTERN PROVINCE  LOW ENROLLMENT  
WESTERN PROVINCE  STABLE

### LEGEND:

-  High Coverage
-  Moderate Coverage
-  Low Enrollment
-  Critical Insurance Gap

## 9. RIGHT PANEL — RSSB EXECUTIVE ANALYTICS

+=====+

|  RSSB EXECUTIVE INTELLIGENCE |

+=====+

**TOTAL ACTIVE BENEFICIARIES: 12.4M**

**CLAIMS TODAY: 18,422**

**APPROVED CLAIMS: 16,911**

**FRAUD ALERTS: 29**

**ACTIVE AUDITS: 14**

### AI FORECASTS:

- ⚠ Malaria treatment costs rising
- ⚠ Oncology expenditure increasing
- ⚠ Emergency epidemic funding risk

### MINISTER REPORTS:

- ✓ National expenditure escalation sent
- ✓ RBC synchronized
- ✓ Fraud investigation active



**NATIONAL FINANCIAL ESCALATION CHANNEL**

- ✓ Ministry of Health
- ✓ Ministry of Finance
- ✓ RBC National Surveillance
- ✓ National Emergency Coordination

- ✓ **National Healthcare Spending**
- ✓ **Epidemic Financial Burden**
- ✓ **Insurance Fraud Escalations**
- ✓ **Hospital Reimbursement Crises**
- ✓ **ICU Expenditure Emergencies**
- ✓ **National Insurance Coverage Gaps**

---

**Subject:**

## Emergency Oncology Expenditure Escalation

**Affected Facilities:**

- **CHUK**
- **King Faisal Hospital**
- **Cancer Referral Unit**

**Priority:**

**NATIONAL FINANCIAL ALERT**

**Timestamp:**

## 11. AI HEALTH FINANCING ENGINE

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## | AI HEALTH FINANCING INTELLIGENCE

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### ACTIVE AI DETECTIONS

- ⚠ Duplicate claim submission detected
- ⚠ Ghost patient fraud suspected
- ⚠ Epidemic expenditure surge predicted
- ⚠ Oncology treatment costs rising
- ⚠ ICU reimbursement overload forecasted
- ⚠ Suspicious private clinic billing identified

### AI SYSTEMS:

- ✓ Fraud Detection ACTIVE
  - ✓ Predictive Cost Forecasting ACTIVE
  - ✓ National Spending Analytics ACTIVE
  - ✓ Smart Claims Verification ACTIVE
  - ✓ Epidemic Financial Modeling ACTIVE
- 

## 12. NATIONAL HEALTH FINANCING FLOW

Hospital Clinical Services



Electronic Claim Submission



AI Billing Validation



RSSB Claims Dashboard



Fraud Detection & Audit



National Financial Reporting



Ministerial Coordination

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13. SMART BENEFICIARY PROFILE

BENEFICIARY INSURANCE PROFILE

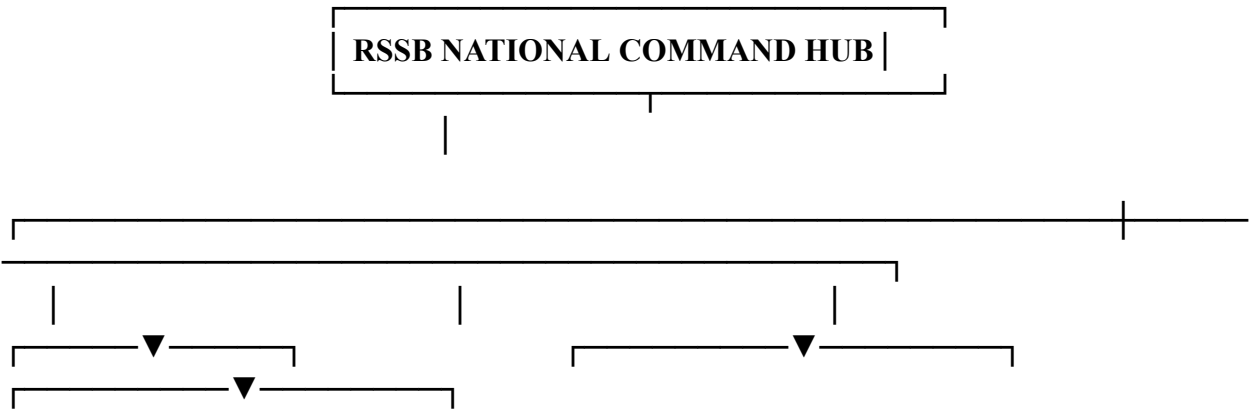
FULL NAME: Uwimana Chantal  
NATIONAL ID: 1199080022113344  
RSSB NUMBER: RSSB-009821  
INSURANCE TYPE: Mutuelle de Santé  
STATUS: ✓ ACTIVE

FAMILY COVERAGE:  
✓ Husband  
✓ 3 Children

RECENT CLAIMS:  
• CHUK Consultation  
• Laboratory Services  
• Pharmacy Dispensation

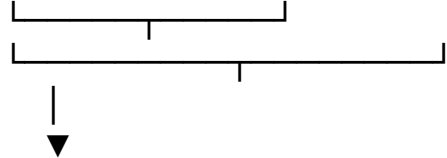
BIOMETRIC STATUS:  
✓ Fingerprint Verified  
✓ Facial Match Confirmed

14. FULL RSSB NATIONAL ARCHITECTURE

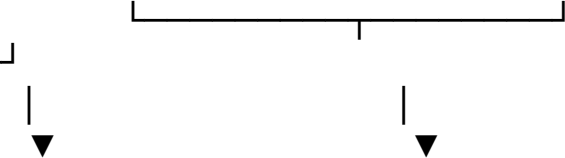


| CLAIMS |  
REIMBURSEMENT |  
| PROCESSING |  
MANAGEMENT |

| FRAUD DETECTION |  
| & AUDITS |

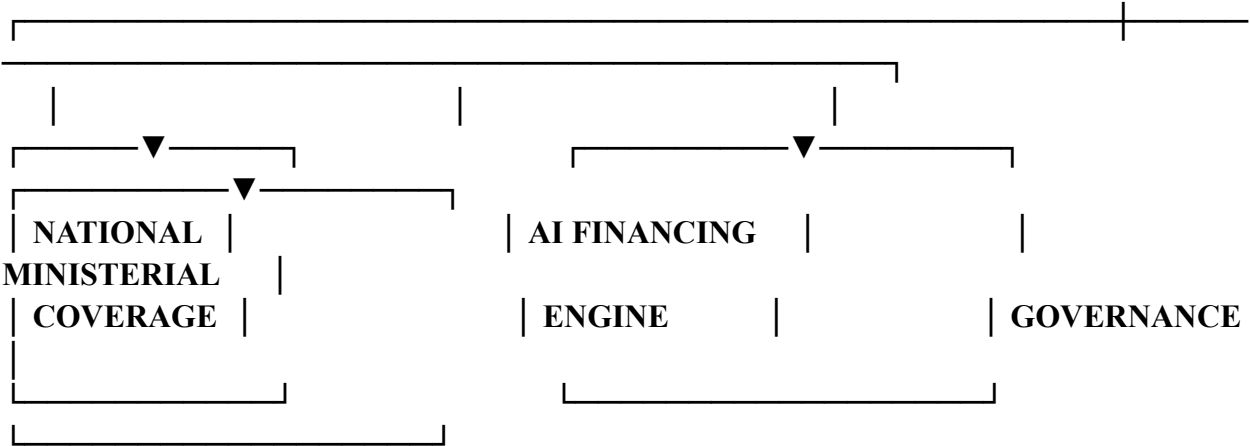


Hospital Claims  
AI Validation  
Insurance Verification



Ghost Patient Detection  
Duplicate Billing  
Forensic Billing

Hospital Payments  
Financial Tracking  
National Expenditure



NATIONAL  
MINISTERIAL  
COVERAGE

AI FINANCING  
ENGINE GOVERNANCE

## Rwanda Social Security Board (RSSB) Smart Healthcare Financing & Claims Intelligence System

### 1. RSSB DIRECTOR — MAIN ENTRY PORTAL

| MEDBRIDGE+ — RSSB NATIONAL HEALTH INSURANCE COMMAND CENTER

| Smart Claims, Mutuelle Management & National Healthcare Financing Platform

Director General: Claudine Uwase

Director ID: RSSB-DIR-0007

Insurance Grid:  ACTIVE

AI Fraud Engine:  ENABLED

|  National Claims Center |  Fraud Detection |  Insurance Verification

| Real-time claim approvals | Suspicious billing detection | Mutuelle eligibility |  
| Hospital reimbursement | Duplicate claims analysis | National coverage validation

| Payment reconciliation | Ghost patient identification | Biometric verification |

|  National Analytics |  Hospital Billing Monitor |  Ministerial Reporting

| Claim heatmaps | Facility financial tracking | Direct MoH escalation |  
| Disease expenditure trends | Outstanding reimbursements | National financial alerts

| AI cost forecasting | Billing code audits | RBC integration |

The diagram illustrates a three-pane financial application layout. It consists of a top header bar, a main content area divided into three vertical panes, and a bottom footer bar. The panes are labeled: 'LEFT SIDEBAR ANALYTICS PANEL' on the left, 'CENTRAL FINANCIAL WORKSPACE' in the middle, and 'RIGHT' on the right. The layout is enclosed in a double-line border with corner markers.

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## NATIONAL INSURANCE VERIFICATION

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Status: ✓ VERIFIED

⚠ INVALID / SUSPICIOUS

1. Jean Bosco  
RSSB Number: RSSB-771201  
Status: ✗ EXPIRED COVERAGE

2. Claudette Uwera  
RSSB Number: RSSB-112933  
Status: ⚠ DUPLICATE REGISTRATION

4. CENTRAL PANEL — NATIONAL CLAIMS PROCESSING

🏥 NATIONAL HEALTHCARE CLAIMS PROCESSING CENTER

| HOSPITAL             | CLAIMS TODAY | APPROVED | STATUS        |
|----------------------|--------------|----------|---------------|
| CHUK                 | 2,182        | 2,001    | ● STABLE      |
| King Faisal          | 1,144        | 1,003    | ● REVIEW      |
| Rwanda Military Hosp | 912          | 870      | ● STABLE      |
| Butare Hospital      | 780          | 642      | ● AUDIT ALERT |

- AI CLAIMS ANALYSIS:
- ⚠ Duplicate billing trend detected
  - ⚠ ICU expenditure spike identified
  - ⚠ Unusual oncology claims increase detected

5. HOSPITAL REIMBURSEMENT CENTER

|                                          |  |  |  |
|------------------------------------------|--|--|--|
| HOSPITAL REIMBURSEMENT & PAYMENT MONITOR |  |  |  |
|------------------------------------------|--|--|--|

| FACILITY            | PENDING PAYMENT | APPROVED | STATUS       |
|---------------------|-----------------|----------|--------------|
| CHUK                | 482M RWF        | 400M RWF | PROCESSING   |
| Kibagabaga Hospital | 119M RWF        | 117M RWF | CLEARED      |
| Ruhengeri Hospital  | 88M RWF         | 61M RWF  | UNDER REVIEW |
| Nyagatare Hospital  | 44M RWF         | 42M RWF  | CLEARED      |

- FINANCIAL ALERTS:
- Delayed reimbursement risk
  - Billing inconsistencies identified
  - Emergency epidemic expenditure rising

6. FRAUD DETECTION & FORENSIC BILLING

|                                              |  |  |  |
|----------------------------------------------|--|--|--|
| AI FRAUD DETECTION & FORENSIC CLAIM ANALYSIS |  |  |  |
|----------------------------------------------|--|--|--|

SUSPICIOUS CLAIMS

| CLAIM ID   | FACILITY        | RISK LEVEL | STATUS        |
|------------|-----------------|------------|---------------|
| CLM-009211 | Butare Hospital | HIGH       | INVESTIGATION |
| CLM-110022 | Private Clinic  | MODERATE   | AUDIT REVIEW  |
| CLM-771003 | CHUK            | LOW        | MONITORING    |



+-----+-----+-----+-----+

**AI FORENSIC DETECTIONS:**

- ⚠ Duplicate patient billing
- ⚠ Ghost patient suspicion
- ⚠ Inflated laboratory pricing
- ⚠ Unusual surgical coding patterns

**7. BILLING AUDIT TERMINAL**

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|  HOSPITAL BILLING AUDIT SYSTEM |

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**AUDITED CLAIM:**

**Claim ID:** CLM-009211

**PATIENT:**

**Name:** Jean Claude Ndayisaba

**Insurance:** Mutuelle

+-----+-----+-----+-----+

| SERVICE | AMOUNT CLAIMED | APPROVED | STATUS |
|---------|----------------|----------|--------|
|---------|----------------|----------|--------|

+-----+-----+-----+-----+

|               |               |               |            |
|---------------|---------------|---------------|------------|
| ICU Admission | 1,200,000 RWF | 1,200,000 RWF | ✓ APPROVED |
|---------------|---------------|---------------|------------|

|         |             |             |            |
|---------|-------------|-------------|------------|
| CT Scan | 450,000 RWF | 450,000 RWF | ✓ APPROVED |
|---------|-------------|-------------|------------|

|                  |             |          |              |
|------------------|-------------|----------|--------------|
| Laboratory Tests | 980,000 RWF | ⚠ REVIEW | ● HIGH PRICE |
|------------------|-------------|----------|--------------|

|                |               |           |             |
|----------------|---------------|-----------|-------------|
| Oxygen Therapy | 1,400,000 RWF | ✗ BLOCKED | ● DUPLICATE |
|----------------|---------------|-----------|-------------|

+-----+-----+-----+-----+

**AUDIT STATUS:**

 **FORENSIC BILLING REVIEW ACTIVE**

**8. NATIONAL COVERAGE HEATMAP**

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## | NATIONAL HEALTH INSURANCE COVERAGE

|

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KIGALI  HIGH COVERAGE  
NORTHERN PROVINCE  STABLE  
SOUTHERN PROVINCE  MODERATE  
EASTERN PROVINCE  LOW ENROLLMENT  
WESTERN PROVINCE  STABLE

### LEGEND:

-  High Coverage
-  Moderate Coverage
-  Low Enrollment
-  Critical Insurance Gap

## 9. RIGHT PANEL — RSSB EXECUTIVE ANALYTICS

+=====+

|  RSSB EXECUTIVE INTELLIGENCE |

+=====+

**TOTAL ACTIVE BENEFICIARIES: 12.4M**

**CLAIMS TODAY: 18,422**

**APPROVED CLAIMS: 16,911**

**FRAUD ALERTS: 29**

**ACTIVE AUDITS: 14**

### AI FORECASTS:

- ⚠ Malaria treatment costs rising
- ⚠ Oncology expenditure increasing
- ⚠ Emergency epidemic funding risk

### MINISTER REPORTS:

- ✓ National expenditure escalation sent
- ✓ RBC synchronized
- ✓ Fraud investigation active

## | NATIONAL FINANCIAL ESCALATION CHANNEL

- ✓ Ministry of Health
- ✓ Ministry of Finance
- ✓ RBC National Surveillance
- ✓ National Emergency Coordination

- ✓ **National Healthcare Spending**
- ✓ **Epidemic Financial Burden**
- ✓ **Insurance Fraud Escalations**
- ✓ **Hospital Reimbursement Crises**
- ✓ **ICU Expenditure Emergencies**
- ✓ **National Insurance Coverage Gaps**

---

**Subject:**

## Emergency Oncology Expenditure Escalation

**Affected Facilities:**

- **CHUK**
- **King Faisal Hospital**
- **Cancer Referral Unit**

**Priority:**

**NATIONAL FINANCIAL ALERT**

**Timestamp:**

## 11. AI HEALTH FINANCING ENGINE

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## | AI HEALTH FINANCING INTELLIGENCE

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  - ✓ Predictive Cost Forecasting ACTIVE
  - ✓ National Spending Analytics ACTIVE
  - ✓ Smart Claims Verification ACTIVE
  - ✓ Epidemic Financial Modeling ACTIVE
- 

## 12. NATIONAL HEALTH FINANCING FLOW

Hospital Clinical Services



Electronic Claim Submission



AI Billing Validation



RSSB Claims Dashboard



Fraud Detection & Audit



National Financial Reporting



Ministerial Coordination

---

13. SMART BENEFICIARY PROFILE

BENEFICIARY INSURANCE PROFILE

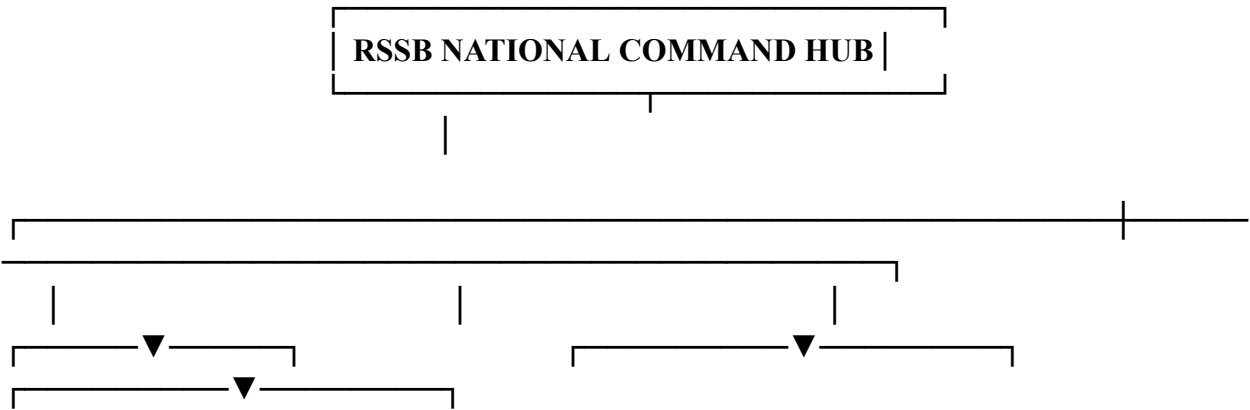
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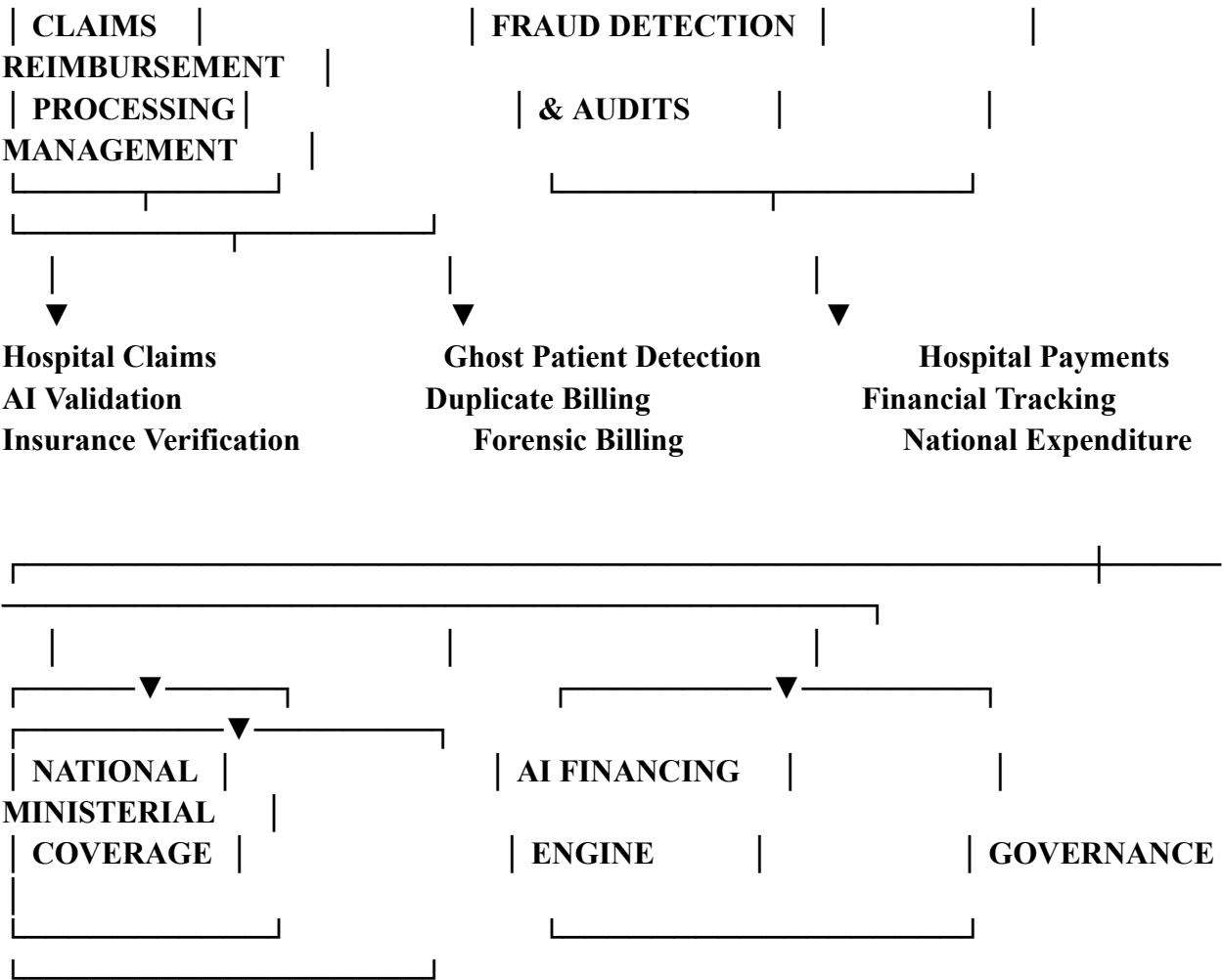
FAMILY COVERAGE:  
✓ Husband  
✓ 3 Children

RECENT CLAIMS:  
• CHUK Consultation  
• Laboratory Services  
• Pharmacy Dispensation

BIOMETRIC STATUS:  
✓ Fingerprint Verified  
✓ Facial Match Confirmed

14. FULL RSSB NATIONAL ARCHITECTURE





 **MEDBRIDGE+ — NISR NATIONAL HEALTH STATISTICS & INTELLIGENCE PORTAL**

**National Institute of Statistics of Rwanda (NISR) — Health Analytics Command Center**

**1. NISR DIRECTOR — MAIN ENTRY PORTAL**

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| MEDBRIDGE+ — NISR NATIONAL HEALTH STATISTICS COMMAND CENTER  
|

| Rwanda National Population, Disease & Healthcare Intelligence Platform  
|

+=====+  
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Director General:   Aimable Niyonsenga

Director ID:        NISR-DIR-0004

National Data Grid:   ● ACTIVE

AI Analytics Engine:   🧠 ENABLED

+-----+-----+-----+

| 📊 National Health Statistics | 🚨 Outbreak Intelligence   | 👥 Population Analytics  
|

| Mortality trends       | Epidemic forecasting       | Demographic mapping       |

| Hospital utilization     | Disease cluster detection   | Birth & death analytics     |

| National healthcare KPIs   | AI risk heatmaps           | Regional population trends   |

+-----+-----+-----+

|                                                                                                        |                                                                                                        |                                                                                                           |  |
|--------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|--|
| +-----+-----+-----+                                                                                    |                                                                                                        |                                                                                                           |  |
|  Facility Performance |  Research & Reporting |  Ministerial Reporting |  |
|                                                                                                        |                                                                                                        |                                                                                                           |  |
| Hospital scorecards                                                                                    | National survey dashboards                                                                             | Direct MoH escalation                                                                                     |  |
|                                                                                                        |                                                                                                        |                                                                                                           |  |
| Workforce analytics                                                                                    | AI statistical modeling                                                                                | RBC synchronization                                                                                       |  |
|                                                                                                        |                                                                                                        |                                                                                                           |  |
| Resource distribution                                                                                  | Exportable policy reports                                                                              | National emergency alerts                                                                                 |  |
| +-----+-----+-----+                                                                                    |                                                                                                        |                                                                                                           |  |

## 2. FULL NISR DASHBOARD LAYOUT

|                 |                             |       |
|-----------------|-----------------------------|-------|
| +=====          |                             |       |
| =====+          |                             |       |
| LEFT SIDEBAR    | CENTRAL ANALYTICS WORKSPACE | RIGHT |
| ANALYTICS PANEL |                             |       |
| +=====          |                             |       |
| =====+          |                             |       |



-  Population Analytics
-  Disease Heatmaps
-  AI Outbreak Alerts
-  Epidemic Intelligence
-  Facility Performance
-  Mortality Trends
-  Healthcare Statistics
-  Demographic Intelligence
-  Minister Notifications
-  Research & Surveys
-  Predictive Statistical Models
-  Regional Risk Maps
-  Minister Reporting

---

### 3. LEFT SIDEBAR — POPULATION ANALYTICS



NATIONAL POPULATION ANALYTICS

### CURRENT POPULATION ESTIMATES

#### 1. Kigali City

**Population: 1,982,211**

**Growth Rate: +3.4%**

**Status: ● STABLE**

## **2. Northern Province**

**Population: 2,114,008**

**Growth Rate: +2.1%**

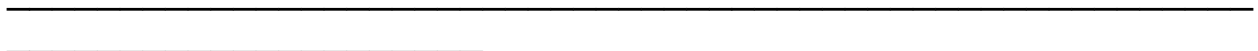
**Status: ● STABLE**

## **3. Southern Province**

**Population: 2,598,441**

**Growth Rate: +2.8%**

**Status: ● HIGH DENSITY**



 **HIGH-RISK DEMOGRAPHIC ALERTS**

**1. Under-5 Mortality Increase**

**Region:** Eastern Province

**Status:**  **CRITICAL**

**2. Maternal Mortality Spike**

**Region:** Rural Districts

**Status:**  **INVESTIGATION ACTIVE**

---

**4. CENTRAL PANEL — NATIONAL HEALTH METRICS**

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|  **NATIONAL HEALTH STATISTICS INTELLIGENCE CENTER**  
|

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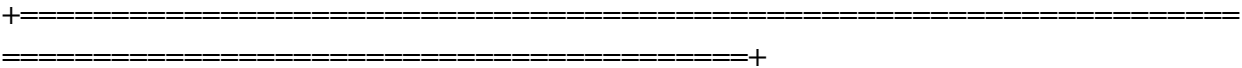
| +-----+-----+-----+-----+ |               |                  |            |
|---------------------------|---------------|------------------|------------|
| INDICATOR                 | CURRENT VALUE | TREND            | STATUS     |
| +-----+-----+-----+-----+ |               |                  |            |
| Maternal Mortality        | 178 /100,000  | ↓ Improving      | ● STABLE   |
| Under-5 Mortality         | 42 /1,000     | ↑ Rising         | ● ALERT    |
| Malaria Cases             | 22,411        | ↑ Seasonal Spike | ● HIGH     |
| Vaccination Coverage      | 91%           | Stable           | ● GOOD     |
| Hospital Occupancy        | 82%           | Increasing       | ● MODERATE |
| +-----+-----+-----+-----+ |               |                  |            |

**AI HEALTH ANALYSIS:**

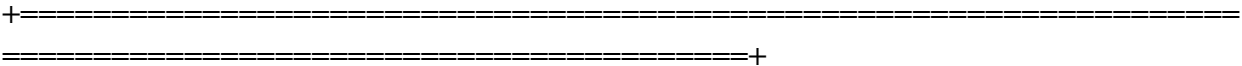
- ⚠ Malaria outbreak risk increasing
- ⚠ Rural maternal mortality clusters detected
- ⚠ Pediatric ICU demand forecast rising

---

5. NATIONAL DISEASE HEATMAP



|  NATIONAL DISEASE SURVEILLANCE HEATMAP  
|



KIGALI  STABLE

NORTHERN PROVINCE  MODERATE MALARIA

SOUTHERN PROVINCE  RESPIRATORY SURGE

EASTERN PROVINCE  MALARIA HOTSPOT

WESTERN PROVINCE  WATERBORNE ALERT

LEGEND:

 Stable

 Moderate Risk

 **High Risk**

 **Critical Outbreak**

---

**6. OUTBREAK INTELLIGENCE CENTER**

+=====+  
=====+

|  **NATIONAL EPIDEMIC & OUTBREAK INTELLIGENCE**  
|

+=====+  
=====+

**ACTIVE OUTBREAKS**

+-----+-----+-----+-----+

| **DISEASE** | **DISTRICT** | **CASES** | **STATUS** |

+-----+-----+-----+-----+

| **Malaria** | **Nyagatare** | **4,421** |  **CRITICAL** |

| Cholera | Rubavu | 112 |  CONTROL ACTIVE |

| Measles | Huye | 71 |  MONITORING |

+-----+-----+-----+-----+

AI OUTBREAK PREDICTIONS:

- ⚠ Flood-related cholera risk forecasted
- ⚠ Mosquito-borne disease expansion detected
- ⚠ Pediatric respiratory cases rising

-----

7. HOSPITAL PERFORMANCE SCORECARD

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=====+

|  NATIONAL HEALTH FACILITY PERFORMANCE  
|

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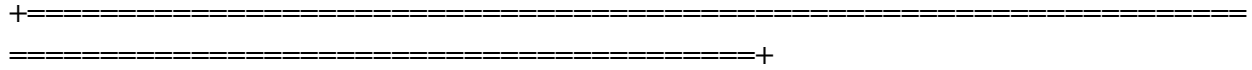
| +-----+-----+-----+-----+ |               |                |                    |
|---------------------------|---------------|----------------|--------------------|
| FACILITY                  | QUALITY SCORE | MORTALITY RATE | STATUS             |
| +-----+-----+-----+-----+ |               |                |                    |
| CHUK                      | 91%           | LOW            | ● HIGH PERFORMANCE |
| King Faisal               | 94%           | LOW            | ● EXCELLENT        |
| Butaro Hospital           | 83%           | MODERATE       | ● STABLE           |
| District Hospitals        | 68%           | RISING         | ● NEEDS REVIEW     |
| +-----+-----+-----+-----+ |               |                |                    |

**QUALITY ALERTS:**

- ⚠ Rural staffing shortages impacting quality
- ⚠ ICU overcrowding identified
- ⚠ Delayed emergency response patterns detected

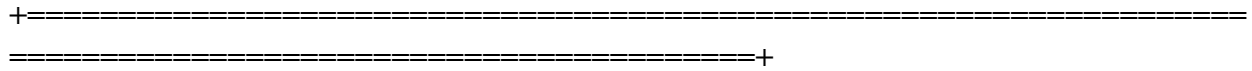
**8. RESEARCH & NATIONAL SURVEY CENTER**





## | NATIONAL RESEARCH & SURVEY ANALYTICS

|



### ACTIVE SURVEYS

✓ National Demographic Survey

✓ Maternal Health Assessment

✓ Child Nutrition Survey

✓ Hospital Workforce Study

✓ Disease Burden Analysis

### EXPORT OPTIONS:

 PDF Reports

 Excel Analytics

 **Interactive Dashboards**

 **GIS Population Maps**

## **AI RESEARCH TOOLS:**

- ✓ **Predictive Statistical Modeling**
  - ✓ **Trend Correlation Analysis**
  - ✓ **Mortality Forecasting**
  - ✓ **Healthcare Demand Forecasting**
- 

## **9. RIGHT PANEL — NISR EXECUTIVE ANALYTICS**

+=====+

|  **NISR EXECUTIVE INTELLIGENCE** |

+=====+

**TOTAL POPULATION:**      **14.2M**

**ACTIVE HEALTH FACILITIES: 612**

**CURRENT OUTBREAKS: 7**

**NATIONAL COVERAGE RATE: 91%**

**AI DETECTIONS:**

⚠ **Maternal mortality clusters**

⚠ **Pediatric malnutrition increase**

⚠ **Rising emergency admissions**

**MINISTER REPORTS:**

✓ **National disease update sent**

✓ **RBC synchronized**

✓ **Emergency epidemic alert active**

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**10. DIRECT REPORTING TO MINISTER**

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|  **NATIONAL STATISTICAL ESCALATION CHANNEL**

|

+=====+  
=====+

#### **DIRECT CONNECTIONS:**

- ✓ **Ministry of Health**
- ✓ **RBC Surveillance Unit**
- ✓ **Ministry of Local Government**
- ✓ **National Emergency Coordination**

#### **REPORT TYPES:**

- ✓ **National Mortality Trends**
- ✓ **Epidemic Intelligence Reports**
- ✓ **Maternal Health Emergencies**
- ✓ **Population Health Forecasts**

✓ **Rural Healthcare Gaps**

✓ **National Resource Allocation**

**LATEST REPORT:**

-----

**Subject:**

**Eastern Province Malaria Surge Escalation**

**Affected Districts:**

• **Nyagatare**

• **Kirehe**

• **Kayonza**

**Priority:**

 **NATIONAL EPIDEMIC ALERT**

Timestamp:

10:21 AM

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11. AI NATIONAL STATISTICS ENGINE

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=====+

| 🧠 AI NATIONAL HEALTH STATISTICS ENGINE  
|

+=====+  
=====+

ACTIVE AI DETECTIONS

- ⚠ **Maternal mortality risk clusters detected**
- ⚠ **Population migration affecting healthcare demand**
- ⚠ **Malnutrition increase predicted**
- ⚠ **ICU occupancy overload forecasted**
- ⚠ **Climate-linked disease spread identified**
- ⚠ **Rural healthcare access gaps detected**

#### **AI SYSTEMS:**

- ✓ **Predictive Analytics ACTIVE**
  - ✓ **Disease Modeling ACTIVE**
  - ✓ **GIS Heatmapping ACTIVE**
  - ✓ **Mortality Forecasting ACTIVE**
  - ✓ **Population Intelligence ACTIVE**
-

## **12. NATIONAL HEALTH INTELLIGENCE FLOW**

**Hospital & Community Data**



**National Data Aggregation**



**AI Statistical Analysis**



**NISR Intelligence Dashboard**



**Outbreak & Trend Detection**



**Ministerial Reporting**



**National Policy Decision**

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### 13. SMART DEMOGRAPHIC PROFILE SYSTEM

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#### | POPULATION DEMOGRAPHIC INTELLIGENCE |

+=====+  
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**DISTRICT:**           Nyagatare

**TOTAL POPULATION:**    621,442

**UNDER-5 POPULATION:**   88,221

**MATERNAL AGE GROUP:**   132,110

#### **HEALTH INDICATORS:**

✓ **Vaccination Coverage: 84%**

✓ **Skilled Birth Attendance: 71%**

⚠ **Malaria Incidence: HIGH**

⚠ **Child Malnutrition: RISING**

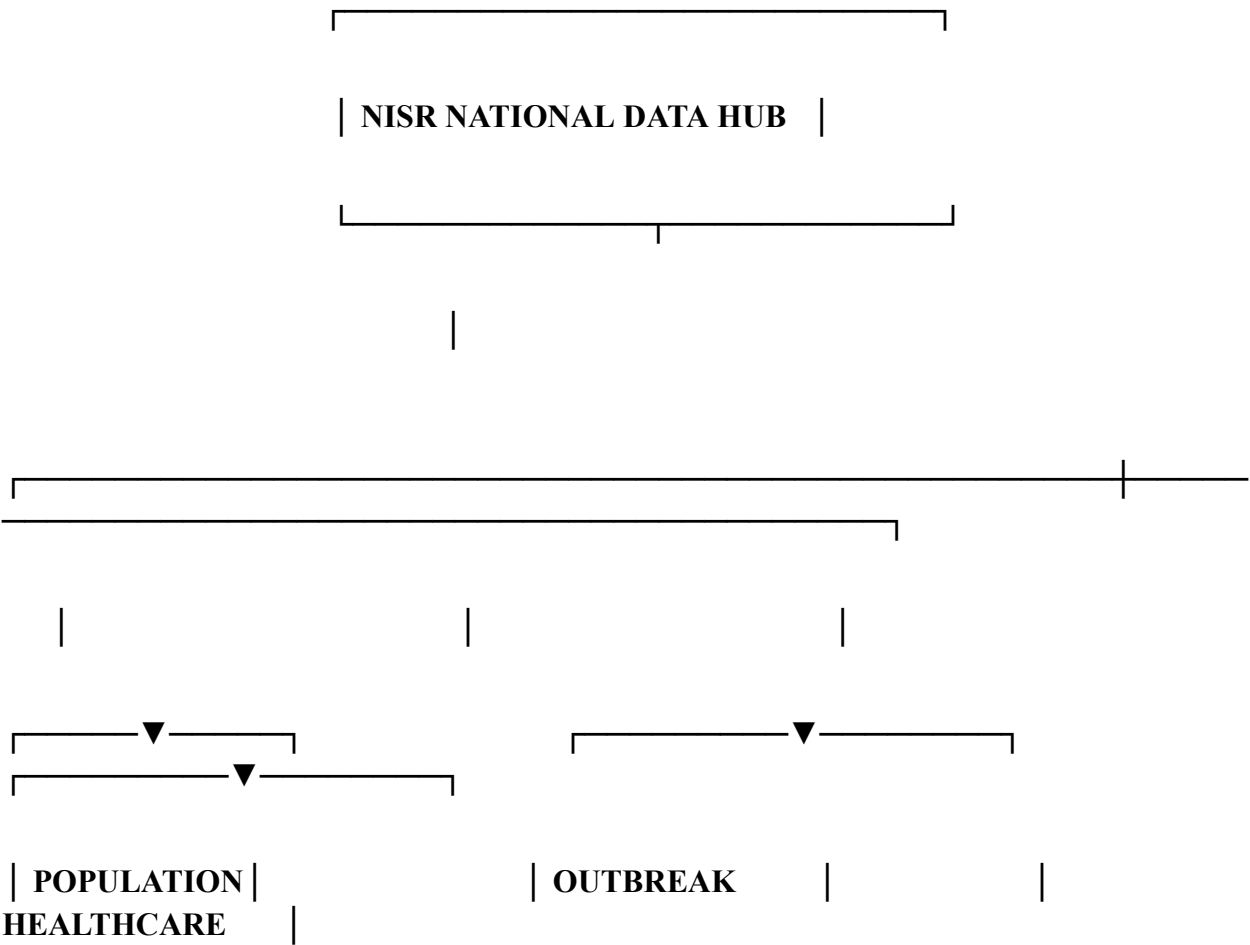
**AI INSIGHTS:**

⚠ **High pediatric healthcare demand forecasted**

⚠ **Waterborne disease risk elevated**

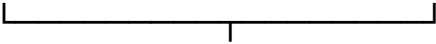
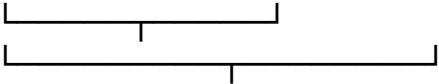
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**14. FULL NISR NATIONAL ARCHITECTURE**



| ANALYTICS |  
| PERFORMANCE |

| INTELLIGENCE |



|

|

|



Demographic Mapping

Epidemic Forecasting

Facility Scorecards

Birth & Death Trends

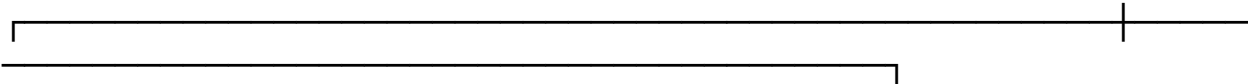
Disease Heatmaps

Workforce Analysis

Migration Analytics

AI Risk Detection

Resource Allocation



|

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|



| AI HEALTH |

| NATIONAL |

| MINISTERIAL

| STATISTICS |  
| GOVERNANCE |

| RESEARCH |

MEDBRIDGE+ — NATIONAL IMMUNIZATION & EPI INTELLIGENCE SYSTEM

Rwanda Expanded Program on Immunization (EPI) Command Dashboard

1. NATIONAL EPI COMMAND CENTER

| MEDBRIDGE+ — NATIONAL VACCINATION TRACKER & EXPANDED  
PROGRAM ON IMMUNIZATION (EPI) |  
| Rwanda National Immunization Intelligence & Outbreak Prevention System  
|

Director: Dr. Sandrine Mukayisenga  
Program ID: EPI-RW-0021  
Cold Chain Status: ● ACTIVE  
AI Immunization Engine: 🧠 ENABLED

|                                     |                                    |                                        |
|-------------------------------------|------------------------------------|----------------------------------------|
| <span>📊</span> Vaccination Coverage | <span>🚨</span> Outbreak Prevention | <span>🧊</span> Cold Chain Intelligence |
| National vaccine tracking           | Herd immunity monitoring           | Temperature monitoring                 |
| Child immunization records          | Measles hotspot detection          | Vaccine transport analytics            |
| Booster compliance analytics        | AI outbreak forecasting            | Freezer stability monitoring           |

|                                    |                                   |                                      |
|------------------------------------|-----------------------------------|--------------------------------------|
| <b>Child Protection Coverage</b>   | <b>National EPI Analytics</b>     | <b>Ministerial Reporting</b>         |
| <b>Under-5 immunization status</b> | <b>Rural dropout heatmaps</b>     | <b>Direct MoH escalation</b>         |
| <b>HPV adolescent programs</b>     | <b>Vaccine stock intelligence</b> | <b>RBC synchronization</b>           |
| <b>School vaccination programs</b> | <b>AI compliance forecasting</b>  | <b>Emergency immunization alerts</b> |

2. FULL EPI DASHBOARD LAYOUT

|                     |                                       |                              |
|---------------------|---------------------------------------|------------------------------|
| <b>LEFT SIDEBAR</b> | <b>CENTRAL IMMUNIZATION WORKSPACE</b> | <b>RIGHT ANALYTICS PANEL</b> |
|---------------------|---------------------------------------|------------------------------|

|                              |                                       |                               |
|------------------------------|---------------------------------------|-------------------------------|
| <b>Dashboard Home</b>        | <b>National Coverage Monitor</b>      | <b>EPI Reports</b>            |
| <b>Vaccination Coverage</b>  | <b>Immunization Heatmaps</b>          | <b>AI Outbreak Alerts</b>     |
| <b>Cold Chain Monitoring</b> | <b>Child Vaccination Registry</b>     | <b>Coverage Deficits</b>      |
| <b>Outbreak Surveillance</b> | <b>Vaccine Inventory Tracking</b>     | <b>Minister Notifications</b> |
| <b>National Analytics</b>    | <b>Predictive Immunization Models</b> | <b>Regional Risk Scores</b>   |
| <b>Minister Reporting</b>    |                                       |                               |

3. NATIONAL VACCINATION COVERAGE PANEL

|                                                                                  |
|----------------------------------------------------------------------------------|
| <b>NATIONAL VACCINATION TRACKER &amp; EXPANDED PROGRAM ON IMMUNIZATION (EPI)</b> |
|----------------------------------------------------------------------------------|

| VACCINE ANCHOR     | TARGET COVERAGE | ACTUAL COVERAGE | DEFICIT DETECTED |
|--------------------|-----------------|-----------------|------------------|
| BCG (Tuberculosis) | 99%             | 98.2%           | NEGLIGIBLE       |
| DTP-HepB-Hib       | 95%             | 94.1%           | RURAL DROPOUT    |
| MMR (Measles)      | 95%             | 88.5%           | CRITICAL HOLES   |
| PCV (Pneumococcal) | 95%             | 92.0%           | STOCK DELAYS     |
| HPV                | 93%             | 94.5%           | TARGET EXCEEDED  |

- AI IMMUNIZATION INSIGHTS:
- ⚠ Measles herd immunity collapse risk detected
  - ⚠ Booster dropout increasing in rural sectors
  - ⚠ Pneumococcal stock delays affecting outreach

#### 4. IMMUNIZATION HEATMAP SYSTEM

🌐 NATIONAL IMMUNIZATION HEATMAP

- KIGALI

● HIGH COVERAGE
- NORTHERN PROVINCE

● BOOSTER DROPOUT RISK
- SOUTHERN PROVINCE

● MEASLES IMMUNITY GAP
- EASTERN PROVINCE

● LOW BOOSTER COMPLIANCE
- WESTERN PROVINCE

● STABLE

- LEGEND:
- Safe Herd Immunity
  - Moderate Deficit
  - Booster Compliance Risk
  - Critical Outbreak Vulnerability

5. CHILD IMMUNIZATION REGISTRY

👶 CHILD IMMUNIZATION REGISTRY

CHILD PROFILE:

Name: Uwase Aline  
Gender: Female  
Age: 18 Months  
National ID: CHILD-009221  
Guardian: Mukamana Chantal

| VACCINE       | DATE GIVEN  | STATUS      | NEXT DOSE        |
|---------------|-------------|-------------|------------------|
| BCG           | 02 Jan 2025 | ✓ COMPLETED | NONE             |
| Pentavalent-1 | 02 Feb 2025 | ✓ COMPLETED | NEXT SCHEDULED   |
| Pentavalent-2 | 02 Mar 2025 | ✓ COMPLETED | NEXT SCHEDULED   |
| MMR           | MISSED      | ✗ OVERDUE   | URGENT FOLLOW-UP |

AI FOLLOW-UP ALERTS:

- ⚠ Child overdue for measles vaccine
- ⚠ SMS reminder sent to guardian
- ⚠ Community health worker follow-up scheduled

6. OUTBREAK SURVEILLANCE CENTER

🚨 IMMUNIZATION OUTBREAK SURVEILLANCE

ACTIVE IMMUNITY ALERTS

| DISTRICT  | VACCINE GAP        | RISK LEVEL              | STATUS              |
|-----------|--------------------|-------------------------|---------------------|
| Huye      | Measles Coverage   | <span>●</span> CRITICAL | OUTBREAK RISK       |
| Nyagatare | Booster Compliance | <span>●</span> HIGH     | ACTIVE MONITORING   |
| Rubavu    | HPV Dropout        | <span>●</span> MODERATE | COMMUNITY FOLLOW-UP |

- AI SURVEILLANCE:
- ⚠ Herd immunity threshold breached in Huye
  - ⚠ Under-2 booster compliance decreasing
  - ⚠ School vaccination campaign recommended

7. COLD CHAIN MONITORING CENTER

|                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------|
|  NATIONAL VACCINE COLD CHAIN INTELLIGENCE |
|------------------------------------------------------------------------------------------------------------------------------|

| STORAGE CENTER      | TEMPERATURE | VACCINE STATUS | ALERT                          |
|---------------------|-------------|----------------|--------------------------------|
| Kigali Central      | 2°C         | ✓ SAFE         | <span>●</span> STABLE          |
| Huye Regional Store | 9°C         | ⚠ RISK         | <span>●</span> FREEZER FAILURE |
| Rubavu Depot        | 4°C         | ✓ SAFE         | <span>●</span> STABLE          |
| Nyagatare Facility  | 7°C         | ⚠ UNSTABLE     | <span>●</span> INVESTIGATION   |

- AI COLD CHAIN ALERTS:
- ⚠ Vaccine potency risk detected
  - ⚠ Freezer instability identified
  - ⚠ Emergency transport relocation recommended



8. SCHOOL HPV PROGRAM MONITOR

🎓 NATIONAL SCHOOL HPV IMMUNIZATION PROGRAM

| DISTRICT | TARGET GIRLS | VACCINATED | STATUS           |
|----------|--------------|------------|------------------|
| Kigali   | 18,220       | 18,014     | 🟢 HIGH SUCCESS   |
| Musanze  | 8,422        | 8,100      | 🟢 STABLE         |
| Huye     | 7,991        | 6,112      | 🟡 LOW COMPLIANCE |

PROGRAM ALERTS:

- ⚠️ School outreach gaps detected
- ⚠️ Community resistance identified
- ⚠️ Awareness campaign escalation recommended

9. RIGHT PANEL — EPI EXECUTIVE ANALYTICS

📊 NATIONAL EPI EXECUTIVE INTELLIGENCE

TOTAL IMMUNIZED CHILDREN: 2.8M  
ACTIVE OUTBREAK RISKS: 6  
COLD CHAIN ALERTS: 3  
OVERDUE CHILDREN: 22,114

AI DETECTIONS:

- ⚠️ Measles outbreak risk rising
- ⚠️ Booster dropout increasing

⚠ Rural immunization inequity detected

**MINISTER REPORTS:**

- ✓ National outbreak escalation sent
- ✓ RBC synchronized
- ✓ Emergency vaccination campaign active

---

**10. DIRECT REPORTING TO MINISTER**

+=====

=====+

|  NATIONAL IMMUNIZATION ESCALATION CHANNEL

|

+=====

=====+

**DIRECT CONNECTIONS:**

- ✓ Ministry of Health
- ✓ RBC Immunization Division
- ✓ National Emergency Coordination
- ✓ District Vaccination Officers

**REPORT TYPES:**

- ✓ Herd Immunity Collapse Risks
- ✓ Vaccine Stock Emergencies
- ✓ Cold Chain Failures
- ✓ School Immunization Gaps
- ✓ Pediatric Outbreak Alerts
- ✓ Rural Coverage Deficits

**LATEST REPORT:**

-----

**Subject:**

**Huye District Measles Herd Immunity Collapse**

**Affected Areas:**

- Huye
- Gisagara
- Nyaruguru

**Priority:**

 **NATIONAL IMMUNIZATION ALERT**

**Timestamp:**

**11:14 AM**

## 11. AI IMMUNIZATION INTELLIGENCE ENGINE

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# AI IMMUNIZATION ANALYTICS ENGINE

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## ACTIVE AI DETECTIONS

- ⚠️ **Measles hotspot prediction detected**
- ⚠️ **Under-2 booster dropout risk increasing**
- ⚠️ **Vaccine hesitancy clusters identified**
- ⚠️ **Cold chain instability forecasted**
- ⚠️ **Rural outreach inefficiency identified**
- ⚠️ **School vaccination gap predicted**

## AI SYSTEMS:

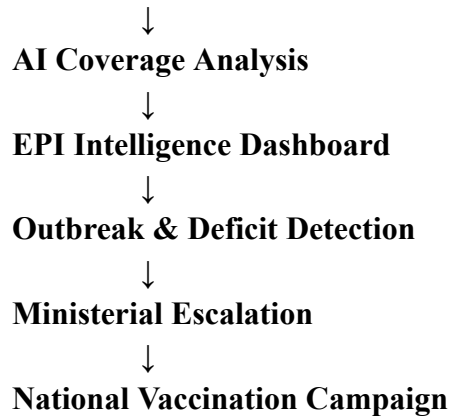
- ✓ Herd Immunity Modeling ACTIVE
- ✓ Outbreak Forecasting ACTIVE
- ✓ Vaccine Demand Prediction ACTIVE
- ✓ Child Follow-Up Intelligence ACTIVE
- ✓ Cold Chain AI Monitoring ACTIVE

## 12. IMMUNIZATION INTELLIGENCE FLOW

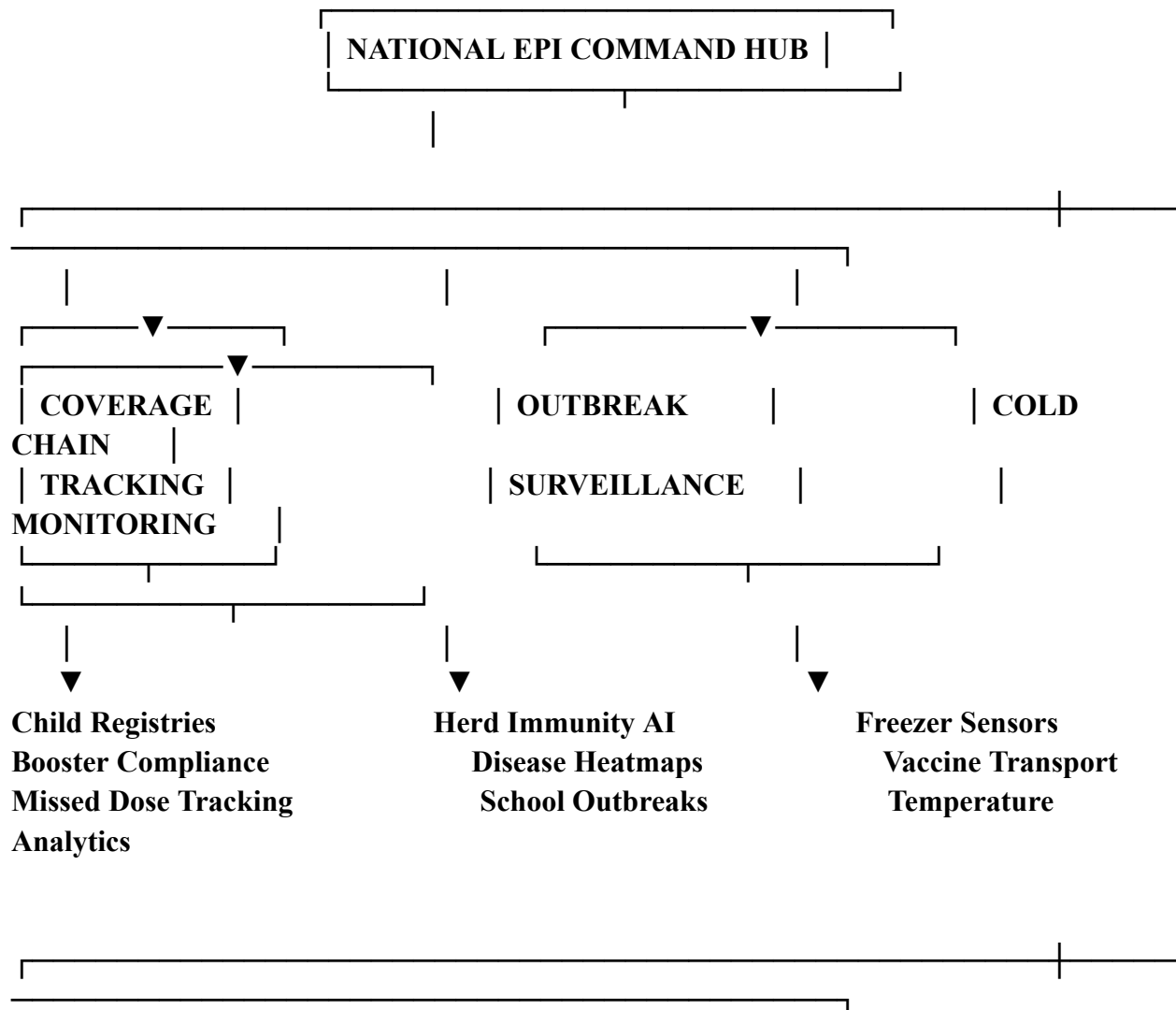
## Health Facility Vaccination Data



## National Immunization Registry



### 13. FULL NATIONAL EPI ARCHITECTURE





|                                                                                                               |                                                                                                             |                                                                                                                  |
|---------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
|  <b>Workforce Governance</b> |  <b>National Analytics</b> |  <b>Ministerial Reporting</b> |
| <b>Staffing compliance</b>                                                                                    | <b>National workforce heatmaps</b>                                                                          | <b>Direct MoH escalation</b>                                                                                     |
| <b>Rural deployment tracking</b>                                                                              | <b>Nurse-midwife ratios</b>                                                                                 | <b>RBC synchronization</b>                                                                                       |
| <b>Workforce shortages</b>                                                                                    | <b>AI workforce forecasting</b>                                                                             | <b>National emergency alerts</b>                                                                                 |

2. FULL NCNM DASHBOARD LAYOUT

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>LEFT SIDEBAR</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>CENTRAL GOVERNANCE WORKSPACE</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>RIGHT</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>ANALYTICS PANEL</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|  <b>Dashboard Home</b><br> <b>Licensing &amp; Renewal</b><br><b>Alerts</b><br> <b>Discipline &amp; Ethics</b><br> <b>CPD &amp; Accreditation</b><br><b>Notifications</b><br> <b>National Analytics</b><br> <b>Minister Reporting</b> |  <b>National Workforce Registry</b><br> <b>Midwifery Practice Monitoring</b><br> <b>Facility Staffing Compliance</b><br> <b>Internship &amp; Training Oversight</b><br> <b>Predictive Workforce Models</b> |  <b>National Reports</b><br> <b>AI Compliance</b><br> <b>Workforce Trends</b><br> <b>Minister</b><br> <b>Regional Heatmaps</b> |

3. NATIONAL LICENSE REGISTRY

|                                                                                                                                          |
|------------------------------------------------------------------------------------------------------------------------------------------|
|  <b>NATIONAL NURSE &amp; MIDWIFE LICENSE REGISTRY</b> |
|------------------------------------------------------------------------------------------------------------------------------------------|

ACTIVE LICENSES

| PROFESSIONAL         | LICENSE TYPE       | STATUS          | EXPIRY DATE |
|----------------------|--------------------|-----------------|-------------|
| Chantal Uwimana      | Registered Nurse   | ✓ ACTIVE        | 14 Jan 2027 |
| Alice Mukandayisenga | Registered Midwife | ✓ ACTIVE        | 03 Mar 2026 |
| Eric Habimana        | ICU Nurse          | ✓ ACTIVE        | 11 Jul 2027 |
| Olive Uwase          | Pediatric Midwife  | ⚠ EXPIRING SOON | 22 Jun 2025 |

#### AI LICENSING ALERTS:

- ⚠ 221 licenses expiring within 60 days
- ⚠ Rural facilities using temporary staff licenses
- ⚠ Internship supervision gaps identified

## 4. MIDWIFERY PRACTICE MONITORING

### 🧑‍⚕️ NATIONAL MIDWIFERY PRACTICE MONITOR

| FACILITY            | MONTHLY DELIVERIES | MIDWIVES AVAILABLE | STATUS     |
|---------------------|--------------------|--------------------|------------|
| CHUK                | 1,201              | 42                 | ● STABLE   |
| Kibagabaga Hospital | 821                | 24                 | ● STABLE   |
| Nyagatare District  | 611                | 11                 | ● SHORTAGE |
| Ruhango Hospital    | 411                | 7                  | ● CRITICAL |

#### MATERNAL CARE ALERTS:

- ⚠ Midwife-to-delivery ratio unsafe in Ruhango
- ⚠ Rural emergency obstetric coverage declining
- ⚠ Night shift shortages increasing

5. PROFESSIONAL DISCIPLINE & ETHICS CENTER

PROFESSIONAL DISCIPLINE & ETHICS INVESTIGATION CENTER

ACTIVE INVESTIGATIONS

| CASE ID   | FACILITY          | INCIDENT              | STATUS        |
|-----------|-------------------|-----------------------|---------------|
| NCNM-8821 | District Hospital | Medication Negligence | ACTIVE        |
| NCNM-7721 | Maternity Ward    | Scope Violation       | REVIEW        |
| NCNM-9912 | Health Center     | Ethical Misconduct    | INVESTIGATION |

FORENSIC FINDINGS:

- ⚠ Delayed emergency maternal response detected
- ⚠ Unauthorized clinical procedure identified
- ⚠ Documentation inconsistencies flagged

6. CPD & TRAINING ACCREDITATION CENTER

CONTINUOUS PROFESSIONAL DEVELOPMENT (CPD) & TRAINING ACCREDITATION

| TRAINING PROGRAM | PARTICIPANTS | COMPLIANCE RATE | STATUS |
|------------------|--------------|-----------------|--------|
|------------------|--------------|-----------------|--------|

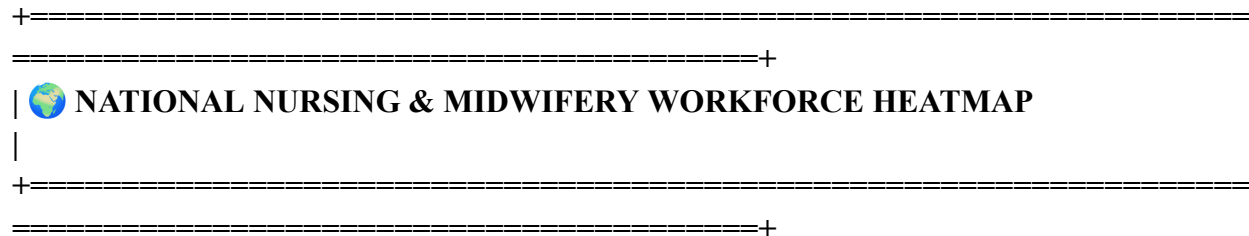


|                               |     |                                                                                                      |  |
|-------------------------------|-----|------------------------------------------------------------------------------------------------------|--|
| Emergency Obstetrics   2,114  | 91% |  GOOD              |  |
| Neonatal Resuscitation  1,821 | 84% |  MODERATE          |  |
| Infection Prevention   4,422  | 96% |  EXCELLENT         |  |
| Mental Health Care   811      | 62% |  LOW PARTICIPATION |  |
| +-----+-----+-----+           |     |                                                                                                      |  |

#### AI TRAINING ALERTS:

- ⚠ Rural CPD participation declining
- ⚠ Maternal emergency training gaps detected
- ⚠ High neonatal risk facilities need refresher courses

## 7. NATIONAL WORKFORCE HEATMAP

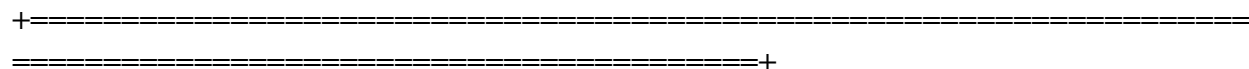


|                   |                                                                                                       |
|-------------------|-------------------------------------------------------------------------------------------------------|
| KIGALI            |  ADEQUATE STAFFING |
| NORTHERN PROVINCE |  MODERATE SHORTAGE |
| SOUTHERN PROVINCE |  MIDWIFE DEFICIT   |
| EASTERN PROVINCE  |  CRITICAL SHORTAGE |
| WESTERN PROVINCE  |  HIGH WORKLOAD     |

#### LEGEND:

-  Adequate Coverage
-  Moderate Pressure
-  Workforce Deficit
-  Critical Staffing Crisis

## 8. INTERNSHIP & STUDENT OVERSIGHT CENTER



| 🎓 **NURSING & MIDWIFERY INTERNSHIP OVERSIGHT**

|  |
|--|
|  |
|  |

**ACTIVE INTERNS**

|                |                     |                      |                   |  |
|----------------|---------------------|----------------------|-------------------|--|
|                |                     |                      |                   |  |
| STUDENT        | TRAINING FACILITY   | SUPERVISOR           | STATUS            |  |
| Aline Mukamana | CHUK                | Chantal Uwimana      | ✓ ACTIVE          |  |
| Jean Bosco     | Kibagabaga Hospital | Alice Mukandayisenga | ✓ ACTIVE          |  |
| Sandrine Uwera | Ruhango Hospital    | Eric Habimana        | ⚠ SUPERVISION GAP |  |
|                |                     |                      |                   |  |

**OVERSIGHT ALERTS:**

- ⚠ Internship supervision ratio exceeded
- ⚠ Rural student mentorship shortages
- ⚠ Clinical competency delays identified

**9. RIGHT PANEL — NCNM EXECUTIVE ANALYTICS**

|                               |
|-------------------------------|
|                               |
| 📊 NCNM EXECUTIVE INTELLIGENCE |
|                               |

**TOTAL LICENSED NURSES:** 18,422  
**TOTAL LICENSED MIDWIVES:** 4,211  
**ACTIVE INVESTIGATIONS:** 28  
**CPD COMPLIANCE RATE:** 89%

**AI DETECTIONS:**

- ⚠ Rural staffing crisis rising
- ⚠ Midwife burnout increasing
- ⚠ Maternal emergency skill gaps identified

**MINISTER REPORTS:**

- ✓ Workforce shortage escalation sent

- ✓ RBC synchronized
- ✓ Emergency staffing alert active

---

## 10. DIRECT REPORTING TO MINISTER

=====+  
=====+  
|  NATIONAL PROFESSIONAL GOVERNANCE ESCALATION CHANNEL  
|  
+=====+  
=====+

### DIRECT CONNECTIONS:

- ✓ Ministry of Health
- ✓ Rwanda Biomedical Center
- ✓ National Licensing Authorities
- ✓ Hospital Governance Units

### REPORT TYPES:

- ✓ National Workforce Shortages
- ✓ Maternal Care Emergencies
- ✓ Ethical Misconduct Investigations
- ✓ Licensing Compliance Gaps
- ✓ Rural Midwife Deficits
- ✓ Emergency Workforce Deployment

### LATEST REPORT:

-----

#### Subject:

Eastern Province Midwife Workforce Crisis

#### Affected Districts:

- Nyagatare
- Kirehe
- Gatsibo

#### Priority:

 NATIONAL MATERNAL HEALTH ALERT

**Timestamp:**  
**12:11 PM**

## 11. AI WORKFORCE GOVERNANCE ENGINE

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# | 🧠 AI NURSING & MIDWIFERY GOVERNANCE ENGINE

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## ACTIVE AI DETECTIONS

- ⚠️ **Midwife burnout risk identified**
- ⚠️ **Unsafe staffing ratios forecasted**
- ⚠️ **Rural deployment imbalance detected**
- ⚠️ **Maternal emergency skill gaps predicted**
- ⚠️ **Internship supervision deficits identified**
- ⚠️ **CPD non-compliance clusters detected**

## AI SYSTEMS:

- ✓ **Workforce Forecasting ACTIVE**
- ✓ **Ethical Compliance ACTIVE**
- ✓ **Licensing Intelligence ACTIVE**
- ✓ **Maternal Safety Analytics ACTIVE**
- ✓ **Rural Deployment Modeling ACTIVE**

## 12. NATIONAL WORKFORCE GOVERNANCE FLOW

## Health Facility Workforce Data



↓  
**Professional Risk Detection**  
↓  
**Ministerial Escalation**  
↓  
**National Workforce Intervention**

---

### 13. SMART PROFESSIONAL PROFILE

|                                                                                                                       |  |
|-----------------------------------------------------------------------------------------------------------------------|--|
| +=====                                                                                                                |  |
| =====+                                                                                                                |  |
|  <b>PROFESSIONAL LICENSE PROFILE</b> |  |
| +=====                                                                                                                |  |
| =====+                                                                                                                |  |

**FULL NAME:** Alice Mukandayisenga  
**PROFESSION:** Registered Midwife  
**LICENSE NUMBER:** MW-RW-88211  
**FACILITY:** Kibagabaga Hospital

**QUALIFICATIONS:**  
✓ Bachelor of Midwifery  
✓ Neonatal Resuscitation Certified  
✓ Emergency Obstetrics Certified

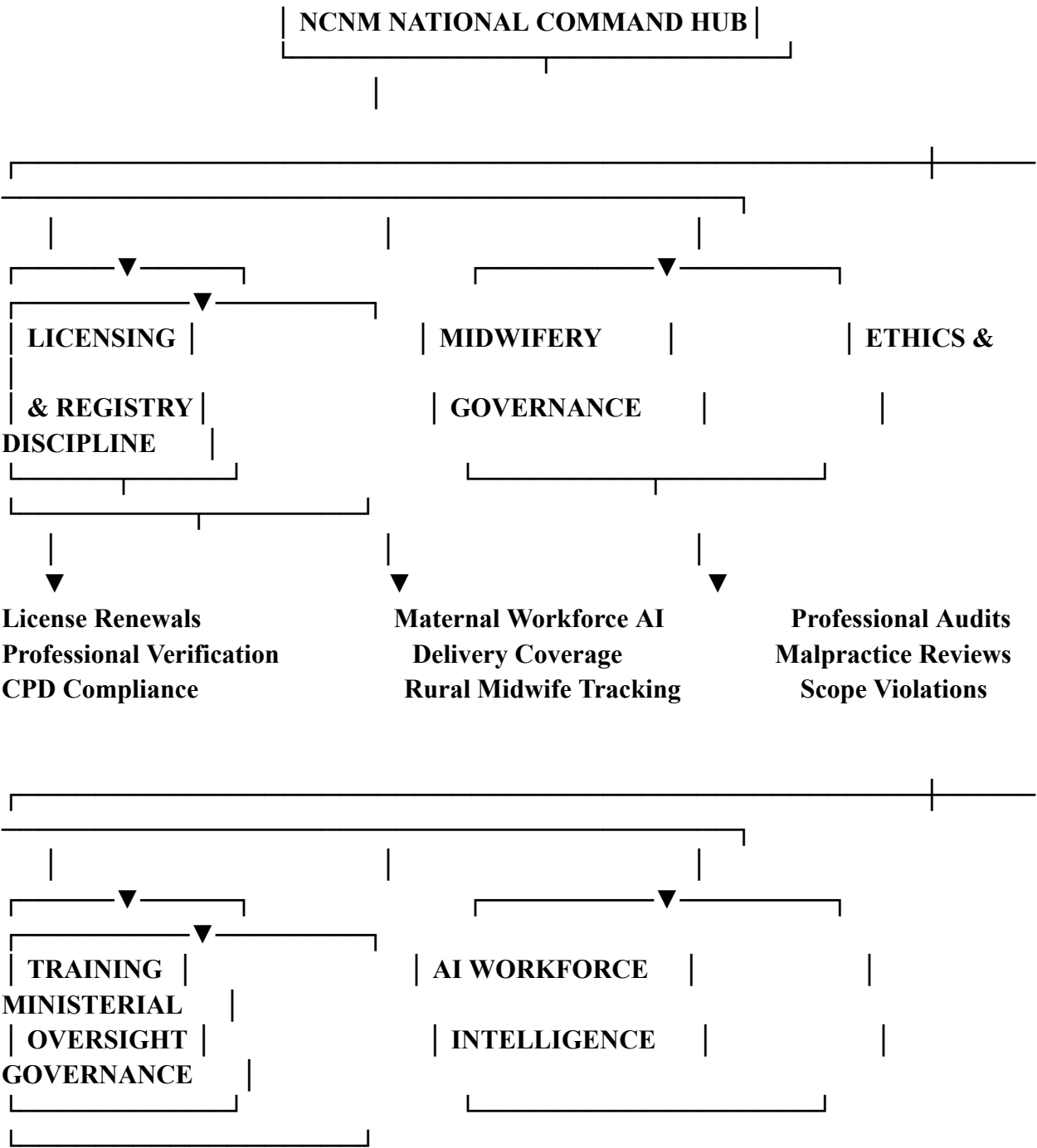
**CPD STATUS:**  
✓ Completed 2025 Requirements

**DISCIPLINE STATUS:**  
✓ No Active Investigations

**AI RISK PROFILE:**  
● **SAFE PRACTICE STATUS**

---

### 14. FULL NCNM NATIONAL ARCHITECTURE



**MEDBRIDGE+ — OPHTHALMOLOGIST DASHBOARD UI/UX BLUEPRINT**

**Eye Care, Vision Diagnostics & Ophthalmic Surgery Workspace**

1. OPHTHALMOLOGIST MAIN DASHBOARD

MEDBRIDGE+ | OPHTHALMOLOGY PORTAL

Dr. Claude Ndayambaje

Rwanda Military Hospital

Ophthalmologist

LEFT SIDEBAR

MAIN OPHTHALMOLOGY WORKSPACE

RIGHT SIDEBAR

New Eye Patients

Emergency Eye Cases

Vision Assessments

Eye Surgery Queue

Eye Imaging & OCT

Follow-Up Visits

Optical Prescriptions

Eye Care Reports

• Patients Today: 58

• Surgeries: 5

• Cataract Cases: 18

• Emergencies: 7

• AI Alerts: 4

2. LEFT PANEL — NEW EYE PATIENTS

NEW EYE PATIENTS

1. Uwase Marie

Patient ID: EYE-2026-00124  
COMPLAINT: Sudden Vision Loss  
TRIAGE:  EMERGENCY

2. Niyonsenga Eric

Patient ID: EYE-2026-00125  
COMPLAINT: Eye Pain & Redness  
TRIAGE:  URGENT

### 3. Mukamana Alice

**Patient ID:** EYE-2026-00126  
**COMPLAINT:** Blurred Vision  
**TRIAGE:** ● **MODERATE**

## COMPLAINT: Blurred Vision

**TRIAGE: 🟡 MODERATE**

### 3. OPHTHALMOLOGY PRIORITY COLORS

| Color                                                                                    | Meaning                      | Priority              |
|------------------------------------------------------------------------------------------|------------------------------|-----------------------|
|  Red    | Vision-threatening emergency | Immediate care        |
|  Orange | Serious eye condition        | Urgent evaluation     |
|  Yellow | Moderate eye issue           | Scheduled review      |
|  Green  | Routine eye consultation     | Normal queue          |
|  Black  | Permanent blindness/trauma   | Critical intervention |

#### 4. WHEN PATIENT PROFILE IS CLICKED

=====+  
=====+  
| **OPHTHALMOLOGY PATIENT PROFILE** |  
+=====+  
=====+

**FULL NAMES:** Uwase Marie

**GENDER:** Female

**AGE:** 54 Years

**PATIENT ID:** EYE-2026-00124

**NATIONAL ID:** 1199880076543210

**PHONE NUMBER:** +250 788 223 114

**ADDRESS:** Gasabo District, Kigali



HEALTH INSURANCE:    RSSB / Mutuelle de Santé

KNOWN CONDITIONS:

- Diabetes Mellitus
- Hypertension

TRIAGE STATUS:         OPHTHALMIC EMERGENCY

\_\_\_\_\_

5. OPHTHALMOLOGY CONSULTATION SUMMARY

|                            |  |
|----------------------------|--|
| +=====                     |  |
| =====+                     |  |
| OPHTHALMOLOGY CONSULTATION |  |
| +=====                     |  |
| =====+                     |  |

CHIEF COMPLAINT:  
Sudden painless loss of vision in left eye

- SYMPTOMS:
- Blurred vision
  - Floaters
  - Difficulty recognizing objects

- PREVIOUS EYE HISTORY:
- Diabetic retinopathy
  - Previous cataract surgery

PROVISIONAL DIAGNOSIS:  
⚠ Retinal Detachment Suspected

\_\_\_\_\_

6. VISION ASSESSMENT PANEL

|                          |  |
|--------------------------|--|
| +=====                   |  |
| =====+                   |  |
| VISUAL ACUITY ASSESSMENT |  |

+=====+  
=====+

**RIGHT EYE (OD):**

**6/9**

**LEFT EYE (OS):**

**Counting Fingers Only ⚠**

**INTRAOCULAR PRESSURE:**

**OD: 14 mmHg**

**OS: 18 mmHg**

**COLOR VISION:**

**Normal**

**AI ANALYSIS:**

**⚠ Severe visual impairment detected**

\_\_\_\_\_

## **7. EYE EXAMINATION PANEL**

+=====+  
=====+

**| EYE EXAMINATION**

**|**

+=====+  
=====+

**ANTERIOR SEGMENT:**

☒ **Clear Cornea**

☒ **Mild Cataract**

**POSTERIOR SEGMENT:**

**⚠ Retinal tear suspected**

**⚠ Vitreous hemorrhage**

**PUPIL RESPONSE:**

**Sluggish in left eye**

**RETINAL STATUS:**

### **Detached Retina Possible**

## 8. OPHTHALMIC IMAGING PANEL

## OPHTHALMIC IMAGING

**IMAGING REQUESTED:**

- ☒ Fundoscopy
- ☒ OCT Scan
- ☒ Retinal Photography
- ☐ CT Orbit Scan

**IMAGING STATUS:**

-  **Uploaded Successfully**

### AI IMAGE ANALYSIS:

- ⚠ **Retinal detachment probability: 92%**
- ⚠ **Diabetic retinal changes detected**

## 9. EYE PROCEDURES PANEL

## | OPHTHALMIC PROCEDURES

| +-----+-----+-----+    |                                               |               |
|------------------------|-----------------------------------------------|---------------|
| PROCEDURE              | STATUS                                        | NOTES         |
| +-----+-----+-----+    |                                               |               |
| Retinal Laser Therapy  | <input checked="" type="checkbox"/> Planned   | Urgent        |
| Cataract Surgery       | <input type="checkbox"/> Follow-Up            | Future review |
| Intravitreal Injection | <input checked="" type="checkbox"/> Scheduled | Today         |

+-----+-----+-----+

=====

10. OPTICAL PRESCRIPTION PANEL

+=====

=====+

| OPTICAL PRESCRIPTION |

+=====

=====+

+-----+-----+-----+

| EYE | SPHERE | CYLINDER |

+-----+-----+-----+

| RIGHT (OD) | -1.50 | -0.50 |

| LEFT (OS) | -2.00 | -1.00 |

+-----+-----+-----+

ADDITIONAL NOTES:

Anti-reflective lenses recommended

=====

11. OPHTHALMIC SURGERY PANEL

+=====

=====+

|  EYE SURGERY |

+=====

=====+

SURGERY TYPE:

Retinal Repair Surgery

SURGEON:

Dr. Claude Ndayambaje

ANESTHESIA:

☒ Local Anesthesia

☐ General Anesthesia

**SURGERY STATUS:**  
● Emergency Surgery Scheduled

**OPERATING ROOM:**  
Ophthalmic Theater 1

**12. AI OPHTHALMOLOGY ASSISTANT**

|                               |  |
|-------------------------------|--|
| +=====                        |  |
| =====+                        |  |
| AI OPHTHALMOLOGY INTELLIGENCE |  |
| +=====                        |  |
| =====+                        |  |

- AI DETECTED RISKS:**
- ⚠ High probability retinal detachment
  - ⚠ Diabetes-related retinal damage
  - ⚠ Risk of permanent blindness

- AI RECOMMENDATIONS:**
- Immediate retinal surgery advised
  - Tight diabetic control required
  - Emergency retina specialist review

**COMMUNITY ALERT:**  
⚠ Increasing diabetic eye disease cases

**13. FOLLOW-UP VISITS PANEL**

|                  |  |
|------------------|--|
| +=====           |  |
| =====+           |  |
| FOLLOW-UP VISITS |  |
| +=====           |  |
| =====+           |  |

**LAST VISIT:**

18 March 2026

LAST PROCEDURE:  
Laser Photocoagulation

FOLLOW-UP STATUS:  
⚠ Vision deterioration observed

NEXT APPOINTMENT:  
25 May 2026

14. OPHTHALMIC MATERIALS & CONSUMABLES

|                           |  |
|---------------------------|--|
| OPHTHALMIC MATERIAL USAGE |  |
|---------------------------|--|

| MATERIAL         | QUANTITY USED | STATUS     |
|------------------|---------------|------------|
| Intraocular Lens | 1 Unit        | ✓ Recorded |
| Eye Drops        | 2 Bottles     | ✓ Recorded |
| Sterile Eye Kit  | 1 Set         | ✓ Recorded |

15. OPHTHALMOLOGY REPORTS & ANALYTICS

|                                                                                                             |  |
|-------------------------------------------------------------------------------------------------------------|--|
|  OPHTHALMOLOGY ANALYTICS |  |
|-------------------------------------------------------------------------------------------------------------|--|

TODAY SUMMARY

|                    |    |
|--------------------|----|
| Eye Consultations: | 58 |
| Cataract Cases:    | 18 |

**Retinal Emergencies:** 7  
**Surgeries Performed:** 5

## COMMON CONDITIONS

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**Cataracts** 24  
**Glaucoma** 11  
**Diabetic Retinopathy** 9

## AI ALERTS

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- ⚠ Increasing diabetic eye disease
  - ⚠ Delayed cataract follow-ups
  - ⚠ Rising glaucoma detection rate
- 

## 16. OPHTHALMOLOGIST ACCESS LIMITATIONS

|                    |  |
|--------------------|--|
| +=====             |  |
| =====+             |  |
| ACCESS LIMITATIONS |  |
| +=====             |  |
| =====+             |  |

### ALLOWED:

- ☒ Perform eye consultations
- ☒ Order ophthalmic imaging
- ☒ Conduct eye surgeries
- ☒ Prescribe optical medications
- ☒ Access eye health records

### PROHIBITED:

- ☒ Modify unrelated clinical files
  - ☒ Access forensic investigations
  - ☒ Delete patient medical records
  - ☒ Access unrelated psychiatric notes
  - ☒ Modify hospital financial systems
-

## 17. NATIONAL EYE HEALTH NETWORK

|                                     |  |
|-------------------------------------|--|
| =====+                              |  |
| =====+                              |  |
| NATIONAL OPHTHALMOLOGY CONNECTIVITY |  |
|                                     |  |
| +=====                              |  |
| =====+                              |  |

### CONNECTED SYSTEMS:

- National Eye Health Registry
- Ophthalmic Imaging Systems
- Hospital Surgical Units
- RSSB Insurance Platform
- RBC Blindness Prevention Program

### SYNC STATUS:

☒ Ophthalmology synchronization active

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## 18. OPHTHALMOLOGIST TABLET / MOBILE VIEW

## 19. FINAL OPHTHALMOLOGIST DASHBOARD STRUCTURE

### LEFT PANEL

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New Eye Patients  
Emergency Cases  
Eye Imaging  
Follow-Up Visits  
Surgery Queue

### CENTER PANEL

-----

Eye Patient Profile  
Vision Assessment  
Eye Examination  
Imaging Results



**Surgical Management**

**RIGHT PANEL**

-----

**Eye Care Reports**

**AI Vision Alerts**

**Surgical Analytics**

**Blindness Monitoring**

**Material Consumption**